

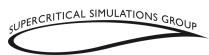
WARNING

THIS MANUAL IS ONLY INTENDED FOR USERS OF THE
SUPERCritical SIMULATIONS GROUP (SSG) B747-8 ADD-ON FOR X-
PLANE11&12. THE INFORMATION CONTAINED IN THIS DOCUMENT
IS NOT SUITABLE FOR ANY OTHER USE.

747-8

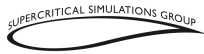
Flight Crew Operations Manual

Software Version: 2.6



SSG B747-8 SERIES

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SSG B747-8 SERIES

Table of Contents

Volume I

Introduction	0.1
Abbreviations	0.2
Revision Record	0.3
Operational Limitations	L
Performance Dispatch	PD
Performance Inflight	PI

Volume II

Airplane General, Emergency Equipment, Doors, Windows	1
Air Systems	2
Anti-Ice, Rain	3
Automatic Flight	4
Communications	5
Electrical	6
Engines, APU	7
Fire Protection	8
Flight Controls	9
Flight Instruments, Displays	10
Flight Management, Navigation	11
Fuel	12
Hydraulics	13
Landing Gear	14
Warning Systems	15

Volume III

Options Menu. Tablet Use	1
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SSG B747-8 SERIES

VOLUME I:

0.1 Introduction

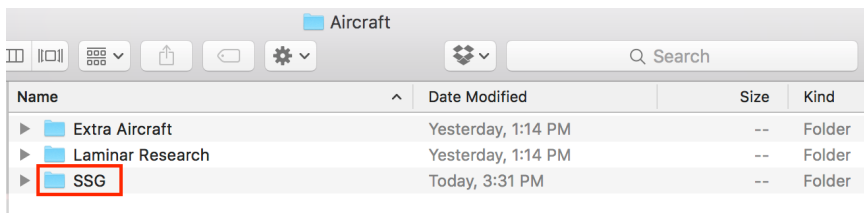
This Flight Crew Operations Manual (FCOM) is exclusively for the Supercritical Simulations Group (SSG) 747-8 Series Edition (Version 2) aircraft designed for the X-Plane (XP) desktop flight simulator. The information within this FCOM applies to both the passenger and freighter version of the aircraft, which are designated by Boeing as the “Intercontinental” (747-8i) and “Freighter” (747-8F) respectively. This FCOM is in no way a complete reference for the real 747-8 aircraft. Therefore, this FCOM cannot be used as a reference for operating any real aircraft.

0.1.1 General

The SSG 747-8 version 2.6 add-on is designed to work within X-plane 12 for Mac, Windows and some Linux.

0.1.4 Installation

For installation into XP, SSG recommends creating a folder called “SSG” for present and future aircraft released by SSG as shown in the screenshot below.



To install the add-on, unzip the downloaded file and simply copy/move the aircraft folder from the unzipped files under X-Plane12/Aircraft/

Install the navigation database as described in 0.1.8 below.

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To fly the aircraft simply select it as you would normally from the aircraft menu and for first time installations (and updates) you will have to go through the activation process. Please note that the Intercontinental will be listed under “Airliner” in the main XP menu while the Freighter will be listed under “Cargo”.

The activation of the ACF is an on-line method that requires an active Internet connection method that are described below.

This is the initial dialog box that appears once the aircraft is first loaded:



The character sequence should look like the one shown below.

XX-XXXXXX-XXXXXX-XXXXXX-XXXXXX

Note that you can click on the “Paste” button if you have previously copied the serial number to avoid having to retype it.

After a successful activation process, restart the ACF or reload using the Developer menu by selecting either “Reload the Current Aircraft and Art” options.

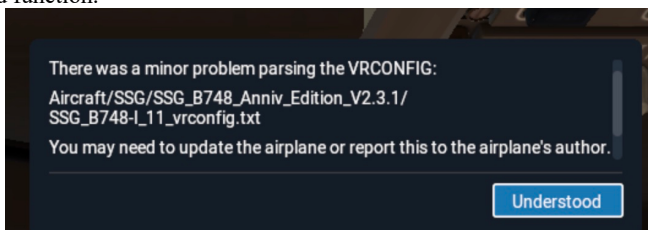
SSG B747-8 SERIES

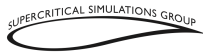


As a reminder, you will have to re-enter this serial number with any updates or if you change computers or reinstall XP, so please keep it in a safe place. If you have any issues with serial numbers or the installation of the aircraft, please contact us via the support electronic mail address shown above.

We recommend reloading the aircraft with textures until no more warnings are shown. This process takes some attempts until all the components get installed.

Note that some users may see a warning from XP about the “VRCONFIG” file (see screenshot below). This will only appear upon an initial installation and can be ignored, it is not a bad function.





SSG B747-8 SERIES

0.1.5 Support Information

For additional support, please visit our forum at X-Plane.ORG at the following URL:
<https://forums.x-plane.org/index.php?/forums/forum/206-boeing-747-8/>

News and update information from SSG are also available on our web site at the following URL:

<http://www.supercriticalsimulation.com>

Or contact our support :

support@supercriticalsimulation.com

0.1.6 System Requirements

X-Plane 11 v2.5.1

X-Plane 12 v2.6.x

Windows , Mac and Linux

8th generation I5 or higher / Rizen 3 or higher

Dedicated video card with vulkan/metal support 8 GB VRAM Minimum

Also, hardware such as a joystick with 3 axes plus 4 buttons is recommended.

0.1.7 Joystick/Controller Configuration

Included in this package is a list of custom command assignments that can be set up within XP to configure commands by joysticks, yokes, throttles, and other controllers for ACF. This list is in a text filed called SSGCUSTOMLIST.TXT and is included into the aircraft folder/documents.

0.1.9 Navigation Database Installation

The SSG 747-8 comes with a navigation database for use with the FMC supplied by Navigraph. There are some considerations for users to operate the aircraft most effectively that require some explanation. This is particularly important for those who wish to fly on line or follow real world flight plans. The most important thing to understand is that the FMC has its own database and it can be updated, but XP also has databases of navigation data (which can also be updated through various methods). Not everyone wants to invest the time and resources to keep updating these databases, but problems can occur when these two sets of navigation data are not at the same revision level.

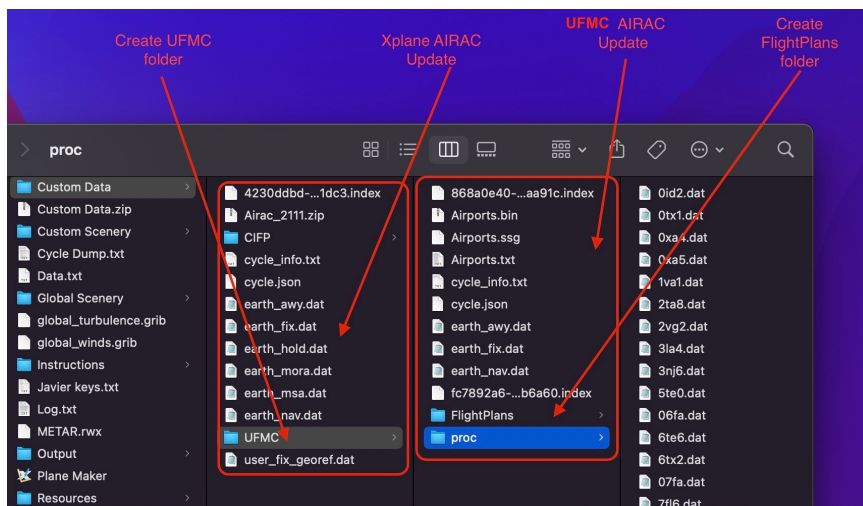
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NOTE: If an AIRAC navigation database is not installed, the FMC will show a message to this effect in the scratchpad. For this add-on to work properly, you need to install a navigation database.

If you have not already installed an UFMC navidata package follow this guide.

- 1 - Unzip the file “AIRAC_23xx_manual_install.zip”
- 2 - Copy the content to “X-plane 12/Custom Data/”

The folders Structure must be as follow:



SSG B747-8 SERIES

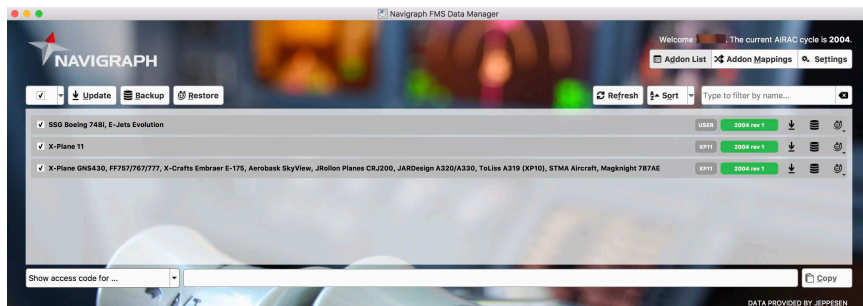
0.1.10 Navigation Database Update Instructions

It is also possible to install navigation dataset updates available through a paid subscription from Navigraph, which can be installed in 2 ways.

To update the Navigraph database manually, go to www.navigraph.com and log in. Then click the link for the “FMS Data Manager.” On the right-side panel select “Manual Install.” Once there, scroll down the page and select the “SSG Boeing 748i,” version. Then click the download button and decompress the downloaded archive. Then take the “ssg_native_200x” folder and drag the contents into X-Plane’s “Custom Data/UFMC/” folder. If the folder does not exist yet, it will need to be created. This is the link for manually installed files:

<https://www.navigraph.com/FmsDataManualInstall.aspx>

Another option is to use Navigraph’s FMS Data Manager (DM). Open the DM and note if the current generation is loaded (green buttons versus yellow ones). If an update is required, select the left-most tab under the Navigraph logo to “Select All.” Then select the “Update” button. When all of the files are updated you will see a screen like the one shown below.



If you have not loaded the SSG data set, here is how. Click the Addon Mappings tab. Once there, select “Please select an addon.” That will open a window with a long list of addon file sets to choose from. Select “SSG Boeing 748i, E-Jets Evolution.” Then that set will be added to your list and stored within the Navigraph app. Save that selection and return to the Addon List where you will now see “SSG Boeing...” added to your list.

SSG B747-8 SERIES

NOTE: For a better experience, it is recommendable to install the AIRAC update for X-plane as well. Just follow the same procedure then before downloading the X-plane 12 or X-plane 11 set data and install it into the X-Plane/Custom Data/ folder (not UFMC folder).

0.1.11 Updater Instructions

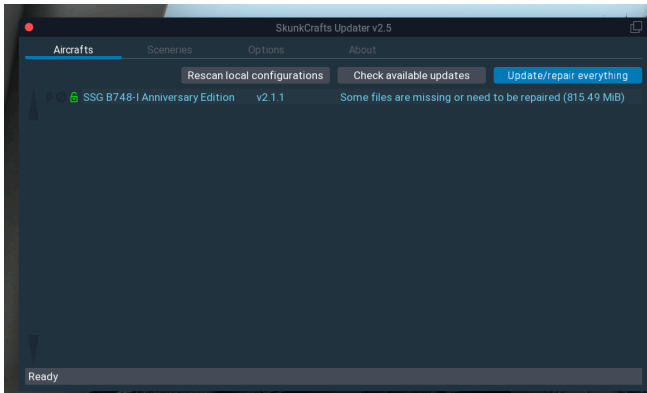
The SSG 747-8 is now compatible with the SkunkCrafts updater plugin to facilitate and automate the update process to keep it up to date. It also enables smaller and faster downloads because only the changed files are replaced.

Instructions:

1. Download the SkunkCrafts Updater plugin package via this link (current version is 2.5):

<https://forums.x-plane.org/index.php?/forums/topic/144828-updater-download-page-v121r2-for-x-plane-12-available/>
2. Unzip and place the entire SkunkCrafts folder into your X-Plane/Resources/plugins/ folder.
3. Run X-Plane by opening any aircraft in the Flight Configuration Menu, such as the default Cessna. Do NOT load the aircraft that you are going to update.
4. In the Plugin menu in X-Plane, run the SkunkCrafts plugin (see screenshot below). It will indicate which aircraft have files that can be updated – Then click “Check available updates” followed by “Update/repair everything” to install the new files.

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5. In the menu bar, click “Developer” and select “Regenerate Icons for Current Aircraft” to account for additional liveries, changes, etc.

You will normally need to reactivate the product in the standard manner.

0.1.12 Credits

3D Modeling and Animation:

Ricardo Bolognini
George Garrido

Systems and Flight Dynamics:

Ricardo Bolognini
Javier Cortes

SSG B747-8 SERIES

L: Operational Limitations

L.1 Airplane General

Runway slope limitation	+/- 2%
Maximum takeoff and landing tailwind component	15 knots
Maximum crosswind component	30 knots
Maximum takeoff and landing altitude	9,500 feet

Freighter

Maximum operating altitude	42,100 ft
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Intercontinental

Maximum operating altitude	43,100 ft
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L.2 Weight Limitations

Freighter

Maximum Taxi Weight	990,000 kgs
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Intercontinental

Maximum Taxi Weight	449,056 kgs
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Freighter

Maximum Takeoff Weight (MTOW)	987,000 lbs
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Intercontinental

SSG B747-8 SERIES

Maximum Takeoff Weight (MTOW) 447,695 kgs

Freighter

Maximum landing weight 763,000 lbss

Intercontinental

Maximum landing weight 312,071 kgs

Freighter

Maximum zero fuel weight (ZFW) 727,000 lbs

Intercontinental

Maximum zero fuel weight (ZFW) 295,288 kgs

L.3 Autoflight

The autopilot must not be engaged below 200 feet after takeoff.

Without LAND 2 or LAND 3 annunciated, the autopilot must be disengaged below 100 feet radio altitude (RA).

L.4 Engines and APU

Intentional selection of reverse thrust in flight is prohibited.

Backing the airplane with use of reverse thrust is prohibited.

APU is rated only up to 20,000 feet and cannot be started in flight.

L.5 Flight Controls

Use of speedbrakes in flight with flaps extended past 20 is not recommended.

Do not extend flaps above 20,000 feet.

SSG B747-8 SERIES

Flap Limit Speeds:

Flaps 1	-	285 kts
Flaps 5	-	265 kts
Flaps 10	-	245 kts
Flaps 20	-	235 kts
Flaps 25	-	210 kts
Flaps 30	-	185 kts

L.6 Landing Gear

Gear Limit Speeds:

Retract	Extend	Extended
270 kts (Mach 0.82)	270 kts (Mach 0.82)	320 kts (Mach 0.82)

Maximum Tire Speed:

235 MPH (376 KPH)

L.7 Capacity

Freighter

Total cargo capacity:

30,288 cu ft (857.7 cu m);

Intercontinental

Passengers (Typical 3-class configuration):

467

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PD: Performance Dispatch

Fuel Distribution

TOTAL X1000	Reserve 1	Main 1	Main 2	CWT	Main 3	Main 4	Reserve 4	Stabilizer
5.0	0.0	1.3	1.3	0.0	1.3	1.3	0.0	0.0
10.0	0.0	2.5	2.5	0.0	2.5	2.5	0.0	0.0
15.0	0.0	3.8	3.8	0.0	3.8	3.8	0.0	0.0
20.0	0.0	5.0	5.0	0.0	5.0	5.0	0.0	0.0
25.0	0.3	6.0	6.3	0.0	6.3	6.0	0.3	0.0
30.0	1.5	6.0	7.5	0.0	7.5	6.0	1.5	0.0
35.0	2.8	6.0	8.8	0.0	8.8	6.0	2.8	0.0
40.0	4.0	6.0	10.0	0.0	10.0	6.0	4.0	0.0
42.3	4.5	6.0	10.6	0.0	10.6	6.0	4.5	0.0
Reserve Tanks Full								
45.0	4.6	6.7	11.3	0.0	11.3	6.7	4.6	0.0
50.0	4.6	7.9	12.5	0.0	12.5	7.9	4.6	0.0
55.0	4.6	9.2	13.8	0.0	13.8	9.2	4.6	0.0
60.0	4.6	10.4	15.0	0.0	15.0	10.4	4.6	0.0
65.0	4.6	11.7	16.3	0.0	16.3	11.7	4.6	0.0
70.0	4.6	12.9	17.5	0.0	17.5	12.9	4.6	0.0
75.0	4.6	14.2	18.8	0.0	18.8	14.2	4.6	0.0
80.0	4.6	15.4	20.0	0.0	20.0	15.4	4.6	0.0
82.0	4.6	15.9	20.5	0.0	20.5	15.9	4.6	0.0
Main Tanks 1 & 4 Full								
85.0	4.6	15.9	22.0	0.0	22.0	15.9	4.6	0.0
90.0	4.6	15.9	24.5	0.0	24.5	15.9	4.6	0.0
95.0	4.6	15.9	27.0	0.0	27.0	15.9	4.6	0.0
100.0	4.6	15.9	29.5	0.0	29.5	15.9	4.6	0.0
105.0	4.6	15.9	32.0	0.0	32.0	15.9	4.6	0.0
110.0	4.6	15.9	34.5	0.0	34.5	15.9	4.6	0.0
115.0	4.6	15.9	37.0	0.0	37.0	15.9	4.6	0.0
120.0	4.6	15.9	39.5	0.0	39.5	15.9	4.6	0.0
125.0	4.6	15.9	42.0	0.0	42.0	15.9	4.6	0.0
127.1	4.6	15.9	43.2	0.0	43.2	15.9	4.6	0.0
Main Tanks 2 & 3 Full								
130.0	4.6	15.9	43.2	2.7	43.2	15.9	4.6	0.0
135.0	4.6	15.9	43.2	7.7	43.2	15.9	4.6	0.0
140.0	4.6	15.9	43.2	12.7	43.2	15.9	4.6	0.0
145.0	4.6	15.9	43.2	17.7	43.2	15.9	4.6	0.0
150.0	4.6	15.9	43.2	22.7	43.2	15.9	4.6	0.0
155.0	4.6	15.9	43.2	27.7	43.2	15.9	4.6	0.0
160.0	4.6	15.9	43.2	32.7	43.2	15.9	4.6	0.0
165.0	4.6	15.9	43.2	37.7	43.2	15.9	4.6	0.0
170.0	4.6	15.9	43.2	42.7	43.2	15.9	4.6	0.0
175.0	4.6	15.9	43.2	47.7	43.2	15.9	4.6	0.0
178.1	4.6	15.9	43.2	50.8	43.2	15.9	4.6	0.0
Center Tank Full								
180	4.6	15.9	43.2	50.8	43.2	15.9	4.6	1.9
185	4.6	15.9	43.2	50.8	43.2	15.9	4.6	6.9
190	4.6	15.9	43.2	50.8	43.2	15.9	4.6	10

(rounded to nearest 100kg)

SSG B747-8 SERIES

Long Range Trip Fuel and Time Required

Air Dist (NM)	TRIP FUEL (1000 LB)												Time (HR: MIN)
	BRAKE RELEASE GROSS WEIGHT (1000 LB)												
	450	500	550	600	650	700	750	800	850	900	950	1000	
800	28.0	29.8	31.7	33.8	36.2	38.2	40.6	43.2	45.4	48.2	51.2	54.2	1:50
1200	39.3	41.8	44.5	47.5	51.0	54.0	57.4	61.2	64.4	68.3	72.6	76.9	2:39
1600	50.3	53.7	57.1	60.9	65.5	69.5	73.9	78.8	83.0	87.8	93.3	98.8	3:29
2000	61.4	65.3	69.5	74.1	79.6	84.8	90.0	95.9	101.2	107.0	113.5	120.2	4:18
2400	72.3	76.8	81.7	87.1	93.4	99.7	105.7	112.6	119.1	125.7	133.2	141.0	5:08
2800	83.1	88.0	93.6	99.8	106.9	114.1	121.2	129.0	136.4	144.1	152.5	161.3	5:57
3200	93.8	99.1	105.4	112.3	120.1	128.3	136.0	145.0	153.4	162.1	171.3	181.1	6:47
3600	104.2	110.2	117.1	124.6	133.1	142.1	151.3	160.6	169.9	179.7	189.8	200.3	7:37
4000	114.5	121.2	128.5	136.7	145.9	155.7	165.8	176.0	186.1	196.8	207.9	219.2	8:27
4400	124.7	132.0	139.7	148.6	158.5	168.9	179.9	191.2	201.9	213.5	225.5	237.7	9:17
4800	134.6	142.6	150.8	160.3	170.9	182.0	193.7	205.9	217.5	229.8	242.7	255.9	10:08
5200	144.5	153.1	161.8	171.9	183.0	194.8	207.2	220.2	232.7	245.8	259.9	273.6	10:58
5600	154.1	163.4	172.8	183.2	195.0	207.4	220.4	234.2	247.7	261.4	275.9	290.8	11:49
6000	163.6	173.6	183.6	194.4	206.8	219.7	233.4	247.9	262.4	276.8	292.0	307.7	12:41
6400	173.0	183.6	194.2	205.4	218.4	231.9	246.2	261.4	276.5	291.9	307.7	324.1	13:32
6800	182.2	193.4	204.7	216.4	229.8	243.9	258.8	274.5	290.3	306.6	323.1	340.2	14:24
7200	191.3	203.1	215.0	227.3	241.0	255.7	271.1	287.5	303.9	320.9	338.4	356.0	15:16
7600	200.2	212.6	225.1	238.0	252.0	267.3	283.3	300.2	317.2	334.9	353.2	371.4	16:09
8000	209.0	222.0	235.1	248.6	263.1	278.7	295.2	312.7	330.3	348.6	367.6		17:02

NOTE: Based on 340/M0.84 climb long range cruise speed and .84/290/250 descent

Short Trip Fuel and Time Required

AIR DIST (NM)		LANDING WEIGHT (1000 LB)									TIME (HRS:MIN)
		460	500	540	580	620	660	700	740	780	
50	FUEL (1000 LB)	4.7	4.9	5.2	5.4	5.6	5.9	6.2	6.4	6.9	0:29
	ALT (FT)	11000	11000	9000	9000	9000	9000	9000	9000	9000	
100	FUEL (1000 LB)	7.1	7.4	7.7	8.0	8.3	8.6	9.0	9.4	10.0	0:22
	ALT (FT)	19000	17000	17000	17000	17000	15000	17000	17000	15000	
150	FUEL (1000 LB)	9.3	9.7	10.1	10.4	10.9	11.3	11.7	12.2	12.9	0:29
	ALT (FT)	31000	23000	23000	23000	23000	23000	23000	21000	21000	
200	FUEL (1000 LB)	11.0	11.5	11.9	12.4	12.9	13.5	14.0	14.7	15.5	0:36
	ALT (FT)	35000	35000	35000	35000	33000	31000	31000	29000	29000	
250	FUEL (1000 LB)	12.6	13.2	13.7	14.3	14.9	15.5	16.1	16.9	17.7	0:42
	ALT (FT)	41000	41000	37000	35000	35000	35000	35000	35000	35000	
300	FUEL (1000 LB)	14.2	14.8	15.5	16.1	16.8	17.5	18.2	19.1	20.1	0:48
	ALT (FT)	41000	41000	37000	39000	37000	37000	35000	35000	35000	
350	FUEL (1000 LB)	15.7	16.4	17.2	17.9	18.8	19.6	20.4	21.4	22.5	0:54
	ALT (FT)	41000	41000	41000	39000	37000	37000	35000	35000	35000	
400	FUEL (1000 LB)	17.3	18.1	18.9	19.7	20.7	21.6	22.6	23.7	24.9	1:00
	ALT (FT)	41000	41000	41000	39000	39000	37000	35000	35000	35000	
450	FUEL (1000 LB)	18.8	19.7	20.7	21.6	22.6	23.6	24.7	26.0	27.2	1:06
	ALT (FT)	41000	41000	41000	39000	39000	37000	35000	35000	35000	
500	FUEL (1000 LB)	20.4	21.4	22.4	23.4	24.5	25.7	26.9	28.2	29.6	1:13
	ALT (FT)	41000	41000	41000	39000	39000	37000	35000	35000	35000	

SSG B747-8 SERIES

Long Range Cruise Control

WEIGHT (1000 LB)		PRESSURE ALTITUDE (1000 FT)									
		25	27	29	31	33	35	37	39	41	42
1000	%N1	87.3	88.5	90.0	92.3						
	MACH	.846	.849	.848	.846						
	KIAS	359	346	331	316						
	FF/ENG	7982	7729	7599	7525						
950	%N1	86.1	87.3	88.5	90.4	93.1					
	MACH	.839	.848	.849	.848	.844					
	KIAS	356	345	331	317	302					
	FF/ENG	7622	7413	7219	7069	7138					
900	%N1	84.7	86.1	87.3	88.8	90.9	93.9				
	MACH	.826	.844	.848	.849	.846	.843				
	KIAS	349	344	331	317	303	288				
	FF/ENG	7213	7083	6895	6674	6658	6778				
850	%N1	83.1	84.7	86.0	87.4	89.0	91.4				
	MACH	.806	.834	.847	.849	.848	.845				
	KIAS	340	339	331	317	304	289				
	FF/ENG	6756	6719	6588	6324	6239	6529				
800	%N1	81.6	83.2	84.6	86.0	87.3	89.1	92.3			
	MACH	.787	.816	.840	.848	.849	.847	.843			
	KIAS	332	331	328	317	304	290	275			
	FF/ENG	6331	6289	6523	6022	5869	5820	5922			
750	%N1	80.0	81.5	83.1	84.7	85.9	87.3	89.8	93.8		
	MACH	.768	.794	.824	.844	.848	.849	.846	.842		
	KIAS	323	321	321	316	304	290	276	263		
	FF/ENG	5932	5868	5874	5723	5568	5452	5466	5642		
700	%N1	78.4	79.9	81.4	83.2	84.5	85.7	87.7	90.9		
	MACH	.749	.774	.800	.831	.846	.849	.848	.845		
	KIAS	315	312	311	310	303	290	277	264		
	FF/ENG	5636	5585	5455	5390	5294	5139	5088	5157		
650	%N1	76.7	78.2	79.7	81.4	83.0	84.3	85.9	88.5	91.9	93.9
	MACH	.730	.753	.779	.806	.836	.847	.849	.848	.844	.842
	KIAS	306	303	301	300	299	290	277	277	251	245
	FF/ENG	5230	5202	5064	4976	4980	4869	5088	5088	4841	4942
600	%N1	74.9	76.3	77.8	79.6	81.1	82.7	84.3	86.4	89.1	90.7
	MACH	.708	.732	.755	.782	.811	.839	.847	.849	.847	.845
	KIAS	296	294	292	290	289	287	290	265	252	246
	FF/ENG	4862	4829	4673	4586	4584	4570	5139	4417	4424	4461
550	%N1	72.9	74.4	75.8	77.6	79.1	80.7	82.6	84.7	86.8	88.0
	MACH	.684	.709	.733	.757	.783	.813	.840	.848	.849	.848
	KIAS	285	284	282	280	278	277	274	265	253	247
	FF/ENG	4498	4458	4307	4204	4193	4189	4201	4219	4159	4158
500	%N1	70.7	72.3	73.8	75.5	77.0	78.5	80.5	82.8	84.9	85.9
	MACH	.659	.682	.707	.731	.756	.783	.813	.840	.848	.849
	KIAS	274	273	272	270	267	266	264	262	253	247
	FF/ENG	4135	4079	4050	3940	3907	3888	3913	3935	3884	3851
450	%N1	68.3	70.0	71.4	73.2	74.7	76.3	78.2	80.5	82.9	84.0
	MACH	.629	.654	.677	.703	.728	.753	.780	.810	.839	.845
	KIAS	261	261	259	258	257	254	253	251	250	246
	FF/ENG	3742	3716	3684	3585	3552	3518	3531	3564	3599	3587

NOTE: Shaded area shows approximate optimum altitude.

SSG B747-8 SERIES

PI: Performance Inflight

Takeoff Speeds (Flaps 10) - Dry Runway

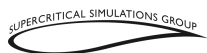
Max Takeoff Thrust

WEIGHT (1000 LB)	FLAPS 10		
	V1	VR	V2
1000	166	182	196
950	162	177	192
900	157	172	188
850	151	165	183
800	146	159	178
750	140	153	173
700	134	146	167
650	127	139	161
600	120	132	155
550	112	125	149
500	103	117	142
450	94	109	136

Takeoff Speeds (Flaps 20) - Dry Runway

Max Takeoff Thrust

WEIGHT (1000 LB)	FLAPS 20		
	V1	VR	V2
1000	155	169	180
950	151	164	177
900	146	159	173
850	141	153	168
800	136	147	164
750	130	142	159
700	124	135	154
650	117	129	148
600	110	122	143
550	103	115	137
500	94	107	131
450	86	100	125



SSG B747-8 SERIES

VREF (KIAS)

WEIGHT (1000 LB)	FLAPS	
	30	25
1000	186	190
950	181	185
900	176	179
850	171	174
800	165	168
750	161	164
700	155	158
650	149	152
600	143	145
550	136	139
500	130	132
450	123	125

SSG B747-8 SERIES

Airplane General, Emergency Equipment, Doors, Windows

Airplane General

Maximum Design Taxi Weight (MTW). Maximum weight for ground maneuver as limited by aircraft strength and airworthiness requirements. (It includes weight of taxi and run-up fuel.)

Maximum Design Takeoff Weight (MTOW). Maximum weight for takeoff as limited by aircraft strength and airworthiness requirements. (This is the maximum weight at start of the takeoff run.)

Maximum Design Landing Weight (MLW). Maximum weight for landing as limited by aircraft strength and airworthiness requirements.

Maximum Design Zero Fuel Weight (MZFW). Maximum weight allowed before usable fuel and other specified usable agents must be loaded in defined sections of the aircraft as limited by strength and airworthiness requirements.

Operating Empty Weight (OEW). Weight of structure, powerplant, furnishing systems, unusable fuel and other unusable propulsion agents, and other items of equipment that are considered an integral part of a particular airplane configuration. Also included are certain standard items, personnel, equipment, and supplies necessary for full operations, excluding usable fuel and payload.

Maximum Payload. Maximum design zero fuel weight minus operational empty weight.

Maximum Seating Capacity. The maximum number of passengers specifically certificated or anticipated for certification.

Maximum Cargo Volume. The maximum space available for cargo. **Usable Fuel.** Fuel available for aircraft propulsion.

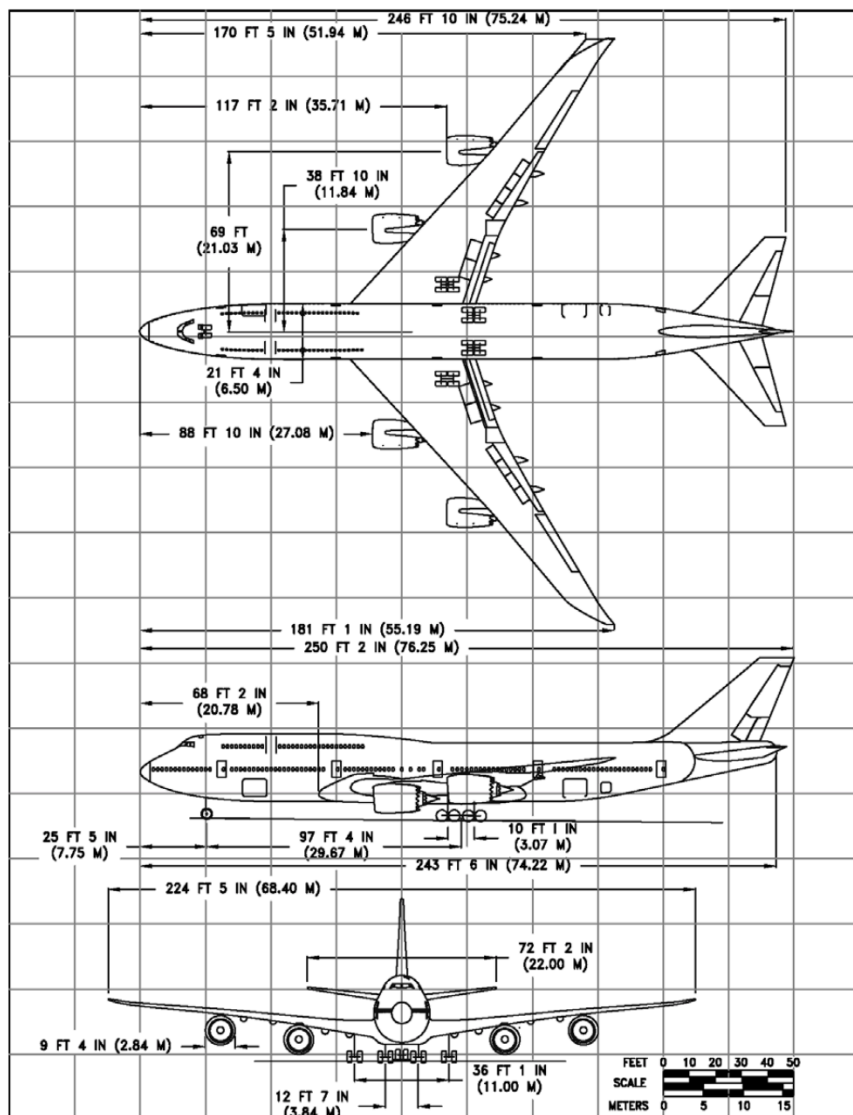
SSG B747-8 SERIES

CHARACTERISTICS	UNITS	747-8F	747-8F
MAX DESIGN	POUNDS	978,000	990,000
TAXI WEIGHT	KILOGRAMS	443,613	449,056
MAX DESIGN	POUNDS	975,000	987,000
TAKEOFF WEIGHT	KILOGRAMS	442,253	447,696
MAX DESIGN	POUNDS	761,000	763,000
LANDING WEIGHT	KILOGRAMS	345,184	346,091
MAX DESIGN	POUNDS	725,000	727,000
ZERO FUEL WEIGHT	KILOGRAMS	328,854	329,762
OPERATING	POUNDS	434,600	434,600
EMPTY WEIGHT (1)	KILOGRAMS	197,131	197,131
MAX STRUCTURAL	POUNDS	290,400	292,400
PAYLOAD (1)	KILOGRAMS	131,723	132,630
TYPICAL CARGO – MAIN DECK	CUBIC FEET	24,462	24,462
CONTAINERS	CUBIC METERS	693	693
MAX CARGO - LOWER DECK	CUBIC FEET	5,850	5,850
CONTAINERS (LD-1)	CUBIC METERS	166	166
MAX CARGO - LOWER DECK	CUBIC FEET	520	520
BULK CARGO	CUBIC METERS	14.7	14.7
USABLE FUEL CAPACITY	U.S. GALLONS	59,734 (2)	59,734 (2)
	LITERS	226,118	226,118
	POUNDS	400,218	400,218
	KILOGRAMS	181,536	181,536

NOTES:

- ESTIMATED WEIGHTS FOR ENGINE/AIRFRAME CONFIGURATION SHOWN. OPERATING EMPTY WEIGHT REFLECTS STANDARD ITEM ALLOWANCES. ACTUAL OEW AND PAYLOAD WILL VARY WITH AIRPLANE AND AIRLINE CONFIGURATION. CONSULT USING AIRLINE FOR VALUES.
- 747-8F IS NOT DESIGNED WITH TAIL FUEL TANKS

SSG B747-8 SERIES



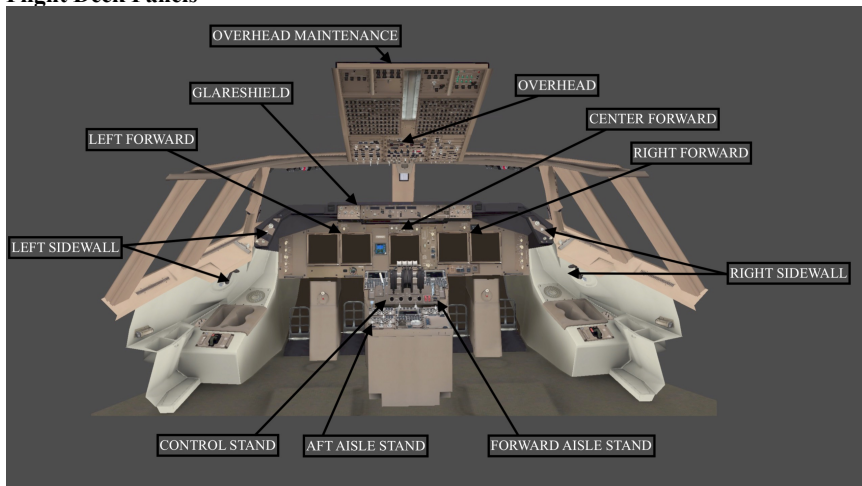
SSG B747-8 SERIES

Cockpit Panels

Complete Panel

The SSG 747-8 has most of the same panel components as the real aircraft; however, it mostly includes functionality for those that are used during normal pilot operations. Other components that are used in emergency situations and for maintenance are not implemented yet, even if they are modeled and are visible in the cockpit. The diagram below depicts the entire set of cockpit panels.

Flight Deck Panels

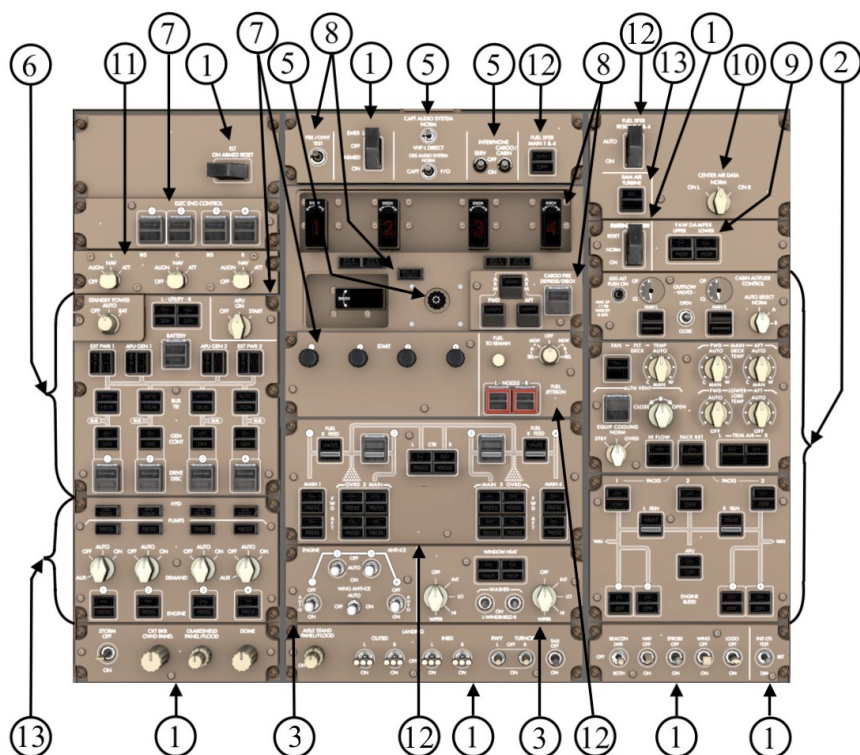


Overhead Panel

While not all aircraft systems are fully simulated yet, the controls and indicators for the overhead panel are located in their proper places compared to the real aircraft. The diagram below shows the overhead panel layout. The numbers in these illustrations refer to the chapter of the FCOM where the relevant systems are described.

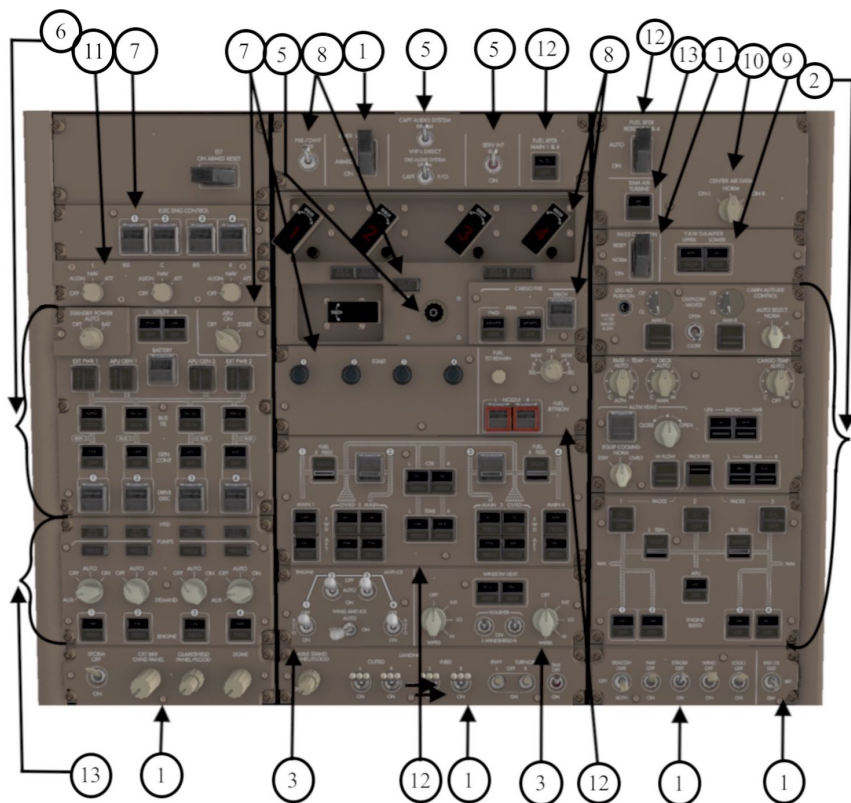
Overhead Panel (Freighter)

SSG B747-8 SERIES



SSG B747-8 SERIES

Overhead Panel (Intercontinental)



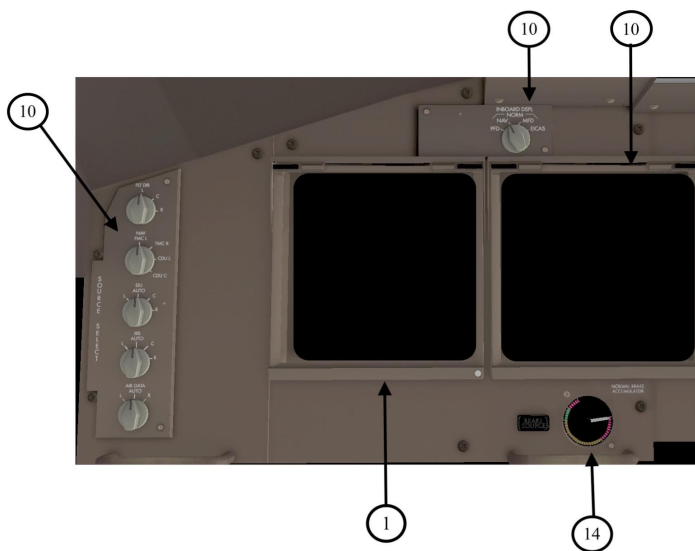
SSG B747-8 SERIES

NOTE: Some buttons on the overhead panel have clear plastic guards to protect them from inadvertent operation by the pilots. To lift these guards in the SSG 747-8 simply click over the button.

Forward Panel

The forward panel display screens on the SSG 747-8 are presented using the same arrangement as on the real aircraft.

Left Forward Panel



The forward panel is composed of the following displays and controls:

- Primary Flight Display (PFD)
- Navigation Display (ND)
- Display selector switch

NOTE: Other controls are non-functional

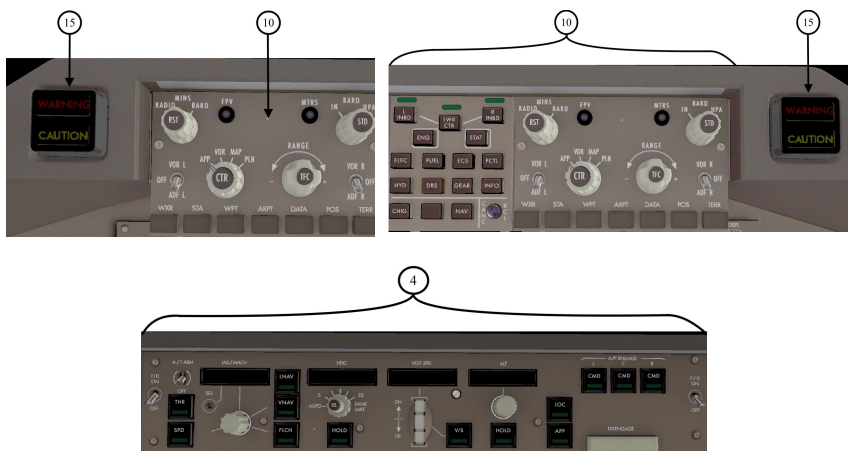
The displays themselves are interchangeable, but typically the PFD is on the left and the ND is on the right for the Captain's side. On the First Officer's side, this arrangement is mirrored.

SSG B747-8 SERIES

Glareshield Panel

The glareshield panel includes various selectors to control displays for the pilots, as well as autoflight functions. It is divided into segments as shown below:

Glareshield Panel



Mode Control Panel (MCP)

The Mode Control Panel (MCP) includes the following controls and indicators:

- Autothrottle selector
- Flight Director (FD) selector
- Airspeed selectors
- Heading selectors
- Vertical Speed (V/S) knob
- Autopilot selectors
- Altitude selector

SSG B747-8 SERIES

Mode Control Panel



Display Select Panel

The various synoptic pages on the Lower MFD can be selected with the Display Select Panel. On the 747-8 aircraft, the top 3 buttons with green LED lights allow for switching of various display information to the various pilot screens. This function is partially implemented in the SSG 747-8. The buttons allow the pilots to display system synoptics on the left inboard or right inboard displays in front of the pilots, and not just on the Lower MFD display. The green light above each button indicates the particular screen that is being assigned. A typical arrangement is to use the Lower MFD to display system synoptics. If synoptics are displayed on either of the inboard displays, the ND can be displayed again by pressing the NAV button.

NOTE: On the 747-400, Boeing called the Center Instrument Panel displays the Upper and Lower EICAS. On the 747-8 this was changed to EICAS as the top screen and MFD as the lower one because it can be used to display information other than those of an EICAS, such as the electronic checklist. Furthermore, either inboard display, which is typically used as an ND is also an MFD. To avoid confusion, in this manual, the lower center display on the aisle stand is called the “MFD” or “Lower MFD” to distinguish it from the other displays.

SSG B747-8 SERIES

Display Select Panel



The lower buttons have the following functions:

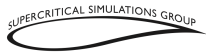
ENG	Secondary engine information
STAT	Status
ELEC	Electrical system synoptic
FUEL	Fuel system synoptic
ECS	ECS synoptic
FCTL	Flight controls synoptic
HYD	Hydraulic system synoptic
DRS	Aircraft door synoptic
GEAR	Landing gear synoptic
INFO	Information page (not implemented in the SSG 747-8)
CHKL	Electronic checklist
NAV	Navigation page

At the bottom right is the CANC/RCL button, which stands for “CANCEL” and “RECALL” and allows pilots to manage various EICAS messages that can form lists of several pages.

Center Instrument Panel

The Center Instrument Panel contains the following:

- Integrated Standby Flight Display (ISFD)
- Engine Indication and Crew Alerting System (EICAS)
- Landing gear lever
- Autobrake selector
- Heading reference switch

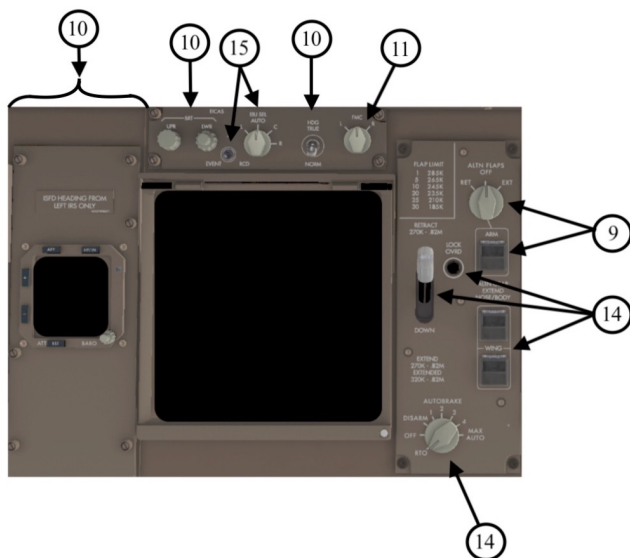


SSG B747-8 SERIES

NOTE: Other controls are non-functional

SSG B747-8 SERIES

Center Instrument Panel



Forward Aisle Stand

The Forward Aisle Stand contains the following:

- Left Control Display Unit (CDU)
- Lower Multifunction Display (MFD)
- Right Control Display Unit CDU

SSG B747-8 SERIES

Forward Aisle Stand



Control Stand

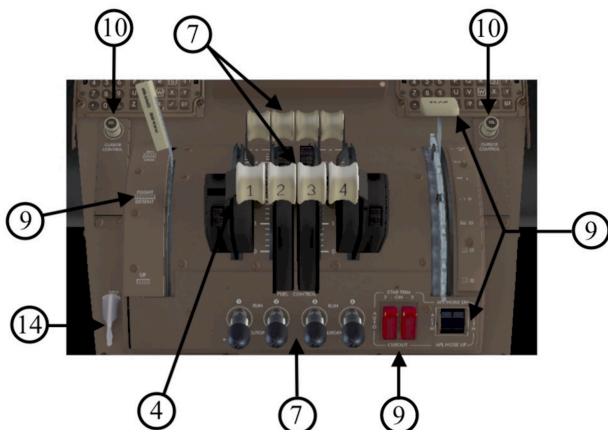
The Control Stand contains the following:

- Parking brake lever
- Speedbrake lever
- Throttles with autothrottle cutoff switches and TO/GA switches
- Fuel control switches
- Flap handle
- Cursor control

NOTE: Other controls are non-functional

SSG B747-8 SERIES

Control Stand



Aft Aisle Stand

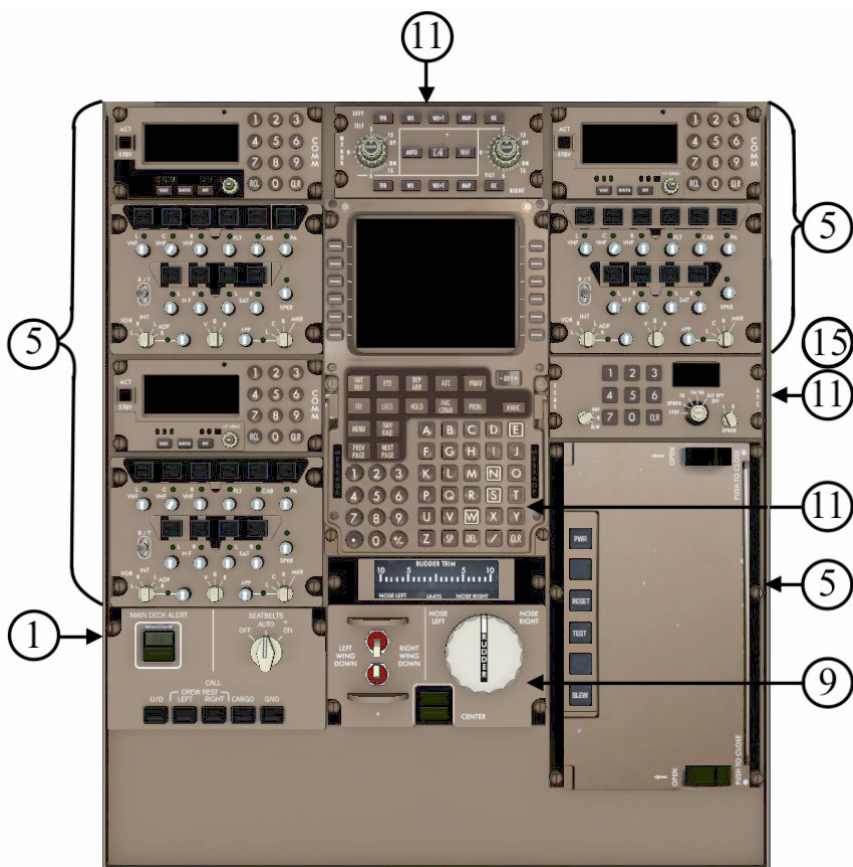
The Aft Aisle Stand contains the following:

- Weather radar
- Communication radios
- Transponder controls
- Center CDU
- Seat belt sign and chime panel
- Trim panel

NOTE: Other controls are non-functional

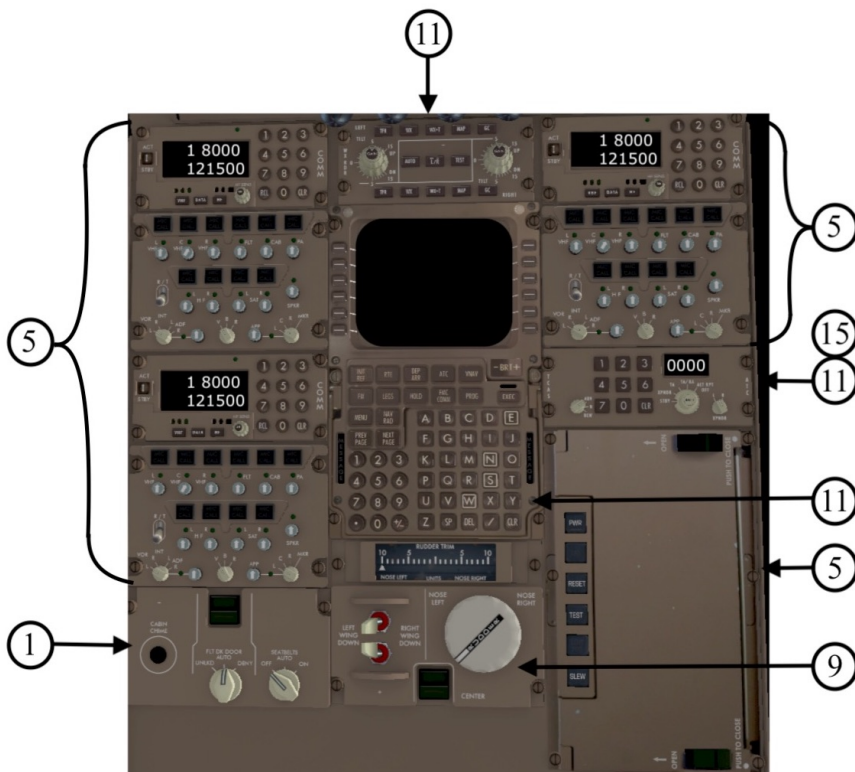
SSG B747-8 SERIES

Aft Aisle Stand (Freighter)



SSG B747-8 SERIES

Aft Aisle Stand (Intercontinental)



Sidewall Panel

The Sidewall Panel contains the following:

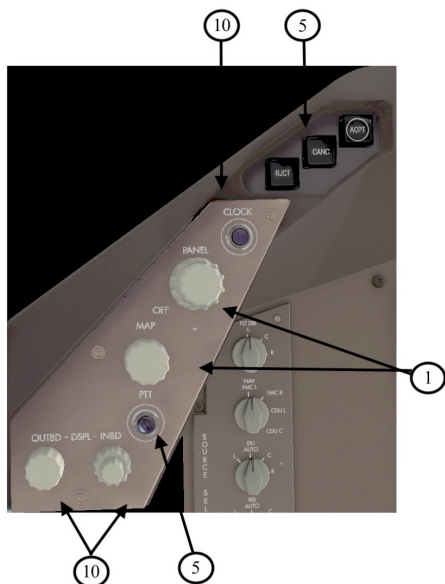
- Data link reject/cancel/accept switches (Note that these have an alternate function in the SSG 747-8)
- Screen brightness controls

On the First Officer's side, this arrangement is mirrored.

NOTE: Other controls are non-functional

SSG B747-8 SERIES

Left Sidewall Panel



Exterior Lighting

The illustrations below show the various types of exterior lights that are fitted on the SSG 747-8.

SSG B747-8 SERIES

Overhead Exterior Light Panel

This panel contains individual switches for each light on the 747-8. Notice that all switches for the different lights are of a different shape to make it easier for pilots to recognize them by feel. It should be noted that the “IND LTS TEST” switch turns on all the overhead button lights to test for any burnt out bulbs. Also, the wingtip marker lights are controlled by the same switch as the navigation lights.

Exterior Light Panel



Interior Lighting

The SSG 747-8 panel incorporates lighting effects such as rheostats for backlit text on the control panels, backlit gauges, a dome light, spot lighting on various areas of the panel, and modified gauges to change display screen brightness via rheostats. It should be noted that a rheostat is simply a knob that can be used to control an electrical device in increments (i.e. gradually).

Overhead Light Panel



The PANEL rheostat on the left side panel controls the illumination of panel text and markings. The dome, panel backlighting, and spots can be controlled separately. For example, the dome light can be turned up for pretakeoff activities, then decreased and spots used for flight using mostly backlighting, and then use spots after landing, which is quite similar to how the real 747-8 interior lighting is used by flight crews.

There are 7 lighting knobs, and your enjoyment of the SSG 747-8 panel will be in direct relation with your ability to appreciate and use them. They are not difficult to use, and work as follows:

SSG B747-8 SERIES

Rheostat	Location	Controls
DSPL - INBD	Left glareshield	PFD, ND, and EICAS brightness
MAP	Left glareshield	Map light to light the chart clip area.
PANEL	Left glareshield	Backlighting for lettering on glareshield, main panel, and aisle stand
CPT BREAKER OVHD PANEL	Overhead	Backlighting for lettering on overhead panels
GLARESHIELD PANEL FLOOD	Overhead	Spot lighting for glareshield and under the glareshield
DOMES	Overhead	Overall brightness for cockpit (Turn this down to low at night)
AISS STAND PANEL / FLOOD	Overhead	Spot lighting for aisle stand and radios

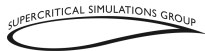
NOTE: The STORM switch on the overhead panel can be used to turn all lights UP or DOWN, typically in conditions where lightning is a factor (which is why they are called “storm” lights”) to preserve night vision for the pilots.

Passenger Signs

The SSG 747-8 also has a seat belt sign switch to alert passengers (in the Freighter they are called “supernumeraries”) with two settings, one is “automatic” and the other is “on”. When in automatic mode, the seatbelt sign turns on and off according to system logic, primarily when passing through 10,000 feet.

Passenger Signs (Freighter)





SSG B747-8 SERIES

1. MAIN DECK ALERT (not implemented in the SSG 747-8)
2. SEAT BELTS selector

Activating the SEAT BELTS selector also triggers a chime that can be used to alert supernumeraries during various phases of flight.

SSG B747-8 SERIES

Passenger Signs (Intercontinental)



AFT AISLE STAND

1. CABIN CHIME button
2. SEAT BELTS selector

Activating the SEAT BELTS selector also triggers a chime that can be used to alert flight attendants during various phases of flight. The cabin chime button simply triggers the chime.

Doors

Freighter

The 747-8 Freighter has one entry door on the left side of the aircraft, designated as “L1”. There is also a smaller door on the right side of the upper deck. Doors can be opened via a menu in the cockpit tablet. At present, opening the entry doors via this menu results in the following doors opening:

- L1
- Right upper deck

The cargo doors are on the lower right side of the aircraft and can also be opened via the cockpit tablet menu. There are 3 lower compartment cargo doors designated as follows:

- Forward cargo door
- Rear cargo door
- Bulk cargo door

SSG B747-8 SERIES

There are also 2 main deck cargo doors designated as follows:

- Nose cargo door
- Side cargo door

A door status synoptic can be called up on the MFD as well, showing which doors are opened. Open doors appear in yellow.

Intercontinental

The 747-8 has several opening doors, which are numbered 1 to 5 on the left and right side of the aircraft. Therefore, the first door on the left of the aircraft is called "L1". Doors can be opened via a menu in the cockpit tablet.

The cargo doors are on the right side of the aircraft and can also be opened via the tablet menu. There are 3 cargo doors designated as follows:

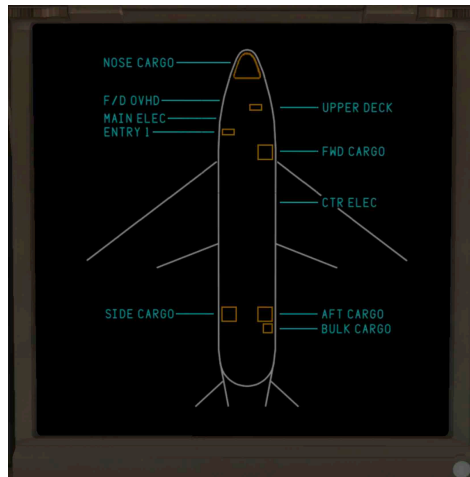
- Forward cargo door
- Rear cargo door
- Bulk cargo door

A door status synoptic can be called up on the MFD as well showing which doors are opened. Open doors appear in yellow. The "M" and "A" designations indicate whether the doors have been set by the flight attendants to deploy the evacuation slides automatically when a door is opened. Unless it is an emergency situation, the doors are set to "manual" before they are opened. They typically are set to automatic after the doors are closed and the jetway or airstairs are removed to permit passenger evacuations, if necessary.

Be aware that on the real aircraft, a door can only be opened once the aircraft is depressurized, typically when all engines are shut down and the aircraft is parked and chocked. On the SSG 747-8, as a protective measure, any open doors will automatically close if you start moving and they cannot be opened until the aircraft is stopped once again.

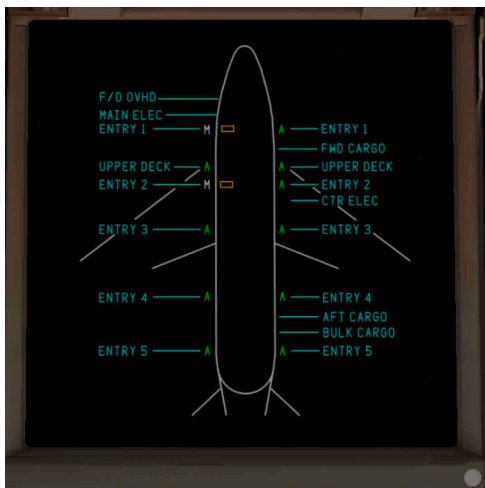
SSG B747-8 SERIES

Doors Synoptic (Freighter)



SSG B747-8 SERIES

Doors Synoptic (Intercontinental)



EICAS Messages

EICAS Alert Messages

Freighter

Message	Level	Message Logic
DOOR AFT CARGO	Caution	Aft cargo door is not closed and secure.
DOOR FWD CARGO	Caution	Forward cargo door is not closed and secure.
DOOR R UPPER DK	Caution	Upper deck door is not closed and secure.
DOOR ENTRY L1	Caution	Main deck entry door is not closed and secure.
DOOR SIDE CARGO	Caution	Side cargo door not closed, latched and locked condition sensed.
DOOR NOSE CARGO	Caution	Nose cargo door is not closed, latched and locked condition sensed.

SSG B747-8 SERIES

Intercontinental

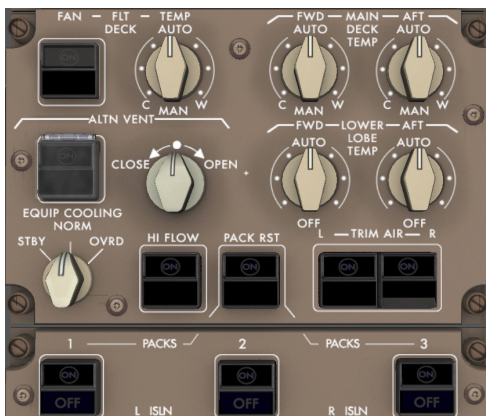
Message	Level	Message Logic
DOOR AFT CARGO	Caution	Aft cargo door is not closed and secure.
DOOR FWD CARGO	Caution	Forward cargo door is not closed and secure.
DOOR ENTRY L1, L2, L5, R1, R5	Advisory	Main deck entry door is not closed and secure.
EMER LIGHTS	Advisory	The emergency lights switch is not armed.

SSG B747-8 SERIES

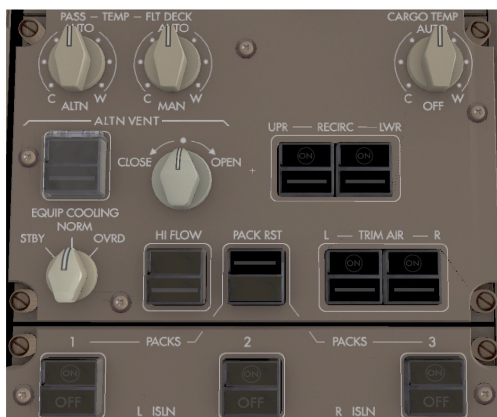
2. Air Systems

Controls and Indicators

ECS Panel (Freighter)



ECS Panel (Intercontinental)



SSG B747-8 SERIES

OVERHEAD PANEL

This panel includes the following

1. Pack switches
2. Zone temperature control switches

NOTE: Other overhead controls on this panel are non-functional on the SSG 747-8.

Air Conditioning System Description

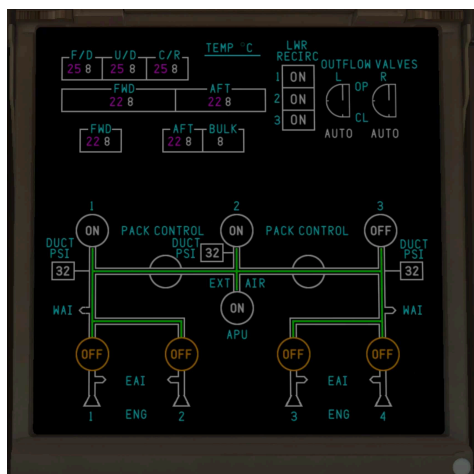
The real 747-8 has 3 packs, which are air cycle machines that convert so called “bleed air” tapped from the compressor in the Auxiliary Power Unit (APU), or the engines, into conditioned air to pressurize the aircraft and maintain selected cabin and cargo compartment temperatures. You can see the pack doors operate on the belly of the airplane as they are switched on or off.

ECS Synoptic

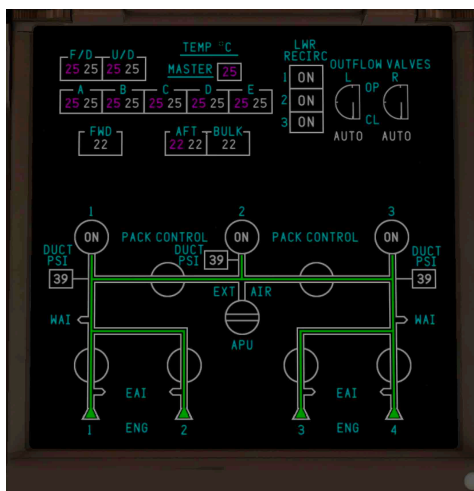
The ECS synoptic shows the 3 packs, normal flow between the packs, and the various sources (primarily the engines and APU). The top portion shows the various cabin and cargo compartment zones as well as selected and actual temperatures. The top right section shows the position of the outflow valves that help to regulate cabin pressure (which are visible on the aircraft exterior as shown with red arrows in the illustration below), as well as the status of various recirculation fans.

SSG B747-8 SERIES

ECS Synoptic (Freighter)



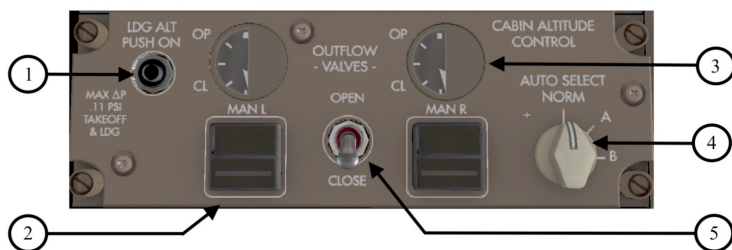
ECS Synoptic (Intercontinental)



SSG B747-8 SERIES

Pressurization System Description

Cabin Altitude Control



OVERHEAD PANEL

1. Landing altitude switch
2. Landing altitude selector
3. Outflow valve manual switch
4. Outflow valve position indicator
5. Cabin altitude AUTO selector (not implemented)
6. Outflow valves manual control

On the bottom left portion of the EICAS, the current cabin altitude (CAB ALT) is displayed, along with the rate and change in pressure differential between the cabin and the outside air. This is completely automatic on the SSG 747-8 and landing altitude is obtained from the destination airport entered into the FMC.

The pressurization schedule in automatic mode will cause the outflow valves to close completely at around 7,300 feet. However, if the aircraft departs from an airport at a higher altitude (e.g. 8,000 feet) the system will accommodate that and leave the outflow valves open until the aircraft starts to climb, and they will close completely at approximately 10,000 to 12,000 feet.

When starting the aircraft up in a “cold and dark” configuration, the system is in automatic mode, which will be active for the duration of the flight. If required, one or both of the outflow valve manual switches can be selected by pressing them (the light goes off) to go into manual mode and open/close the outflow valves with the manual control switch.

SSG B747-8 SERIES

Bleed Air System Description

Bleed air is also used for engine starts, and the 747-8 has several sources of bleed air, one for each engine and one for the APU. The pneumatics controls are located on the overhead panel. With the panel, the bleed air from the APU and engines can be selected. Each engine bleed valve can be controlled individually, and so can the isolation valves.

In addition to pressurization and temperature control, the pneumatics on the 747-8 are used for engine starting, leading edge and nacelle anti-ice, and some other systems.

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
BLEED 1-4 OFF	Advisory	Engine bleed air is off and the engine is operating and the engine bleed air valve is closed.
CABIN ALT AUTO	Caution	Both outflow valve manual switches are on.
CABIN ALTITUDE	Warning	Cabin altitude is excessive.
PACK 1+2+3	Caution	All packs are shut down.

EICAS Memo Messages

Message	Level	Message Logic
PACK 1, 2, 3, OFF	Memo	Pack switch is off.
PACKS 1+2 OFF, 1+3 OFF, PACKS 2+3 OFF	Memo	Pack switches are off.
PACKS OFF	Memo	All pack switches are off.

SSG B747-8 SERIES

3. Anti-Ice, Rain

System Description

The 747-8 has a variety of built-in equipment to protect it against various types of icing. The primary systems for this purpose are:

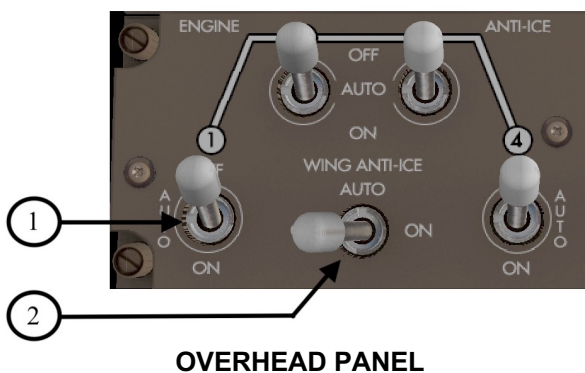
- Engine anti-ice (EAI)
- Wing anti-ice (WAI)
- Cockpit window heat

Overhead Anti-Ice and Window Control Panels

Ice and rain protection functions that can be selected on the SSG 747-8 overhead include:

- EAI (OFF/AUTO/ON)
- WAI (OFF/AUTO/ON)
- Window heat

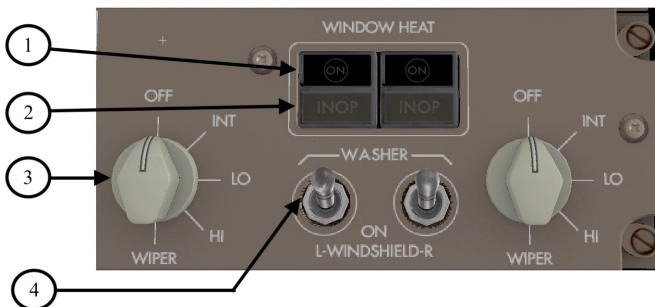
Anti-Ice Panel



SSG B747-8 SERIES

1. EAI switch
2. WAI switch

Windshield Heat and Washers Panel



OVERHEAD PANEL

1. Window heat switch
2. Window heat inoperative light
3. Windshield wiper selector
4. Windshield washer switches (not implemented)

Typically, window heat is kept on at all times the aircraft is flown. This is required to maintain the flexible properties of the windows in case of a bird strike.

The anti-ice system for the pitot and static probes and ports is turned on when other anti-ice systems are turned on. Also, if an icing condition is detected a message will appear on the EICAS.

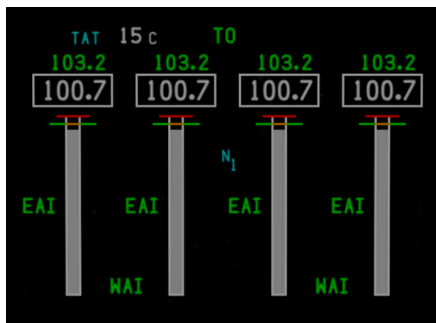
The windshield wiper settings are as follows:

- INT: Intermittent
- LO: Low
- HI: High

Please note that the windshield wiper switches are independent on the SSG 747-8 like on the real aircraft with controls for the left and right sides.

SSG B747-8 SERIES

EAI/WAI Indication on EICAS



The primary indication that these anti-ice systems are active are the “EAI” and “WAI” labels on the EICAS engine tapes, as appropriate.

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
ANTI-ICE	Advisory	Any engine anti-ice switch, or the wing anti-ice switch is on, and TAT is more than 10°C, and icing conditions do not exist.
ICING	Advisory	Any engine anti-ice switch or the wing anti-ice switch is off, and icing conditions exist.

SSG B747-8 SERIES

4: Automatic Flight

System Description

The real 747-8 has a very sophisticated autoflight system, and the SSG 747-8 incorporates most of these automatic functions.

Mode Control Panel (MCP)



GLARESHIELD PANEL

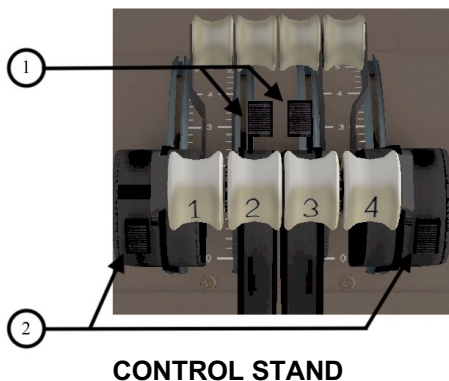
The MCP includes the following controls and indicators (from left to right):

- Left F/D switch
- A/T ARM switch
- Thrust (THR) switch
- Speed (SPD) switch
- IAS/MACH window
- IAS/MACH select switch
- IAS/MACH selector
- LNAV switch
- VNAV switch
- Flight Level Change (FL CH) switch
- Heading (HDG) window
- Heading select (SEL) switch
- Heading selector (inner knob)
- Bank limit selector (outer knob)
- Heading HOLD switch
- Vertical speed (VERT SPD) window
- Vertical speed selector (up and down)
- Vertical speed (V/S) switch
- Altitude (ALT) window

SSG B747-8 SERIES

- Altitude selector (knob)
- Altitude HOLD switch
- Localizer (LOC) switch
- Approach (APP) switch
- Autopilot (A/P) engage switches (3 of them)
- Autopilot DISENGAGE bar
- Right F/D switch (inactive in this version)

Autothrottle Disconnect and TO/GA Switches

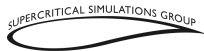


1. TO/GA switches
2. Autothrottle disconnect switches

Autopilot and FD

The FD can be activated by flipping the “F/D” switch up to “ON”. The Command (CMD) button will activate the autopilot independently of the FD switch, and the DISENGAGE bar will deactivate both the FD and autopilot.

For pilots who may be unfamiliar with FDs, it is useful to think of them as providing the guidance that the autopilot would follow using the information entered by the pilots on the MCP. The autopilot can then follow the FD commands once it is engaged; or the pilots can follow the FD commands for smooth and efficient flying. Flying without an FD is called “raw data” flying, but it is typical for airline pilots to use some sort of guidance for added safety.



SSG B747-8 SERIES

The Lateral Navigation (LNAV) button is used to make the autopilot follow the FMC route.

When pushed, LNAV illuminates green to indicate it is engaged or armed. LNAV status is shown on the PFD.

Vertical Navigation (VNAV) and Flight Level Change (FLCH) modes work as expected in XP. The CMD button will engage the autopilot independently from the FD. Pushing CMD will turn the autopilot on, but not the autothrottle. To turn the autopilot off it is necessary to use the Autopilot Disengage bar.

Pushing the Heading Hold (HDG HOLD) button will hold the current heading. The center Select (SEL) button on the heading selector switch is used to acquire the MCP selected heading by engaging the heading mode and can be turned to select the desired heading. The heading selector switch also has an outer knob that can be used to select the bank limit for the autopilot, ranging from 5 to 25 degrees of bank, (or to an automatic setting).

The Vertical Speed (VERT SPD or V/S) mode works as expected after takeoff, with negative indications available. These are displayed with a “+” or “-“ sign, as applicable.

Present altitude can be captured by one press of the HOLD button, otherwise you have to dial in your desired altitude and reach it through a vertical flight mode, FLCH, V/S, or VNAV through the FMCs.

LOC captures the LOC or ILS courses, and the Approach (APP) button also captures the glideslope if it is available, such as with an ILS or GLS. However, pressing the APP button alone will capture both localizer and glideslope without the LOC button illuminating.

The DISENGAGE bar will deploy downward, and when pressed will disengage both the autopilot. This will not disengage the autothrottle or FD, so that presently selected autoflight parameters will still be sent to the FD if hand flying is desired, for example. When hand flying an approach in this manner, the localizer and glideslope cues will still be reflected on the FD.

Incidentally, the preferred method of disengaging the autopilot on the real aircraft is the yoke mounted “autopilot disengage” button, which is visible on the SSG 747-8, but is non-functional.

NOTE: To allow for easier “heads-up” flying, SSG has provided another autopilot disconnect switch that is located in front of the left pilot seat on the glareshield. It is an

SSG B747-8 SERIES

invisible button where the Data Link switches are located. The leftmost button is the autopilot disconnect switch.

**The AP is disengaged if the Yokes are grabbed a bit left-right or up-down.
To reengage the AP, just press L, C o R AP button.**

Autothrottle

To arm the autothrottle, The A/T ARM switch needs to be selected “up” before the Thrust (THR) or Speed (SPD) button are engaged to activate the auto thrust system. Simply select a target speed in the window before pushing the SPD button and the autothrottle will accelerate to that speed. If you want the autothrottle to capture your current airspeed, push the SPD button and the value in the window will change to your current airspeed if not in VNAV mode. If in the VNAV mode it updates the VNAV FMC selected speed.

Throttles respond as soon as you select A/T ARM and THR. If the autothrottle is not armed, the throttles will not be controlled by the autothrottle. When the autothrottle is armed, the MCP speed changes to the current aircraft airspeed and SPD mode is engaged. The Takeoff / Go-Around (TOGA) button will set the throttles to the maximum N1 value computed by the FMC.

The MCP autothrottle controls also include a small SEL button that allows switching from Indicated Air Speed (IAS) to Mach Number (MACH) for the MCP speed window. Typically, speeds are flown in terms of MACH above transition altitude (where the Flight Levels start) and in IAS below it.

NOTE: To allow for easier “heads-up” flying, SSG has provided another TO/GA switch in front of the left pilot seat on the glareshield. It is the ACPT button

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
AUTOPILOT DISC	Warning	All engaged autopilots have disengaged.
AUTOTHROT DISC	Caution	Autothrottle has disconnected.

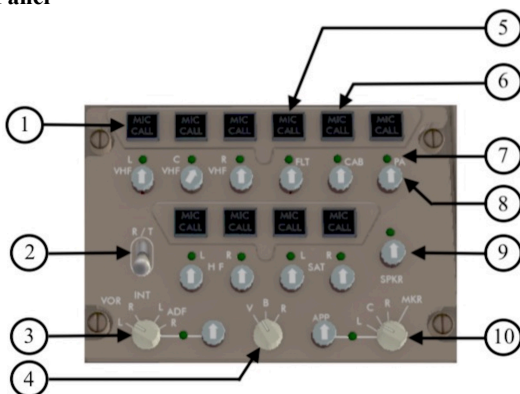
SSG B747-8 SERIES

5. Communications

Radios

The SSG 747-8 has 2 communication (COM) radios that are selectable via the radio panel. The radio selector is used to select the frequencies digitally on the lower half (the standby area) and the swap (arrow) button switches it to the active window.

Audio Control Panel



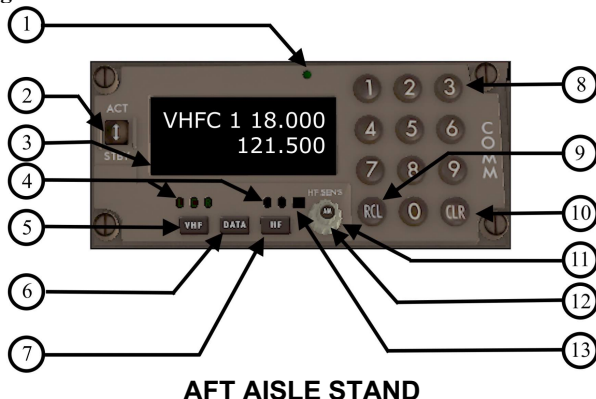
AFT AISLE STAND AND OBSERVER'S PANEL

The following switches are implemented in the SSG 747-8:

1. MIC CALL turns the ATIS or RADIO CHANNEL ON/OFF for on-line ATC communications (such as IVAO, VATSIM, etc.)
2. VHF button switches between the left and right radios as indicated by “L” and “R” on the display.
3. Switches ON/OFF the Morse code identification tones received from the radio beacons.
4. VOR/ADF receiver selector to monitor the identification tones from nav aids on the left and right side.

SSG B747-8 SERIES

Radio Tuning Panel



AFT AISLE STAND

The following buttons and switches are implemented in the SSG 747-8:

2. Frequency transfer switch
3. Active (ACT) / Standby (STBY) frequency window
4. VHF selected radio lights - Indicates whether left or right receiver is active
5. VHF select switch - Selects the left and right receiver (independent on Captain's and First Officer's side.)
8. Numeric keypad
10. Clear (CLR) key

Communication radios have 8.33 kHz spacing and will allow tuning frequencies 3 digits after the decimal (e.g. 118.015) for use in regions where it is required (e.g. Europe).

Please note that on the 747-8 navigation (NAV) radios (NAV1 and NAV2) for VOR and ILS tuning, as well as ADF radio tuning for Non-Directional Beacons (NDBs) is accomplished via the FMC "NAV RAD" page.

Transponder

The SSG 747-8 has a single transponder with TCAS capability. The transponder code (or "squawk") is entered using the numeric buttons. The transponder can be switched from STANDBY (OFF) mode, to XPNDR mode (which is altitude encoding) and then Traffic Advisory only (TA ONLY) or Traffic Advisory / Resolution Advisory (TA/RA) modes. Both of these modes turn on TCAS capabilities to show other traffic on the ND and are

SSG B747-8 SERIES

functionally equivalent on the SSG 747-8. The TEST and ALT RPTG OFF modes are inoperative.

Transponder Panel



AFT AISLE STAND

1. Transponder code switches
2. TCAS airspace selector (not implemented)
3. Transponder mode selector
4. Transponder code display
5. Transponder (XPNDR) selector (not implemented)
6. Identification (IDENT) switch – pressing this switch can be used by pilots to “IDENT” at (virtual) ATC’s request.

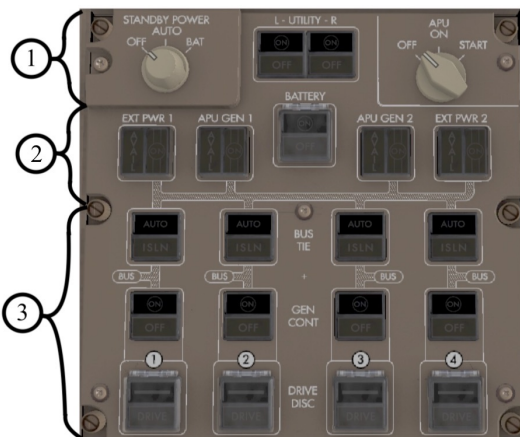
SSG B747-8 SERIES

6. Electrical

System Description

The SSG 747-8 has 4 engine-driven generators that provide electrical power to the aircraft. There is also an APU generator, and a standby battery. In the simulator, the SSG 747-8, also has provisions for external (ground) power to be hooked up to the aircraft, if needed. This is available on startup even if the source (e.g. GPU) is not visible or not at a gate.

Electrical Panel



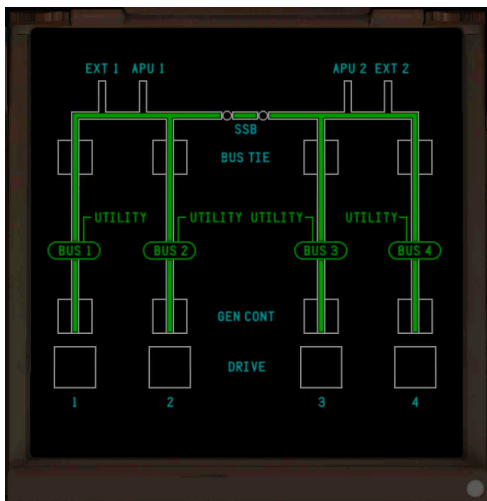
OVERHEAD PANEL

1. Standby power, battery and utility bus controls
2. APU generators and external power controls
3. AC bus and generator controls

The overhead electrical panel allows for the switching of the battery on and off (it has a plastic guard that needs to be lifted beforehand by clicking above the button), as well as the bus ties to connect the engine-driven generators to the main electrical busses. The APU Generators (1 and 2) and engine Generator Controls (GEN CONT) can also be selected. Standby power is available from the on-board batteries when the system is selected to AUTO and can power cockpit systems for a limited time in case of a total failure of the generators or when ground power is unavailable.

SSG B747-8 SERIES

Electrical System Synoptic



The electrical system synoptic that is accessible on the MFD is a representative display showing the generators that are working, and it will show if engine-driven generator bus ties are open or closed.

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
BATTERY OFF	Advisory	Battery switch is off.
ELEC AC BUS 1, 2, 3, 4	Caution	AC bus is not powered.
ELEC GEN OFF 1, 2, 3, 4	Advisory.	Generator control breaker is open.
ELEC UTIL BUS L, R	Advisory	One or more galley or utility busses are not powered.

SSG B747-8 SERIES

7. Engines, APU

System Description

The 747-8 is equipped with 4 powerful GE GENx-2B engines (a derivative of the GENx-1B engines available on the 787). This engine allows for the advances in efficiency that convinced Boeing to create a follow-on to the successful 747-400 series. GENx engines have increased power, reduced fuel consumption, and are more environmentally friendly than the GE, Rolls-Royce, and Pratt & Whitney engines installed on older 747-400s .

Engine Start Panel



OVERHEAD PANEL

1. Engine START switches. Note that these do not have white lights like on the 747-400.

Fuel Control Switches

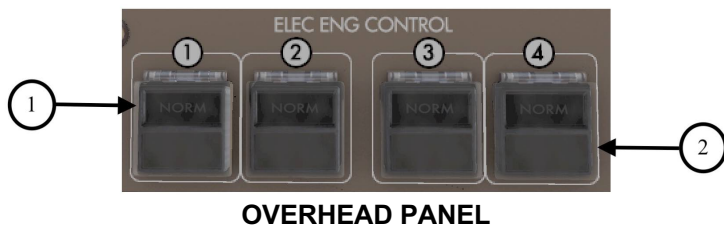


CONTROL STAND

1. FUEL CONTROL switches

Electronic Engine Control Panel

SSG B747-8 SERIES



1. Electronic (ELEC) Engine (ENG) CONTROL switches (EEC)

The EEC selects normal engine control mode and electronic engine control sets thrust using N1 RPM as the controlling parameter. Note that these are guarded switches.

Engine Indicators

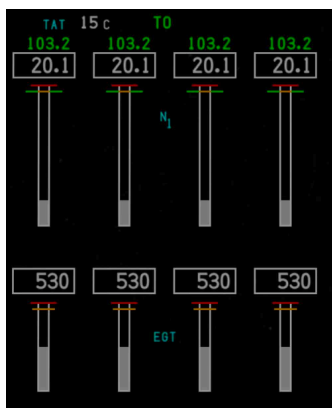
Engine performance and status is primarily indicated on two screens. The EICAS engine indicators are always visible, and the Lower MFD has secondary engine parameters that can be accessed through one of the synoptics for the MFD (which can be moved to other displays).

Primary Engine Display (EICAS)

The displays show the primary engine indications and are permanently displayed on the EICAS.

SSG B747-8 SERIES

EICAS Engine Indications

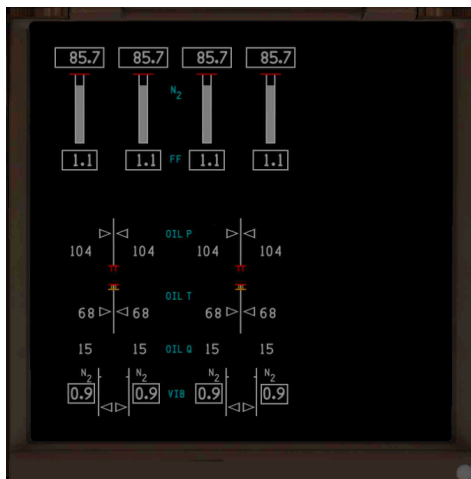


The information displayed in this presentation includes (from top to bottom):

- Full Authority Digital Engine Control (FADEC) Mode
- Selected N1 for FADEC mode (N1 is the primary engine fan visible at the front of the engine while N2 is the second (internal) engine spool, which turns independently from the N1 spool.)
- Current N1 value (in the white boxes)
- Individual vertical tapes showing the N1 percentage value as a white bar
- Exhaust Gas Temperature (EGT) tape

SSG B747-8 SERIES

Secondary Engine Synoptic



The secondary engine synoptic is accessed on the MFD and shows the following information for each engine:

- Current N2 value (in the white boxes)
- Individual vertical tapes showing the N2 percentage value as a white bar
- Fuel Flow (FF)
- Oil Pressure
- Oil Temperature
- Oil Quantity
- Engine vibration

Engine Starting

The GENx engines are started using pneumatic power (primarily bleed air) by opening a starter valve that lets high pressure air into the engine to drive the fans and provide enough energy to start the engine compression cycle, at which point fuel is introduced and ignited. Once the engine has started, the engine is self-sustaining and the starter and igniter are no longer required. This process is completely automatic on the 747-8 using start switches on the overhead panel, with fuel being controlled with the fuel control

SSG B747-8 SERIES

switches on the center aisle stand. This same process is used for engine starting on the SSG 747-8.

NOTE: One difference between the 747-400 and the 747-8 is that there are no longer white lights in the center of the starter switches to show the starters are operating. These lights are also absent on the SSG 747-8 overhead panel.

Thrust Reverser Operation

The thrust reversers can be engaged and disengaged using the standard XP “Shift” + “/” key combination.

Pilots have an indication on the EICAS that each individual engine is in reverse by the “REV” indication above the N1 tape white boxes.

The real aircraft has systems that prevent the thrust reversers from being operated during flight; the SSG 747-8 also includes these safeguards. Therefore, pilots cannot accidentally apply reverse thrust while airborne.

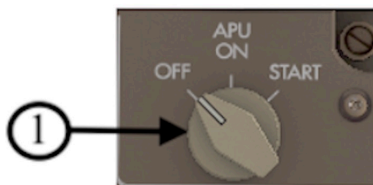
Note that reverse thrust is typically “all or nothing” in XP, while the real aircraft allows for idle reverse (the most commonly used) or any intermediate setting up to full reverse.

Auxiliary Power Unit (APU)

System Description

The APU is a small turbine powerplant mounted in the 747’s tail that provides pneumatic and electrical power for the aircraft while it is on the ground. It runs on the same fuel as the aircraft’s engines and allows the aircraft to function quite independently while on the ramp, without the need for external power, air, or start carts. The APU is not designed for use in flight on the 747-8 (that is, it cannot be started in flight but it can operate up to 20,000 feet of altitude under certain special circumstances)

APU Selector



OVERHEAD PANEL

SSG B747-8 SERIES

APU Operation

APU operation is quite simple: a single switch on the overhead panel can be set to START and then released. After it is released, it moves to the ON position during the APU starting sequence, which is fully automatic. The APU inlet door operation is also automatic and is visible in XP on the right side of the empennage tailcone.

The status synoptic on the MFD provides a means to monitor the APU's status by indicating its N1 and N2 percentages. There are also status messages on the EICAS associated with operation of the APU.

Shutdown of the APU is also quite easy, simply turn the overhead APU panel switch to the OFF position and the APU will shut down after a certain amount of time to allow for generator cool down.

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
APU FUEL	Advisory	Insufficient fuel to start/run the APU
ENG 1, 2, 3, 4 AUTOSTART	Caution	Autostart did not start the engine during a ground start, or EGT start limit has been exceeded.
ENG 1, 2, 3, 4 EEC MODE	Advisory	EEC in alternate control mode.
ENG 1, 2, 3, 4 FAIL	Caution	Engine failure or flameout. Inhibited on the ground.

EICAS Memo Messages

Message	Level	Message Logic
APU RUNNING	Memo	APU selector is on and APU N1 RPM exceeds 95%.

SSG B747-8 SERIES

8. Fire Protection

System Description

The SSG 747-8 has a sophisticated fire protection system for the engines (although not for the APU, which is simplified versus the one in the real aircraft.)

Engine Fire Panel



OVERHEAD PANEL

1. Engine fire switch (one per engine)
2. Engine fire warning light
3. Bottle discharged light

Fuel Control Switches



CONTROL STAND

1. Fuel control switch fire warning light (one per engine)

In case of engine fire, a master warning light comes on accompanied by a fire bell sound. The relevant engine fire warning light and fuel control switch fire warning light comes on (it is red in color) to inform the pilots as to which engine has the fire indication. The

SSG B747-8 SERIES

procedure is to bring that engine's thrust lever to idle, and then move its fuel control switch to CUTOFF. Then, the pilots should pull the relevant fire switch on the overhead panel for that engine (this disconnects the engine from its fuel supply and most other systems) and then twist the handle left or right to select either extinguisher Bottle A or Bottle B.

Pilots should allow time for the bottle to discharge completely, which is indicated by the bottle discharged light for Bottle A or Bottle B as selected. If that does not extinguish the fire, the other extinguisher bottle should be discharged. It should be noted that in the real aircraft, once the fire switch is pulled, that engine can no longer be restarted.

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
FIRE ENG 1, 2, 3, 4	Warning	Fire is detected in the engine.

SSG B747-8 SERIES

9. Flight Controls

System Description

For added lift and drag, each wing trailing edge also has 2 sets of flaps, one inboard and one outboard. They always move together in normal operations.

Additional lift is provided by 3 sets of leading edge flaps per wing. On the 747-8, these extend further to provide a bigger air gap than on previous 747 models. The SSG 747-8 now also retracts most leading edge flaps during application of thrust reversers, as in the real aircraft. This feature is designed to protect the leading edge flaps from foreign object damage. There is no indication in the cockpit of this temporary retraction and redeployment of the leading edge flaps and is completely automatic.

For additional drag, and to aid in turns, each wing has 2 sets of spoiler panels. There are 3 basic settings with various individual panels moving to different degrees. These are:

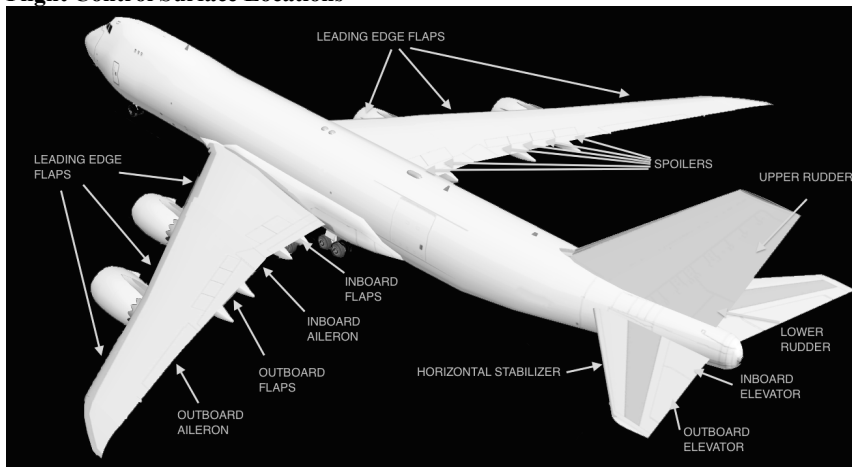
1. Flight spoilers
2. Speedbrakes (airborne)
3. Speedbrakes (ground)

Yaw is controlled via 2 sets of rudders on the vertical stabilizer. On the 747-8, the lower rudder is divided into 2 segments to provide additional travel. As on most swept jets, a yaw damper is installed to avoid dangerous oscillations. The yaw damper (with 2 separate systems) has the added benefit of applying the necessary rudder needed during turns so pilots do not need to use their rudder pedals during normal flight operations. This feature has been implemented on the SSG 747-8.

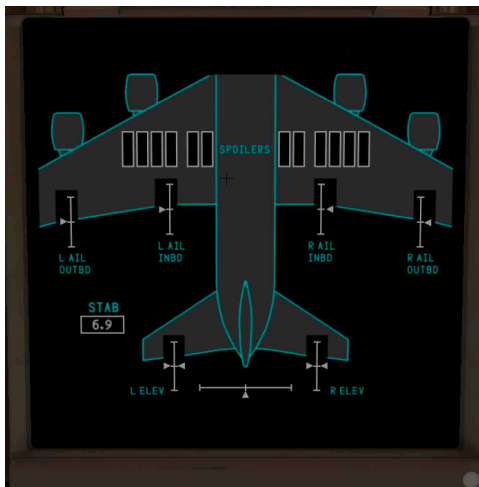
Pitch control is achieved via 2 sets of elevators on each side of the horizontal stabilizer. These move together in normal operations. Additional pitch trim is provided by the horizontal stabilizer, which moves as an entire unit up and down acting as the stabilizer trim.

SSG B747-8 SERIES

Flight Control Surface Locations



Flight Controls Synoptic



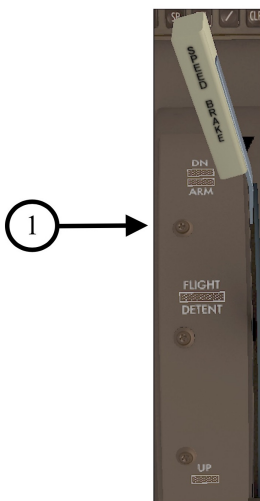
The flight controls synoptic can be selected on the MFD. The synoptic display shows the position of the various flight controls, including the primary flight controls, stabilizer trim and spoilers.

SSG B747-8 SERIES

Speedbrakes/Spoilers

The 747-8 has 4 outboard sets of flat panels on the wing and 2 inboard that can act as speedbrakes or spoilers as needed.

Speedbrake Lever



CONTROL STAND

1. Speedbrake lever

The speedbrake handle on the center aisle stand has several positions that can be selected up and down by grabbing the handle. The speedbrakes are armed by hovering the mouse over the lower part of the handle to get 2 vertical arrows; clicking on the handle then arms the speedbrake by moving the handle slightly upward. A SPEEDBRAKE ARMED message appears on the EICAS to confirm this feature has been engaged. On landing, the spoiler panels deploy in a staggered sequence on the SSG 747-8 like on the real aircraft. Apparently this sequence was implemented on the real aircraft after flight testing showed the nose would come down too fast if all panels deployed simultaneously.

The speedbrake handle can be selected in various positions using the standard XP “3” and “4” keys.

SSG B747-8 SERIES

NOTE: In the SSG 747-8 the speedbrake handle cannot be brought past the FLIGHT DETENT line. In the real aircraft, spoiler panel travel is restricted in flight to prevent damage to the wing or control surfaces. As in the real aircraft, only some panels operate as flight spoilers on the SSG 747-8.

Position	Function
DOWN	All spoiler panels are down (i.e. flush with the wing).
ARM	This means the spoilers are armed to deploy on touchdown and help keep the airplane firmly on the ground without bouncing.
FLIGHT DETENT	This is the maximum position to which the speedbrakes can be deployed in flight.
UP	All spoiler panels are up, providing the maximum drag and lift spoiling effect.

Speedbrakes will deploy automatically when applying reverse thrust. This is useful during a rejected takeoff and in case the pilots forget to arm them before landing.

Speedbrakes will go down when applying power. This is useful during a go around situation and after landing, when moving the thrust levers forward to taxi – giving the pilots one less thing to do to clean the aircraft up on the way to the parking spot.

NOTE: While the standard XP “3” and “4” keys allow the pilot to switch between the up, flight detent, and down positions for the spoilers, arming them requires the pilot to move the speedbrake handle manually.

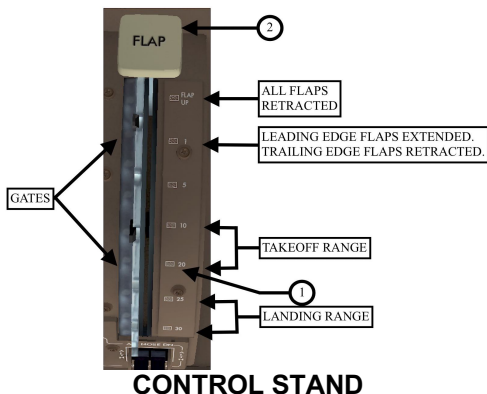
When the speedbrakes have deployed for landing, they will retract automatically once thrust is applied to vacate the runway just like on the real aircraft.

SSG B747-8 SERIES

Flaps

Flaps on the 747-8 can be selected by the pilots in incremental settings from 0 to 30 using the flap handle located on the aisle stand.

Flap Controls



1. Flap gate
2. Flap handle

Note that each gate under the flap handle has a number marked. This indicates the selected flap position.

EICAS Flap Indicator



The flap position is also displayed on the EICAS. Leading edge and trailing edge flaps move together according to system logic, so the same handle operates both sets of flaps as required. The flap setting is shown in magenta while the flaps are in transit and turns green when the selected flap position is reached.

SSG B747-8 SERIES

The 747-8 is equipped with a flap relief system and if the pilots exceed these limits, flaps will retract automatically. This is implemented in the SSG 747-8, and there is an associated EICAS message to tell you that this has happened as well.

Stabilizer Trim

One change on the real 747-8 compared to the 747-400 is that the stabilizer trim display is displayed on the EICAS. The small green triangle shows the position of the stabilizer trim, and a numerical display shows the stabilizer trim setting in a green box at the bottom of the display. Stabilizer trim units range from 0 to 15 and are indicated in terms of “Nose Up” (NU) or “Nose Down” (ND). The green band (indicating the safe stabilizer trim range for takeoff) on the left of the indicator is dynamic on the real aircraft, but is fixed in the SSG 747-8.

EICAS Stabilizer Trim Indicator



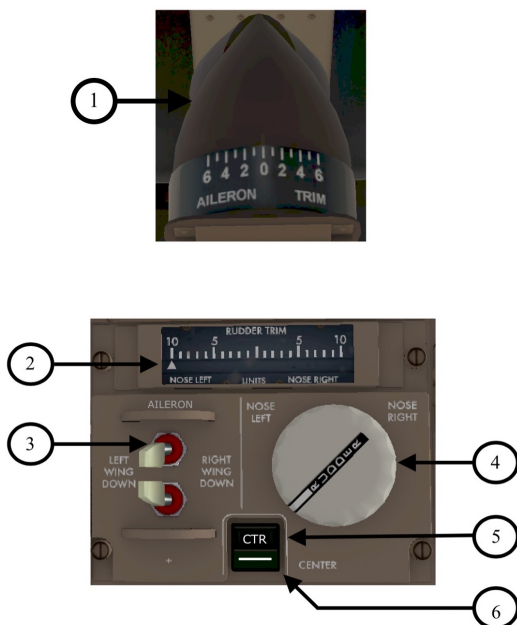
A feature of the real 747-8 is “speed stability trim” and “mach stability trim”, which help pilots by modifying the control response in pitch so that there is less movement required at those speeds than would otherwise be the case. The system automatically operates when the elevators have remained deflected in one direction from neutral for more than 3.5 seconds. This feature is implemented on the SSG 747-8 as autotrim, although not for the exact 3.5 seconds of the real aircraft.

NOTE: The stabilizer trim cutout and alternate stabilizer trim switches on the control stand are used in case of system malfunctions and are not implemented in the SSG 747-8.

SSG B747-8 SERIES

Aileron and Rudder Trim

Aileron and Rudder Trim Controls



AFT AISLE STAND

1. AILERON TRIM indicator
2. RUDDER TRIM indicator
3. AILERON trim switches
4. RUDDER trim selector
5. Rudder trim center (CTR) light (not implemented)
6. Rudder trim centering (CENTER) switch

The SSG 747-8 contains a complete set of aileron and rudder trim controls on the aft aisle stand that work in a very similar fashion to those on the real aircraft.

SSG B747-8 SERIES

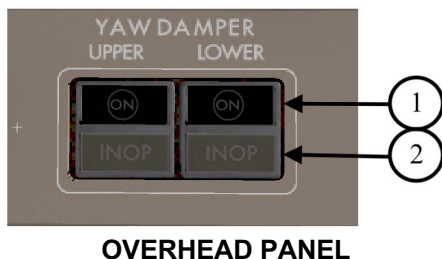
The aileron trim switches move the pilot yokes in the proper direction of the trim selection made by the pilots. The units of trim scale can be read using a placard on the top of the yokes themselves.

Left and right rudder trim can be input by the pilots using the large rudder trim selector and the units of trim can be read off of the indicator strip located above the selector. A centering button is available to instantaneously return the rudder to the centered position.

Yaw Damper

The yaw damper is an automated system that applies rudder to counteract yaw on the 747-8, and most other swept-wing jets.

Yaw Damper Controls



1. YAW DAMPER switches
2. Yaw damper inoperative (INOP) light

The yaw damper is turned on for normal flight by pressing both yaw damper buttons on the overhead panel.

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
FLAP RELIEF	Advisory	Flap load relief occurs.
YAW DAMPER LWR, UPR	Advisory	A yaw damper is inoperative because the button for the individual system has not been selected.
SPEEDBRAKE EXT	Caution	The speedbrakes are extended and one or more of these occur:

SSG B747-8 SERIES

Message	Level	Message Logic
		• The radio altitude is between 15 and 800 feet • The flap lever is in a landing setting • Two or more thrust levers are not at idle

EICAS Memo Messages

Message	Level	Message Logic
SPEEDBRAKE ARMED	Memo	Speedbrakes are armed to deploy automatically.

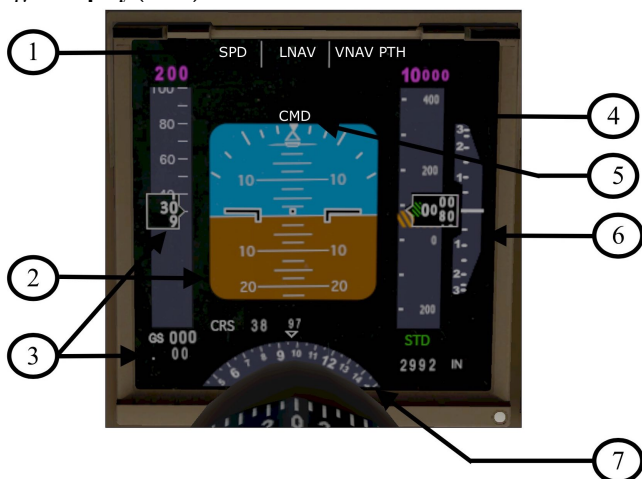
SSG B747-8 SERIES

10. Flight Instruments, Displays

Primary Flight Display (PFD)

The main display of flight instruments for the pilot is the PFD.

Primary Flight Display (PFD)



LEFT AND RIGHT FORWARD PANELS

1. Flight mode annunciations
2. Attitude, steering and miscellaneous displays
3. Airspeed/Mach displays
4. Autopilot, FD system status
5. Altitude displays
6. Vertical speed display
7. Heading and track displays

Within these groupings, the PFD contains the following information.

- On the top are the mode displays for the autothrottle, roll, and pitch modes for the FD and/or autopilot respectively.

SSG B747-8 SERIES

- The gray vertical tape on the left side displays IAS. Note that this is not useable below 30 kts and will indicate 30 kts even if the aircraft is not moving.
- The gray vertical tape on the right side displays altitude in feet (and in meters, which can be selected by the pilot).
- The bottom contains a compass arc showing heading information, as well as the selected magnetic heading from the MCP.
- The altimeter setting is shown on the bottom right in green, in either inches of mercury or in hectopascals (which can be selected by the pilot).
- Mach number and ground speed are shown on the bottom left in white.
- Decision Height (DH) is shown on the top right in white.
- FD bars are shown in magenta on the artificial horizon.
- LOC and ILS bars appear as magenta diamonds on the bottom and right of the artificial horizon.
- LNAV and VNAV deviation are shown with magenta triangles and also can serve as the indicators for IAN approaches (similar to LOC/ILS magenta diamonds).

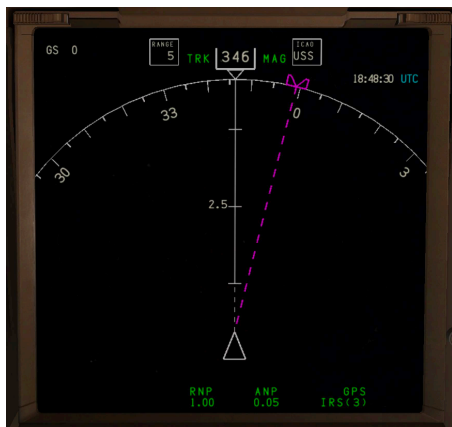
Navigation Display (ND)

The primary navigation display on the 747-8 is the ND and in the default mode provides a representation of the aircraft using a small white triangle to show the aircraft's position, as well as the area ahead of the aircraft.

This presentation can be changed in certain ND modes with a centered position, showing the area around the aircraft, as required. This mode can be useful during non-precision instrument approaches where the runway may at times be behind the aircraft.

SSG B747-8 SERIES

Navigation Display (ND)



The ND contains the following information and is fully independent on the Captain and First Officer's sides:

- Groundspeed (GS) and wind information in the top left corner
- Range display for the ND scale in a white box on the top left
- Magnetic track on a compass arc and the numerical value in a white box at the top of the arc.
- Magnetic heading shown by an upside down white triangle cue at the top of the arc
- Distance to next waypoint on the top right
- Time/timer information on the top right
- Selected heading from the MCP as a magenta dotted line
- Frequency and DME distance information (if applicable) for the 2 selected navaids (VOR or NDB) on the left and right corners
- Other aircraft are typically displayed as white diamonds with altitude relative to the 747-8 indicated in hundreds of feet, as well as an arrow indicating whether they are climbing or descending
- Waypoints, navaids and suitable airports (those with a runway more than 6,000 feet long) as selected on the PFD/ND selector
- Weather radar returns (if selected on the PFD/ND selector)
- A Vertical Situation Display (VSD) can be overlaid on the ND when selected by the pilot by using the CTR button twice

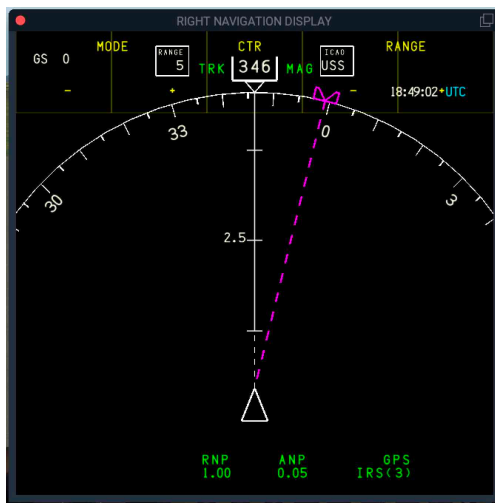
SSG B747-8 SERIES

The First Officer's ND can also be called up as a pop-up on window by pressing the "SHIFT" and "F8" keys simultaneously in Windows and Linux ("SHIFT" + "fn" and "F8" keys simultaneously on a Mac). It is visible both in inside and outside views. It can be moved around the screen by clicking and dragging the top border of this ND once it is displayed, and can also be moved to other connected monitors by clicking on the top right of the pop-up window. To bring the pop-up window back to the main XP screen, click on the top left corner where it says "P_In".

The top part of this ND allows several of its modes to be adjusted as if using the MCP as follows:

- Top left (mode left and right)
- Middle (CTR)
- Right (decrease range and increase range)

By clicking in the middle of this ND it can be closed as well. These zones appear in yellow when hovering the mouse pointer over the pop-up ND as shown below:



The various ND modes are described as follows:

SSG B747-8 SERIES

ND Mode	Description
APP	Approach mode
VOR	VOR mode
MAP	Map mode (this is the most commonly used presentation)
PLAN	Flight planning mode, which is a “North up” presentation that can be used to check the flight plan entered in the FMC.

VSD Display



The VSD is a useful display that is overlaid on the MAP as a white rectangle in both centered and arc modes that shows the aircraft as a triangle on the left along with the current altitude on the left gray scale, and the altitude selected on the MCP (shown in magenta). The black portion of the VSD shows a cross-section depiction of the terrain below and ahead of the aircraft shown in green and scrolling from the right. The bottom scale shows the distance in NM for the various terrain features represented and this will change according to the range selected by the pilots for the ND. The VSD also shows the glidepath to follow during the Approach phase. Furthermore, it shows the waypoints in front of the aircraft and the altitude restrictions.

The vertical axis changes automatically based on the aircraft altitude and the distance scale selected at the horizontal axis. Using orange and red color, warning if the current vertical attitude is lower than the terrain ahead. In this manner, the VSD can show the pilots the nature of the terrain ahead and helps with their situational awareness.

SSG B747-8 SERIES

Airport Map Display



The Airport Map Display can be accessed on the ND by selecting ranges below 5 NM on the ND Range Selector while the aircraft is located on an airport. The display includes an International Civil Aviation Organization (ICAO) ID Box in the top right of the ND where the 4-letter ICAO code of the current airport is displayed.

NOTE: This display is not supposed to be used as a sole reference when operating on an airport. It is intended as a reference to increase situational awareness by the pilots.

Integrated Standby Flight Display (ISFD)

On the real 747-8, the ISFD replaces the three standby instruments that were installed on earlier 747-400 models.

SSG B747-8 SERIES

ISFD



The ISFD has its own independent gyro and takes about 9 minutes to align. This process takes place automatically. The various controls for this display are non-functional in the SSG 747-8.

MCP PFD/ND Selector

The PFD/ND selector on the MCP has a multitude of controls in a small area. These controls are listed as follows:

Control Panel PFD and ND Controls



GLARESHIELD PANEL

SSG B747-8 SERIES

1. Minimums (MINS) selector. This is a radio altimeter warning selector knob. On the real aircraft this can be switched to a barometric altimeter warning, but this is not implemented in the SSG 747-8.
2. RADIO / BARO altitude control (not implemented)
3. Minimums reset (RST) switch (not implemented)
4. Meters (MTRS) switch, which allows pilots to display altitude information in meters as well as feet for operations in countries where altitudes are metric (primarily the People's Republic of China and some former Soviet republics).
5. Barometric standard (STD) switch (inner)
6. Barometric (BARO) selector (middle)
7. Barometric (BARO) reference selector, which allows switching between inches of mercury and hectopascals.
8. Flight Path Vector (FPV) switch.

In addition:

- There are two VOR/NDB selectors on either side of the panel to select between these types of radio beacons for display on the PFD and ND
- The CTR button allows for a change in presentation of the ND with aircraft in the middle to an arc presentation with the aircraft at the bottom of the display. The CTR button also cycles through the various VSD modes.
- The ND selector switch allows pilots to switch between ND modes
- The range selector permits switching of the ND scale from 0.5 to 329 NM.
- The center of the range selector is the button to for the Traffic (TFC) Display to display TCAS targets on the ND.
- The bottom buttons allow for additional information to be presented on the ND as follows:
 - WXR shows radar returns
 - STA shows nav aids
 - WPT shows waypoints
 - ARPT shows airports
 - DATA shows ETA information and FMC altitude constraint for waypoints along the programmed route
 - POS shows VOR radials extended to the aircraft's position along a route
 - TERR shows the Ground Proximity Warning System (GPWS) look-ahead terrain mode

Standby Compass

All airliners have standby magnetic compasses as a backup, and the 747-8 is no exception. It is located at the top of the central window post and within easy view of the

SSG B747-8 SERIES

pilots. The compass also is lit to allow for its use at night and low light conditions. A working standby compass is provided in the SSG 747-8 as well.

Clock Display

The real 747-8 has a clock readout on the upper right corner of the ND instead of the dedicated chronometer on the left of the PFD that can be found on older 747-400s. This new feature also has been implemented in the SSG 747-8. The default mode is Coordinated Universal Time (UTC), sometimes called “Zulu” time.

The timer can be started by pressing the CLOCK button on either pilots’s side glareshield panels. The functions are:

- Press once: Starts timer
- Press again: Stops timer
- Press again: Resets timer

There also is an “ELAPSED TIME” timer for the flight available on the MFD Status synoptic. It is triggered by opening and closing the passenger doors and starts the timer once the aircraft is airborne.

SSG B747-8 SERIES

Electronic Checklist Displays

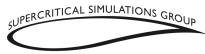
Normal checklists can be displayed on any selected Multifunction Display (MFD) by pressing the CHKL button on the Display Select Panel located on the glareshield.



1. Cursor selection box (magenta) – Highlights cursor selection area for current incomplete line item.
2. Open loop indicator (gray box) – Indicates the line item is an open loop action item. Requires crew confirmation to become complete.
3. Complete indicator (green check mark) – Indicates line item is complete.
4. Checklist line item – Displayed (white) when action is required, line item is incomplete. When action is not required, line item remains white and is complete. Displayed (green) – line item is complete.

The checklist items are selected by clicking the Cursor Control on the Control Stand next to the throttle quadrant. It may be useful to assign a joystick button to this device to facilitate electronic checklist operation.

Once an individual checklist is complete, a green “CHECKLIST COMPLETE” appears below the particular checklist to indicate it is complete. The checklist will then automatically cycle to the next checklist in the sequence.



SSG B747-8 SERIES

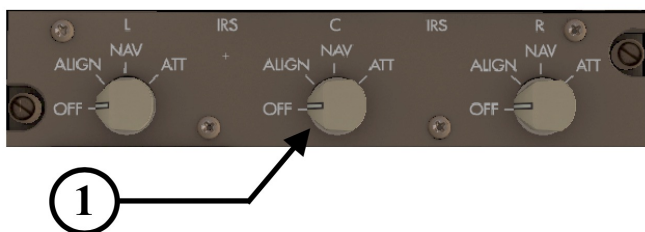
Resetting the checklist (for a through flight, for example where the aircraft is not powered down completely) can be accomplished by selecting “RESET CHECKLIST” from the tablet’s OPT (Option) menu.

SSG B747-8 SERIES

11. Flight Management, Navigation

Inertial Reference System (IRS)

Inertial Reference Unit (IRU) Selectors



OVERHEAD PANEL

1. IRU Mode Selectors

There are 3 IRU Mode Selector switches, one for each IRU (Left, Center and Right) that comprise the overall IRS. The switches control the IRUs, which all are independent from each other, and comprise laser gyro and accelerometer units in the aircraft's electronics bay that measure aircraft movement in all 3 axes. Each IRU interprets these signals and sends position information to the FMCs. Therefore, this source is completely separate from GPS position information and predates GPS use in civil applications (back when the 747-400 was first introduced into service).

The FMC uses a weighted average of all 3 IRU position information to display aircraft position. Although the IRUs are precision equipment, they will drift over time, just like Inertial Navigation Systems (INSs) in the 747 Classics of the old days. In flight, the IRUs can be updated with VOR (old way) and/or GPS (newer way) position information to remain accurate.

The IRUs can only detect movement and have no way to independently determine the position of the aircraft. Therefore, there are 2 processes that are required for them to be useful. One is alignment and the other is position initialization.

Alignment

SSG B747-8 SERIES

It is important to know that the gyro platforms in the IRUs provide attitude and other information to the flight displays. Why install additional gyros when you have these sophisticated platforms already installed? Therefore, when IRU mode selectors are in the OFF position you will get various “flagged” (yellow boxes) or blanked indications on the PFDs and NDs.

The real IRUs use a fascinating method to align, which is to use the True North Pole as a reference and the motion of the earth and gravity (which it can detect) to determine what the aircraft’s relationship is to True North and its latitude and true heading. In the real aircraft there are 2 important things to know regarding alignment:

1. The airplane cannot be moved during this process or it will stop (this is simulated in the SSG 747-8).
2. It takes some time for sufficient motion to be detected for this to work.

To initiate alignment, the IRS switch is rotated from OFF to NAV, not ALIGN. The alignment process takes about 8 to 10 minutes.

NOTE: Normally only the OFF and NAV positions on IRU selectors are used. ALIGN and ATT modes are for abnormal/alternate situations and are not implemented in the SSG 747-8.

Because this is a flight simulator, the tablet OPTIONS menu has a selection that allows you to select a shorter alignment time if you do not wish to wait and are OK with less realism in the alignment process.

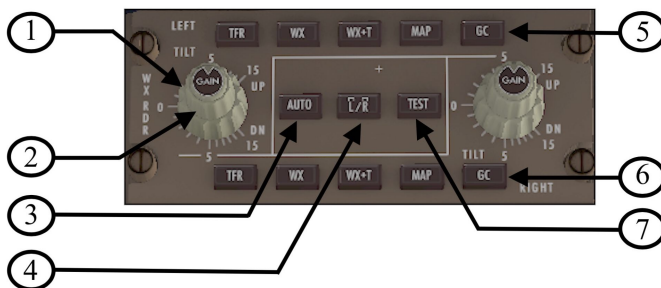
Once the IRU selectors have been set to NAV, the Upper ND displays a message showing “TIME TO ALIGN” at the top left and below it “L” and “C” and “R” for left, center and right IRUs. Each IRU has a timer for the alignment process showing “7+ MIN” at first and counting down the time in minutes until that particular IRU is aligned.

When the alignment process is complete, the flags on the PFD and ND disappear and enter navigation mode so you then get the normal displays showing all the necessary information.

SSG B747-8 SERIES

Weather Radar

Weather Radar Panel



AFT AISLE STAND

2. Gain control

NOTE: Other controls are non-functional

Weather radar returns are shown on the ND by pressing the “WX” button on the MCP PFD/ND Selector. The SSG 747-8 has limited controls for the weather radar. However, the attenuation on the radar display for the ND can be adjusted. This is useful in conditions where there are a lot of radar returns that might mask other ND details.

Flight Management Computer

Like all newer generation 747s starting with the 747-400, the real 747-8 has an extremely sophisticated FMC. A note about terminology is required here. Often the terms “FMS” and “FMC” are used interchangeably; however, an FMS is a larger and more sophisticated system than is implemented in most flight simulators. Therefore, for the purposes of this manual, the system will be referred to as the “FMC” in this FCOM.

The system provides provide guidance information for the autoflight system, as described in Section 4 of this FCOM, as well as other useful functions. Its various components allow for very precise control and management of the aircraft in all flight regimes using inertial guidance capabilities from ring gyros on a stabilized platform, and with Global Positioning System (GPS) position input as well.

The FMC helps the flight crew with navigation, performance optimization and automatic fuel monitoring. The flight functions manage the airplane lateral flight path (LNAV) and vertical flight path (VNAV) via the autopilots. The FMC also tunes the navigation radios

SSG B747-8 SERIES

and sets the desired courses for various procedures such as approaches, SIDs, and STARs, using navigation database supplies the necessary data to fly all routes. Cruise altitudes and altitude restrictions are selected by the pilots based on their ATC clearances and established procedures for the FMC to calculate VNAV behavior. To accomplish this, pilots enter route and flight data into the FMC, which then uses its navigation database and airplane position information to calculate the flight path, and issue pitch, roll, and thrust commands to the aircraft's autopilots and autothrottle to fly the desired autothrottle, autopilot, and FD operating modes (e.g. LNAV and VNAV) in an optimum fashion.

It should be stated up front that the FMC in the SSG 747-8 has some limitations in its capabilities versus the real one, and some features are added to make it more useful in a flight simulator.

Control Display Unit (CDU)

Control Display Unit (CDU)

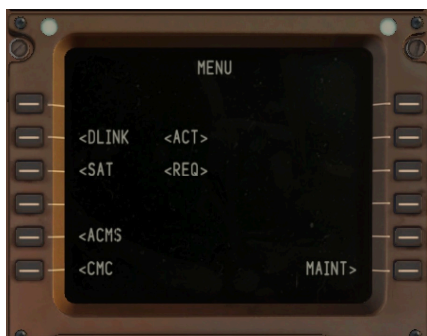


SSG B747-8 SERIES

The primary component of the aircraft's sophisticated FMC that is visible to the pilots is the CDU (see figure above). There are 3 of them on the flight deck. The CDU consists of a keyboard and a display with Line Select Keys (LSKs) on either side of the display.

The SSG 747-8 has two separate CDUs on the captain and first officer side.

NOTE: The Center CDU on the pedestal between the pilots is not used for navigation in the real aircraft. It used for satellite communication, ACARS (textual messaging) and as a central maintenance computer terminal. In the SSG 747-8 it displays the menu options for these functions (see below), but it is not currently functional. At a later time, we may implement some additional features with it.



The two functional CDUs can be called up by pressing the “F8” key in Windows and Linux (“fn” and “F8” keys simultaneously on a Mac) for the captain’s and the “F8” + ALT keys in Windows and Linux (“fn”, “option” and “F8” keys simultaneously on a Mac) for the first officer’s. Alternatively, the CDUs in the 3D panel can be used just as easily.

The dragging (Upper Zone) allows the user to move the “pop-up” FMCs around the screen as needed. By clicking and holding the upper part of the pop-up FMC you can move the CDU around the computer screen. It can also be moved to other connected monitors. This is accomplished by clicking on the top right of the pop-up window. To bring the pop-up window back to the main XP screen, click on the top left corner where it says “P_In”.

Expanding – clicking and dragging the top right corner of the CDU allows the user to increase the size of the FMC to suit their preference and/or monitor size.

LSKs numbered LK1 to LK6 on the left side numbered from top to bottom

SSG B747-8 SERIES

LSKs numbered RK1 to RK6 on the right side also numbered from top to bottom

These LSKs are used to select various lines of information in the screen.

The following keys bring up various pages of FMC data as described in this section:

INIT REF
RTE
DEP / ARR
VNAV
FIX
LEGS
HOLD
PROG
EXEC
MENU
NAV / RAD
PREV PAGE
NEXT PAGE

DEL
CLR

The ATC and FMC COMM menu keys are non-functional in this version of the SSG 747-8.

There are two message light “zones” on the left and right of the CDU. The left one lights up with the message “KEYB” to show that it is accepting direct input from the user’s computer keyboard. To enable the use of the computer keyboard again (say for an XP command key) simply click in the right message zone area. There are two messages that appear in the right area. One is “MSG” which means that there is an FMC message being displayed and serves to alert the pilots. The other is “OFFS”, which informs the pilots that the aircraft is flying a lateral deviation offset that has been entered into the FMC RTE page.

NOTE: These clickable zones are only available on the pop-up CDUs, not the ones built-in to the 3D cockpit. However, the information displayed is the same for both.

The bottom line of the display is reserved for what is called the “scratchpad” and is where entries are made into the FMC. Typically, items selected with the LSKs will be copied down to the scratchpad to reduce the need to retype.

SSG B747-8 SERIES

FMC Functions

Lateral Navigation (LNAV) is provided to the autopilot including FLARE and ROLLOUT modes. FLARE allows for smooth landings, changing the pitch to decrease the vertical speed during touchdown. ROLLOUT ensures the aircraft tracks the correct runway heading during touchdown and runway rollout after landing. This is a very good feature during landings in high crosswinds.

The throttle is fully controlled by the FMC, except in the approach phase, where the pilot can also choose to operate the throttles manually. If the autothrottle is disarmed, then the throttles are controlled manually by the pilot. The pilot can disarm the autothrottle during any flight phase, even when it is being controlled by the FMC (e.g. with VNAV engaged). If the MCP Speed knob is pressed, then the blank speed window on the MCP shows the current speed computed by the FMC (VNAV remains engaged). The pilot can change this speed by rotating the MCP Speed knob. If it is pressed again (and VNAV is engaged) the window is blanked again and the speed setting is computed by the FMC once again.

Features of the FMC in the SSG 747-8 include:

- Optimum N1 limits in takeoff, climb and cruise, and go around
- Built in tables for climb rates and descent rates
- Optimized tables for fuel consumption in climb, cruise and descent
- Top of Climb (T/C), Top of Descent (T/D), fuel at waypoints, Expected Time of Arrival (ETA), Position and overall predictions calculated dynamically
- CLB, CRZ and DES pages
- Step climbs that can be selected from none (0000) to 9,000 feet in size. The cruise altitude changes are showed in the LEGS page
- Implementation of a COST INDEX that affects at climb, cruise, and descent calculated speed
- Full aeronautical database that includes airways, Standard Instrument Departures (SIDs), Standard Terminal Arrivals (STARs), and runway lengths
- Accurate computed fuel prediction for the trip
- Insufficient fuel warning message based on predicted fuel burn
- Vertical Navigation (VNAV) and LNAV modes integrated with the autopilot.
- Provisions for altitude and speed restrictions
- Derated takeoffs
- Calculations for V1, VR and V2 based on real aircraft data
- Ability to save routes, load previously saved routes, and to import standard XP flight plan
- “Route Full” message

SSG B747-8 SERIES

- Manual and automatic control of radio navigation aids
- Ability to pause the simulator some distance before the T/D
- Pop-up for the CDU to facilitate entries and display
- Ability to handle the following types of information and procedures:
 - Intersections
 - DME arcs
 - Altitude waypoints
 - Conditional waypoints
 - Waypoints bearings with distance
 - Holds
 - Intersection of bearings of two waypoints
 - Along track waypoints
 - Route hold page with no holding fix in route
 - Route hold page after pushing POS to hold at the current aircraft position
 - Route hold page with holding fix in route
 - Lat/Lon entries
 - Route lateral offset
 - Required Time of Arrival (RTA)
 - Speed transition in the approach phase until slats/flaps are extended

To adjust the brightness in the CDU screen you can use the brightness button. However, this only works for the CDUs on the 3D panel center console, not on the pop-up.

FMC Operation

When pilots enter the aircraft's cockpit to conduct a flight, the FMC is initialized and is in the preflight phase. When a phase is completed, the FMC internally changes to the next phase in the following sequence:

- Preflight
- Takeoff
- Climb
- Cruise
- Descent
- Approach
- Landing
- Flight complete (end of route)

The FMC never reverts to a previous phase; for example, if it is in Cruise phase and climbs to another cruise level, it accomplishes the climb in the Cruise.

SSG B747-8 SERIES

Preflight Phase

During preflight, the ATC flight plan, routes, and performance data are entered into the FMC. Part of this data is entered manually and the other part is filled in from the FMC's own information sources, such as the known performance of the aircraft, and sensed information that is funneled into the FMC by the aircraft's systems such as fuel quantity, aircraft weight, atmospheric conditions, etc. The flight plan defines the route of flight from origin to destination and initializes the LNAV mode. Flight plan and aircraft information are combined by the FMC to develop the performance data used by the VNAV mode. A previously stored route can be loaded (from any source that can generate flights plans in the XP format, or as a presaved company route), and a newly entered route can be stored; however, only when the aircraft is on the ground.

The preflight data consists of the following:

- Initial position (from XP)
- Route of flight (previously stored or manually entered)
- Performance data (funneled to the FMC)
- Takeoff data (manually entered)

Optional data includes:

- SID (from navigation database)
- STAR (from navigation database)
- Approaches (from navigation database)

The first step is to enter data with the IDENT page, which is selected with the IDENT prompt on the INIT/REF INDEX page.

After the necessary data entered on each page is checked, press RK6 to select the next page. When ACTIVATE is selected on the ROUTE page (RK6), the "Execute" (EXEC) light illuminates, and pressing the EXEC key is required at this point make the route active.

The next step is to use the "Departure/arrival" (DEP/ARR) page to select a runway and a SID. The FMC uses navigation databases for its calculations that are valid for 28 days. Each database set corresponds to the standard navigation chart revision cycle known as Aeronautical Information Regulation and Control (AIRAC). As described in Volume I of this FCOM, the contents of this navigation database can be periodically updated and transferred to the FMC before the active data's expiration date, particularly if you wish to fly using on-line ATC services such as VATSIM and IVAO.

When all required preflight entries are complete, the PREFLT label on the TAKEOFF REF page changes to a dashed line. Now select THRUST LIM pressing RK6 again.

SSG B747-8 SERIES

Takeoff Phase

The takeoff phase starts with selection of the desired takeoff N1 value and finishes when the thrust reduction altitude is reached. At this point climb thrust changes from “Armed” to “Selected”.

Climb Phase

The climb phase starts at the thrust reduction altitude and finishes when the top of climb (T/C) point is reached. The T/C point is where the airplane reaches the cruise altitude that was entered on the PERF INIT page.

Cruise Phase

The cruise phase starts at the T/C point and finishes when reaching the top of descent (T/D) point. The cruise phase can include step climbs and cruise level descents.

Descent Phase

The descent phase starts at the T/D point and finishes with the approach phase.

Approach Phase

The approach phase starts when the first waypoint of the procedure sequence (such as a STAR) is reached, or when the distance to the destination airport is less than 25 NM.

Landing Phase

The landing phase starts when “FLARE” engages and finishes at the End of Route phase.

End of Route

Thirty seconds after aircraft touchdown, the flight complete phase clears the active flight plan and pilot entered information. To prepare for another flight, the FMC needs to be initialized on the INDEX page.

Operational Rules

When using the LNAV and VNAV modes, continuous checking of the pitch, roll, or thrust commands by the pilot is recommended. If any problem is noticed, another roll and pitch mode must be selected.

The FMC will not follow the active waypoint when more than 2.5 NM off the active route (except if it is in LATERAL OFFSET mode). A return to the active route has to be accomplished using DIRECT TO or by changing to HDG mode to intercept the track of the desired course.

SSG B747-8 SERIES

When a waypoint is in the route more than once, the FMC deletes the appropriate one automatically.

When entering airways in a route page, the starting and ending waypoints must be in the navigation database. A route segment must be entered as a DIRECT leg.

DIRECT TO courses are segments of a great circle route. When entering a DIRECT TO waypoint on the LEGS page, this is now the destination waypoint from the aircraft's current position.

Data Entry

Data is entered into the FMC through the scratchpad by pressing the corresponding LSK or RSK. You can also use the rest of the keys by pressing them or use the numerical keys on your computer. If you want to use your computer keyboard, just click in the FMC dragging zone, after which a "KEYBOARD" message appears on the left of the FMC, to inform you that this mode has been entered. To revert to using keyboard commands back to XP you press the MSG area. You can obtain the operation of the mouse by simply clicking any part of the pop-up CDU or the rest of the screen for XP. If the CDU is in front of a button or device in the aircraft panel, clicking it will activate the CDU so it may need to be moved.

Another functionality is that the CLR key remembers the previous scratchpad data you have entered, so if you press the CLR key repeatedly you will see the last entered data. Pressing CLR once will delete the character and if the CLR key is pressed for one second, this will delete the complete scratchpad line and the previous data entered in the scratchpad will appear.

Waypoint Bearing With Distance

For bearing with distance enter in the scratchpad as follows:

DEL180/10

This means, you want to display a point that is on a bearing of 180° and 10 NM away from the DEL VOR.

Now indicate with LK1 the position you want to enter the new waypoint. Then the FMC will show the following:

DEL01

If you enter a second fix with the same name, it will show as follows:

SSG B747-8 SERIES

DEL02

The FMC always displays the first three letters of what you entered and the following numbers, which can range from 01 to 49.

Intersection of Two Waypoint Bearings:

To display these you make the following entry in the scratchpad:

SEA330/OLM020

This means that you want to display a point where the 330° radial of SEA VOR and the 020° radial of OLM VOR intersect. Pressing the LSK position you want, the FMC will show the following:

SEA01

Along Track Waypoints

When you have a route, for example from ELN to EPH (042°) and you want to have a waypoint between these waypoints, then you can enter the following:

ELN/25

This will be a waypoint 25 NM away from ELN, but still with the same course between ELN and EPH (042°)

The FMC will show this in the LEGS page:

ELN01

The same rules apply when creating waypoints as described above.

Lat/Long Entries

The format to enter latitude and longitude positions in the LEGS page is as follows:

N4147.9W08745.1

(This represents a position of N 41°47.9' W 087°45.1')

Note that there are no spaces. When this waypoint is entered, the FMC will display it as follows in the LEGS page:

N41W087

SSG B747-8 SERIES

You can also enter this format: N45W130 or S61E020. This format can be entered in the RTE PAGE as well.

Terminology

The following terms describes some terminology to better understand the use of the FMC.

Active: Flight plan data currently being used to calculate LNAV or VNAV.

Activate: The procedure to change an inactive route to the active route for navigation. It is a two-step process:

1. Select the ACTIVATE prompt
2. Push the execute (EXEC) key

Altitude Restriction: A restriction altitude at one waypoint.

Delete: Remove FMC data and revert to default values, dash or box prompts, or a blank entry using the DELETE key.

Econ: A speed profile calculated to optimize operating cost. The economy speed is based on the cost index. A low cost index results in a lower cruise speed. Maximum range cruise or the minimum fuel speed may be obtained by entering a cost index of zero. This speed profile does not take into account the time. A minimum time speed profile is obtained by entering a cost index of 9999. A low cost index is used when fuel costs are high compared to operating costs. Airliners normally use a low cost index of 70 to 90.

Enter: Writing data in the scratchpad and then pressing any LSK to enter data.

Erase: Removing entered data by selecting the ERASE key before activation.

Execute: Press the EXEC key to make any modified data active.

Initialize: Reset all the data required by the FMC.

Message: The data that the FMC displays in the scratchpad to advise the crew of a system condition.

Modify: Active data that is changed but not yet executed. When a modification is made to the performance mode, MOD is shown in the page title, ERASE shows next to LSK RK6, and the light above the EXEC key illuminates.

SSG B747-8 SERIES

Select: Pressing a key to obtain data, make an action, or to send data to the scratchpad.

Speed restriction: An airspeed limit in a waypoint entered by the flight crew.

Speed transition: An airspeed limit associated with a specified altitude entered in the FMC.

Waypoint: A point on the route. It can be a navigation fix (these are shown on charts and have specific names), latitude and longitude coordinates, or a navigational aid (VOR or NDB station).

Integrated Approach Navigation (IAN)

The SSG 747-8 now incorporates IAN (Integrated Approach Navigation), which is a term developed by Boeing for a suite of features that allows pilots to fly any non-precision approach as they would an ILS approach. Therefore, a VOR, LOC, GLS and RNAV approach can be flown as an ILS approach using a glide path (G/P) generated by the FMC similar to the glideslope of an ILS.

NOTE: IAN is not available yet for NDB approaches in the SSG 747-8.

It is important to note that there is no AUTOLAND capability and you have to land IAN approaches manually. The procedure to fly an IAN approach is very straightforward. The first step is to select the non-precision approach to the desired runway. As with an ILS approach, when you are lower than 6,000 feet MSL and closer than 20 NM from the arrival runway, just press the APP button on the glareshield to arm the autopilot approach mode. The IAN system will set up the approach, which can then be flown in the same manner as an ILS following the same needles on the PFD as on an ILS, but the pointers are magenta triangles rather than diamonds to distinguish them from an ILS.

SSG B747-8 SERIES



Note that it is very important to properly set the proper barometric setting for the conditions at the airport because altimeter is the primary method of ensuring that altitude constraints are adhered to for the approach.

Lateral Navigation (LNAV) Operation

LNAV sends commands to an autopilot to fly to the next waypoint. LNAV engages at or above 50 feet, when laterally within 2.5 nautical miles of the active route leg, if one of the three autopilots is engaged. FMC LNAV guidance normally follows great circle courses between waypoints.

However, when an arrival or approach from the FMC database is entered into the active route, the FMC will supply commands to fly a constant heading, track, or follow a DME arc, as required by the procedure.

Waypoints

Waypoint (navigation fix) identifiers are shown on the FMC and navigation display.

The FMC message NOT IN DATABASE or NOT FOUND is shown if a manually entered waypoint identifier is not in the XP navigation database.

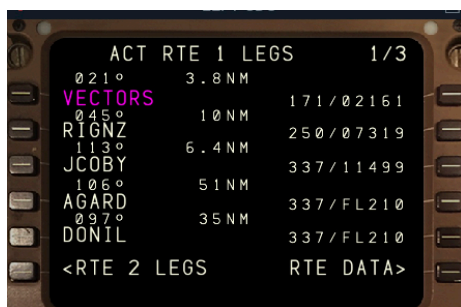
SSG B747-8 SERIES

VORs: Waypoints located at VORs are identified by use of the station identifier.

NDBs: Waypoints located at NDBs are identified by use of the station identifier.

Fix Waypoint Names: Fixes with single names containing five or fewer characters are identified by the fix name.

RTE LEGS Page



Hdg: This is the Magnetic heading to follow for the next waypoint. The magnetic variation of the area is taken into account.

Waypoint: Name of the waypoint that can be any type of navaid. This can even be a GPS coordinate. The maximum number of waypoints in a route is 120 Waypoints and 20 LEGS pages.

Number of pages: Indicates the current page and the total number of pages.

Distance: This displays the calculated distance in NM to the next waypoint (i.e. from one waypoint to the next one.)

Speed/altitude: Speed and altitude associated with a waypoint. This is calculated by the FMC in the INIT/REF page and is later updated during the entire flight.

Cycling through Pages: To see the remaining waypoints you can press NEXT PAGE and PREV PAGE, thus functioning in a cyclical way, meaning that if you reach the last page and click NEXT PAGE, you will return to the first page which (page 1). You can also do the same with PREV PAGE and obtain the same results but going in reverse.

SSG B747-8 SERIES

Modifying: To modify the route, go to RTE PAGE by pressing the RTE Button or in the main LEGS page.

Waypoint Identifier

The active leg is always the first line of the first active RTE LEGS page. The active waypoint is on the active leg. Modified waypoints create a route discontinuity that must be resolved.

All route waypoints are shown. Waypoints on an airway are included on the route legs page. Waypoints are shown in flight sequence. They can also be modified as follows:

- Add waypoint
- Delete waypoint
- Change waypoint sequence

This can be done in the LEGS page or in the RTE PAGE

Distance to the Next Waypoint

Distance (decreasing) from the aircraft's position to the active waypoint (First waypoint or next waypoint), or from waypoint to waypoint.

Waypoint Speed/Altitude Restrictions

Waypoint speed or altitude restrictions are shown in a larger font.

Manual entry is allowed in any phase, but only to waypoints belonging to the Climb phase or Descent phase. They can also be entered by the FMC when restrictions are part of a procedure.

Speed constraints are assumed to be at or below the shown speed.

Data entry is as follows:

- Speed entry has to be an airspeed
- XXX/XXXXX - Airspeed/altitude entered simultaneously
- /XXX, /XXXX, /XXXXX - Altitude only.
- XXX/ Speed only

Altitude restrictions suffixes:

- Blank - Cross at altitude predicted by the FMC
- A - Cross at or above altitude

SSG B747-8 SERIES

- B - Cross at or below altitude (normally AT)

Waypoint Speed/Altitude FMC Predictions

Waypoint speed and altitude predictions are displayed in a smaller font. The predictions are calculated by the FMC during entire flight.

Note that the VNAV PTH mode is present in the Descent phase when there is an altitude restriction below the cruise altitude. And also note that AT may be not Idle during descent because the altitude restrictions.

To modify the route, the following can be accomplished:

- Add and delete waypoints
- Changing the waypoint sequence

RTE Legs Page Modifications

Add a Waypoint

A waypoint can be added to the route at any point and at any moment.

First, the waypoint name is entered into the CDU scratchpad.

Then, find the correct line in the flight plan and press the adjacent LSK. The scratchpad waypoint name is inserted into the selected line. The entered waypoint is connected to the waypoint above and below it via a direct route.

Performance predictions to destination on the PROGRESS page are recalculated assuming the route of flight is direct between waypoints.

Deleting Waypoints

Use the RTE or LEGS page to remove waypoints from a route. However, the active waypoint cannot be deleted. The two valid methods to remove a waypoint are:

- Delete the waypoint with the DELETE function key
- Change the waypoint order.

The data in the route that comes before the deleted waypoint is not affected by these changes.

Delete Entry

SSG B747-8 SERIES

Pressing the DEL key arms the delete function and displays a “DELETE” message on the scratchpad.

With “DELETE” in the scratchpad, pushing the LSK next to a waypoint deletes that waypoint.

Waypoint Order in LEGS Page

A waypoint may be manually typed in the scratchpad or copied from its LSK to the scratchpad. To copy a waypoint from the flight plan, press the waypoint’s LSK on one of the LEGS pages.

Then press an LSK in a route and the waypoint will be inserted at that location, thus changing the order, and creating a “ROUTE DISCONTINUITY” as well.

“Direct To” and “Intercept Course To”

If LNAV is not active, activation will occur in the following situations:

- When the aircraft is within 2.5 miles of lateral offset and on an intercept heading to the active track leg, pushing the LNAV switch activates the LNAV or else it will arm the LNAV mode.
- The aircraft turns to intercept the active leg. If the intercept angle is large, the airplane may overshoot the active leg, but will later rejoin the active leg.

Direct To

As explained before, pressing the LSK next to a waypoint will copy that waypoint into the scratchpad. Then, by pressing LK1, the waypoint will be inserted as a destination waypoint, removing all the waypoints between the new destination waypoint and the previous destination waypoint. In this case, there will be no “ROUTE DISCONTINUITY.”

Vertical Navigation (VNAV) Operation

VNAV is the vertical profile navigation during the climb, cruise, descent and approach phases of flight.

Takeoff and Climb

The FMC takeoff phase starts with the selection of the N1 Mode (TO/GA). Preparation for this phase starts in the Preflight phase and includes entry of TAKEOFF REF page data.

SSG B747-8 SERIES

The Takeoff phase automatically changes to the Climb phase when the FMC switches to climb thrust. The Climb phase continues to the T/C point, which is where the Cruise phase starts.

During takeoff and climb, the pages listed below provide functionality as described:

- TAKEOFF REF - To make last minute changes to the thrust derate or V-speeds
- DEPARTURES - Make last minute changes to the departure runway or SID
- LEGS or RTE - Modify the route and monitor route progress
- PROGRESS - Monitor the progress of the flight
- THRUST LIM - Select climb thrust limits

Takeoff Phase

When selecting the departure runway and SID, in the TAKEOFF REF and DEP ARR pages the pilot must also select the EXEC when the light above it is illuminated to confirm the selection. The modified data are entered in the same manner as during preflight.

The FMC calculates and sets the takeoff thrust when the TO/GA mode switch is pressed. During the takeoff roll, the autothrottle commands the thrust and the FMC commands through the FD bars an acceleration to V_2+20 knots. VNAV is engaged at 400 AGL if the autopilot is engaged.

Usually, LNAV activates at 50 feet radio altitude and commands a roll to fly the active route leg if the FD is set to "ON" and the autopilot is engaged.

At 400 AGL the VNAV mode is engaged and it commands a pitch to fly the climb path. During takeoff, the target airspeed is V_2+20 knots.

Acceleration Height

At the preselected acceleration height, VNAV commands an airspeed increase to the flap retraction speed. Then, as the flaps are retracted, the speed is increased up to 250 knots, or the speed transition associated with the origin airport and entered in the CLIMB Page, whichever is greater.

The FMC then changes the thrust reference mode to the selected climb thrust at the thrust reduction point.

VNAV Climb

The VNAV climb profile uses VNAV SPD or VNAV PTH as the default climb speed or the pilot-selected climb speed to remain within all airspeed and altitude restrictions that are part of the selected SID. The autothrottle uses the selected climb thrust limit.

SSG B747-8 SERIES

If the climb speed profile cannot achieve a particular altitude restriction an “UNABLE NEXT ALTITUDE” message appears in the scratchpad.

At acceleration height or the first movement of the flap handle during flap retraction (depending on what condition is selected in the TAKEOFF page), VNAV commands acceleration to 250 knots or the preselected transition speed.

The VNAV commanded speed always is limited by the aircraft configuration. At acceleration height, VNAV commands a speed that is 5 knots below the flap placard speed, based on the sensed flap handle position.

At the climb thrust reduction point, the FMC commands a reduction to the selected climb thrust. Passing 10,000 feet, VNAV commands acceleration to the economy climb speed (in either kts or %Mach), which is maintained until entering the Cruise phase. It should be noted that waypoint speed restrictions take priority.

During the climb, VNAV complies with the LEGS page waypoint altitude and speed restrictions. A temporary level-off (MCP altitude) for a crossing altitude restriction is accomplished at the commanded speed.

When the climb profile causes an anticipated violation of a waypoint altitude constraint, the FMC displays the message “UNABLE NEXT ALTITUDE” in the CDU scratchpad. A different speed profile that provides a steeper climb angle must be manually selected by the pilot at that point.

Climb Constraints

VNAV then enters the VNAV PTH mode to remain within departure or waypoint constraints. The airspeed maintained during this phase can be any of the following:

- Procedure-based speed restriction
- Waypoint speed restriction
- Default VNAV climb speed
- Manually entered climb speed.

If the FMC predicts the aircraft will not reach an altitude constraint, the message “UNABLE NEXT ALTITUDE” will display in the CDU scratchpad. Speed intervention can be made by using the IAS/MACH selector and manually setting a lower airspeed to provide a steeper climb. Alternatively, climb derates can be deleted on the THRUST LIMIT page.

SSG B747-8 SERIES

If the pilot wishes to clear all altitude restrictions below the MCP selected altitude, this can be accomplished on the CLB page and pressing RK6, labeled as “CLB DIR” for “climb direct”. The restrictions can be cleared one by one by pressing the MCP Altitude knob if the altitude dialed into the MCP is above the restriction altitudes.



Top Of Climb (T/C)

The Top of Climb (T/C) is the point at which the Climb phase reaches the cruising altitude. Approaching this point, the FMC changes from the Climb phase to the Cruise phase as well.

MCP Altitude Intervention

Whenever the aircraft levels off at an MCP-selected altitude that is not in the FMC, VNAV ALT engages. For example, if the FMC cruise altitude is FL370 and the altitude cleared by ATC for the aircraft is lower (i.e. FL200), then that would be the altitude set by the pilot in the MCP. The aircraft autopilot will hold that altitude and the FMC target speed. To resume the climb, the input the cruise altitude into the MCP altitude window and press the altitude intervention button (the MCP Altitude knob), and the climb will restart again.

If a level-off is required, set the MCP altitude window to the desired altitude and the autopilot always will level off at that altitude. Then VNAV changes to VNAV ALT. As stated previously, the climb can be continued by setting the altitude window to a higher altitude and pressing the altitude intervention button (VNAV ALT changes to VNAV SPD or PTH, depending on whether any altitude restriction is associated with any waypoint before T/C). This same logic of operation can be accomplished when descending as well.

In the climb, if the MCP altitude window is set to an altitude above the altitude restriction waypoint, each restriction can be deleted individually by pressing the altitude intervention button.

SSG B747-8 SERIES

To delete the altitude restrictions of all the waypoints before the T/C, the cruise level has to be set in the MCP altitude window, then all waypoint altitude climb restrictions can be deleted pressing RK6 labeled “CLB DIR>” on the CLB page.

Cruise

During the cruise, the FMC sets an ECON cruise speed until reaching the top-of-descent (T/D) point. Pilots can enter another speed in the CRZ page, or use the speed intervention button. They can also select the “LRC Speed” in the VNAV “CRZ” page.

The FMC uses maximum range cruise speed if cost index is set to zero. Step climb fuel and ETA predictions assume the airplane climbs at each predicted step climb point as airplane weight decreases. FMC predicted step climb increments are based on the step size shown on the CRZ page. If a step size of zero is entered, the FMC assumes a constant altitude cruise.

Predicted step altitudes are shown on the waypoints in the RTE LEGS page. The distance and ETA to the next step point (predicted) are shown on the CRZ page.

The cruise phase automatically starts at the top of climb. During cruise, the primary FMC pages are:

- RTE LEGS
- PROGRESS
- CRZ

The RTE LEGS pages are used to modify the route. The PROGRESS pages show flight progress data. The CRZ pages show VNAV related data.

Other pages are:

- POS REF - Verifies the FMC position
- RTE DATA - Contains progress data for each waypoint on the RTE LEGS page
- The CLB page automatically changes to CRZ at the top of climb.
- CLB and DES pages automatically change to CRZ when reaching the cruise altitude. The CRZ page automatically changes to DES at the T/D.

Speed Intervention Button

In VNAV mode, pressing the Speed Intervention Button enables the flight crew to change airplane speed with the IAS/MACH selector.

SSG B747-8 SERIES

Only In a VNAV SPD descent after the T/D, VNAV changes the pitch to follow the speed setting during speed intervention.

VNAV changes to approach phase using the following logic:

- Passing the first waypoint of a STAR approach, or
- The landing runway is the destination waypoint and the direct distance to the runway is 25 NM or less.

Descent

The FMC calculates a descent path based on airspeed and altitude restrictions. The top of descent (T/D) is the point at which the Cruise phase changes to the Descent phase. This is displayed on the LEGS PAGE with the label “T/D”. The descent path starts at the T/D and includes waypoint altitude restrictions. It is also displayed in the ND as a small green circle with the label “T/D.”

The descent may be accomplished at the FMC calculated economy kts/Mach (based on Cost Index) or a manually entered kts/Mach value. VNAV will not command an economy target speed greater than (VMO/MMO).

The FMC takes into account the descent path with a deceleration at the speed transition altitude (typically this is 250 kts below 10,000 feet). The autothrottle provides thrust as required to maintain the path and speed.

The Descent phase starts at the T/D point and continues to the end of descent point (usually a STAR restriction or crossing TRANS altitude). Planning for the Descent phase starts during the cruise.

The Approach phase starts at the end of descent point and continues to touchdown or go-around. The only automatic page change in the descent/approach phases is the VNAV selected page change from Cruise to Descent at the T/D.

During descent, LNAV progress is managed using the RTE LEGS and PROGRESS pages, as in the Cruise phase. VNAV descent management can be selected on the DES page.

Reaching the T/D, VNAV commands pitch to maintain the planned descent path and ECON speed. If there is any altitude restriction during descent, the throttles may change from idle, which is their standard setting for the descent.

SSG B747-8 SERIES

If any level-off occurs because the MCP altitude window is set to a lower altitude, VNAV changes to VNAV ALT. To resume the descent, set the MCP altitude window to a lower altitude and press the Altitude Intervention Button.

Early Descents

An early descent in VNAV is any descent initiated before reaching the top of descent point (T/D). VNAV commands the descent at a rate of 1,200 FPM until the predicted idle descent path is intercepted.

Descend Now

Within 50 NM of the T/D, the DES NOW prompt appears at RK6 on the VNAV DES page to initiate an immediate descent. If the RSK is pressed, the original path and T/D are not changed, and VNAV starts a descent (VNAV PTH) and captures the MCP altitude (VNAV ALT) or the idle descent path (VNAV SPD), whichever is encountered first. The MCP altitude selector must be set below the cruise altitude to initiate a descent within 50 NM of the T/D.

The DES NOW prompt is shown on the descent page when the cruise phase is active and the airplane is within 50 NM of the T/D. Pressing the DES NOW key and selecting "execute" will initiate a VNAV ECON descent of approximately 1,250 FPM at ECON speed.

Cruise Level Modification Descent

Setting an altitude below the current altitude in the MCP altitude window and pressing the Altitude Intervention button (more than 50 NM from the T/D) causes the level cruise to be set to the MCP altitude, and the aircraft starts to descend to the new altitude. A VNAV PTH cruise descent is commanded at cruise speed and a rate of approximately 1,250 FPM.

The autothrottles adjust thrust to maintain the target descent rate and the pitch is used to maintain the commanded speed. The same can be done to increase the cruising altitude.

Approach

During approach, roll and pitch modes usually change to LOC and G/S Modes using the navigation radios automatically tuned by the FMC. The FMC continues to calculate and show present position.

The RTE LEGS and PROGRESS pages are used to monitor the aircraft. Another page is the APPROACH REF page, which is used to specify the approach flap setting and set the approach VREF.

The FMC changes to the Approach phase under the following conditions:

SSG B747-8 SERIES

- The aircraft is in the descent phase and below the transition level
- The aircraft is enroute to a direct-to or intercept-to the airport/runway waypoint and the aircraft is within 25 NM of the runway threshold

The FMC prompts the pilots to select the approach speed on the APPROACH PAGE when an ILS signal is present.

When the FMC is in Approach phase, the following features are available:

- The IAS/MACH command speed can be selected on the MCP while VNAV remains in VNAV PTH descent; VNAV then commands the set speed.
- The MCP altitude can be set below the airplane altitude for a level off without an ALT HOLD, VNAV continues to command a descent until level off, changing to VNAV ALT.
- VNAV remains in VNAV PTH and follows the descent path.

VNAV Modifications

During the cruise phase, two types of climbs can be managed by VNAV: cruise level change and step climbs (they are the same but are calculated and annunciated by the FMC).

Cruise climbs are entered by the flight crew. Planned step climbs has been calculated by the FMC. Optimum step climbs are calculated by the FMC. In all cases, the new climb altitude must be entered into the MCP altitude window before VNAV commands the climb.

Cruise Climb

In the Cruise phase, setting an altitude above the current cruise altitude in the MCP altitude window and pressing the altitude intervention button, causes the cruise altitude to be set to the MCP altitude and the airplane to climb to the new cruise altitude. The reference thrust limit is CLB and the mode annunciation is VNAV SPD.

Calculated Step Climb

When a nonzero value is entered into the STEP SIZE line on the PERF INIT or CRZ pages, the FMC calculates optimum points for step climbs as the aircraft burns fuel. The climb altitude is determined by the value in STEP SIZE. Multiple step climbs are possible based on performance and route length. This is computed only if the T/D is more than 200 NM from the T/C; in other words, if the Cruise Phase is more than 200 NM.

Wind

SSG B747-8 SERIES

The FMC uses wind data to improve performance prediction accuracy. Wind data includes altitude and direction/speed. The FMC adjusts ECON climb speed and T/C using the wind speed from XP.

Sometimes, The FMC calculated ECON climb speed might fluctuate if the T/C is close to a waypoint and if a significant wind velocity has been predicted for that waypoint.

Throttle Management

The thrust management function operates the autothrottle in response to flight crew input or to automatic FMC commands. Reference thrust can be selected on the N1 LIM page.

Automatic FMC autothrottle commands are made ONLY while in VNAV mode. FMC thrust management has the following functionality:

- Calculates reference thrust limits for every phase
- Thrust limits are expressed as N1 limits.
- Calculates a reference thrust for the following thrust settings:
 1. TO - Takeoff
 2. D-TO - Assumed temperature takeoff
 3. R-TO 1 - Reduced Takeoff 1
 4. R-TO 2 - Reduced Takeoff 2
 5. CLB - Climb
 6. R-CLB 1 - Reduced Climb 1
 7. R-CLB 2 - Reduced Climb 2
 8. CRZ - Cruise
 9. CON - Continuous
 10. G/A - Go-around

With VNAV mode, the reference thrust limit changes for the phase of flight. Thrust settings can be selected on the THRUST LIM page. The reference thrust limit displays above EICAS N1 indications.

The flight crew can specify the thrust reduction height where the change from takeoff to climb thrust takes place by making an entry on the TAKEOFF REF page. This can be an altitude from 400 to 9,999 feet, an entry of 10 for Flaps 10, or an entry of 20 for Flaps 20.

Reduced Thrust Takeoff

Reduced thrust takeoffs lower EGT and extend engine life. They are used whenever performance limits and noise abatement procedures permit.

Assumed Temperature Thrust Reduction Takeoff

SSG B747-8 SERIES

Entering an assumed temperature in the N1 LIMIT page that is higher than the actual temperature reduces takeoff thrust and also helps to extend engine life.

Derated Thrust Climb

During climb, the R-CLB derates are gradually removed until reaching 15,000 feet. In the cruise, the thrust reference defaults to CRZ. The reference can be manually selected on the THRUST LIM page.

Two fixed climb thrust derates can be selected on the THRUST LIM page. CLB 1 uses a constant 10% derate of maximum climb thrust to 10,000 feet then increases thrust with altitude in linear fashion to 15,000 feet. CLB 2 consists of a 20% derate of maximum climb thrust to 10,000 feet, then increases thrust in linear fashion to maximum climb thrust at 15,000 feet.

For takeoff thrust reductions from 5% to 15%, CLB 1 is selected. CLB 2 is selected for all takeoff thrust reductions greater than 15%. All of which can be selected by the pilots as they require.

On the ground, the pilots may override the automatic climb derate selection after the takeoff selection is complete.

It should be noted that the use of derated climb thrust reduces engine maintenance costs, but slightly increases the total trip fuel.

Fuel Monitoring

The FMC is able to calculate the fuel that is necessary for the route entered. In the PERF INIT PAGE, pilots can press LK1 to automatically compute the ZFW, enter the reserve fuel and then the COST INDEX (0000 to 9999). The FMC will compute the recommended cruise altitude and the fuel necessary for the trip.

NOTE: The required fuel must be provided to the aircraft using the standard XP menu, there is no automatic refueling in the SSG 747-8.

For a longer trip, the desired Step Climb can be selected (from 1,000 to 9,000 ft), and the necessary fuel will be computed as well. Please note that in some cases, the recommended and optimum cruise altitude may not match because the route has restrictions or is insufficiently long to reach the optimum cruise altitude.

The FMC receives fuel data from the aircraft's systems to calculate fuel. Fuel quantity values are predicted or entered on the PERF INIT page as calculated (CALC) or manual (MAN). In the PROGRESS page 2/2 the fuel is presented as a TOTALIZER value and as

SSG B747-8 SERIES

CALCULATED. The totalizer value is measured by the aircraft's systems and the calculated value is determined by the FMC integrating the instantaneous fuel consumption.

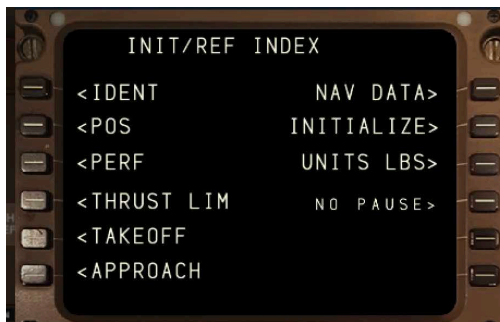
The FMC usually uses the predicted or the calculated fuel value for performance computations.

The scratchpad will display the message "FUEL DISAGREE-PROG 2/2" if the FMC determines that there is a large difference between the totalizer fuel quantity and the calculated one. This could happen as the result of a fuel leak, for example.

The FMC continually estimates the fuel at all the remaining waypoints and the destination airport. The message "INSUFFICIENT FUEL" is displayed in the scratchpad if the estimate is less than half of the fuel reserve value entered on the PERF INIT page.

NOTE: FMC calculated fuel predictions assume a clean configuration and that the aircraft is flown using the ECON speed and altitude profiles. Flight with gear or flaps extended can cause fuel prediction errors.

FMC Preflight



FMC preflight action is required before flight. Completion of the FMC preflight requires data entry in all minimum required data locations.

Preflight Page Sequence

The usual FMC power-up page is the MENU Page and then the identification page. Preflight flow continues in this sequence:

- Identification (IDENT) page

SSG B747-8 SERIES

- Position initialization (POS) page
- Route (RTE) page
- DEPARTURES page (no prompt)
- Performance initialization (PERF INIT) page
- Thrust limit (THRUST LIMIT) page
- Takeoff reference (TAKEOFF) page

Some of these pages are also used during the flight.

Minimum Preflight Sequence

During preflight, a prompt in the RK6 position automatically directs the pilots through the minimum requirements to complete the FMC preflight. Selecting RK6 will show the next page in the flow. If a required entry is missed, a “PRE-FLT” message on the TAKEOFF page informs the flight crew that there is missing data.

For example, a route must be entered and activated. The minimum route data has an origin and destination airports, and a route leg. Performance data requires entry of aircraft weights, fuel reserves, cost index and cruise altitude. Takeoff data requires a takeoff flap setting (10 or 20 degrees).

Supplementary Pages

When a route includes SIDs and STARs, they can be entered into the preflight using the DEP ARR pages.

Route discontinuities need to be removed by the pilots, and the route modifications may be entered in the RTE or LEGS pages. Speed/altitude restrictions are entered only on the LEGS page.

Waypoints, navigation, airport, and runway data is referenced on the REF NAV DATA page.

Fixed takeoff thrust derates can be changed on the N1 LIMIT page.

Preflight Pages

The preflight pages are presented in the sequence used during a typical preflight:

Initialization/Reference Index Page

The initialization/reference index page allows manual selection of FMC pages. It gives access to pages used during preflight and not usually used in flight.

Identification (IDENT)

The IDENT page is used to verify basic airplane data and date of the navigation database.

SSG B747-8 SERIES

Performance (PERF INIT)

The PERF INIT page is used for initialization of data required for VNAV operations and performance predictions.

Thrust Limit (N1 LIMIT)

The N1 LIMIT page is used to select thrust limits and derates.

TAKEOFF REF

The TAKEOFF REF page is used to enter takeoff reference data and confirm the FMC calculated V-speeds or enter new ones.

APPROACH

The APPROACH REF page is used for entry of the approach VREF speed.

NAV DATA

The NAV DATA page is used for FMC flight plans loading.

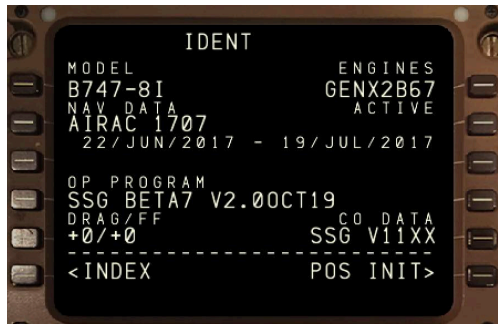
INITIALIZE

Is used to reset all flight data from the beginning.

PAUSE AT T/D

Is used to program a pause in the simulator at 5, 10, or 50 NM before T/D. So a simulator pilot can leave the aircraft unattended on a long haul flight, for example, and then perform the approach and landing.

Identification (IDENT) Page



Most of the data on this page is for flight crew verification. The flight crew verifies FMC data, checks navigation database.

SSG B747-8 SERIES

MODEL (LK1)

Shows the aircraft model.

NAV DATA (LK2)

Shows the navigation database identifier.

INDEX (LK6)

Directs to the INIT/REF INDEX page.

ENGINES (RK1)

Shows the engine model.

ACTIVE

Shows the effective date range for the active navigation database. The active navigation database may be out of date.

Position Initialization (POS INIT) Page



The position initialization page allows airplane present position entry.

Reference Airport (REF AIRPORT) (LK2)

The reference airport entry allows entry of the departure airport. Optional entry. Enter ICAO four letter airport identifiers. This is only for information purposes. The actual departure and arrival airports are entered on the RTE page.

Gate

Not implemented in this version.

Universal Time Coordinate (UTC)

SSG B747-8 SERIES

UTC (GPS) – Displays current time from GPS. (XP Time)

INDEX

Directs to the INIT/REF INDEX page.

GPS Position (GPS POS)

Displays the GPS present position.

SET IRS POS (RK5)

Allows for entry of GPS position to align the IRSs.

ROUTE

Directs to the ROUTE page.

Route 1 Page

NOTE: Route 2 Page is separate and allows for the same entries of a different route. This will show in blue dotted lines on the ND until it is made active.



Allows for a route to be entered by the flight crew or loaded through previously saved route file. All routes have two or more pages. The first route page shows origin and destination data. Subsequent route pages show the route segments between waypoints or fixes or airway segments or SIDs and STARs.

ORIGIN

Entries:

- Must be a valid ICAO identifier in the navigation database.
- Is filled automatically when a route is entered.
- Enables direct selection of departure and arrival procedures.

SSG B747-8 SERIES

- Is required for route activation.
- Entry on the ground modifies route; in flight, is not permitted.

RUNWAY

Enter the applicable runway for the origin airport. No pilot entry is allowed; it is automatically filled when selected on the DEPARTURES page.

Destination (DEST)

Entries:

- Must be a valid ICAO identifier in the navigation database.
- Is made automatically when a route is entered.
- Enables selection of departure and arrival procedures.
- May be changed in flight.

Flight Number (FLT NO)

Pilots can enter a company flight number. Entry is optional for activation of the route and only for information purposes. It is then displayed on the PROGRESS page as a reminder.

ACTIVATE

Press the ACTIVATE key (LK6) to set the route entered as the active route. When the EXECUTE key is pressed, the route becomes the active route and the ACTIVATE prompt is replaced with the next required preflight page prompt.

Activation of a route is required for completion of the preflight. After route activation, the ACTIVATE prompt is replaced by RTE to enter the route details.

CO Route

Allows for the loading of a stored company route.

Route Page (RTE)

The subsequent route pages 2/X through X/X, show the route segments in ATC format. Route segments are defined as direct routings, airways, or procedures with start and end points such as waypoints, fixes, nav aids, airports, or runways. More waypoints for each route segment are shown on the RTE LEGS page.

VIA

The VIA column shows the route segment to the waypoint or termination shown in the TO column. Enter the airway or DIRECT route that describes the route segment between the previous waypoint and the segment termination.

SSG B747-8 SERIES

Enter an airway in the VIA column and boxes are shown in the TO column. Alternatively, pilots can enter a navaid in the TO column and the word "DIRECT" will be placed in the VIA column.

Procedures are only entered through selections on the DEPARTURES and ARRIVALS pages. DIRECT is usually entered as a result of entering a TO waypoint first.

Valid airways must do the following:

- Contain the fix entered in the previous TO waypoint, and
- Contain the Next TO waypoint, or
- Intersect the previous VIA route segment.

Invalid VIA entries will result in the message "INVALID ENTRY" to appear in the scratchpad.

Examples of invalid VIA entries are:

- Airways and routes that do not contain the TO waypoint of the previous line.
- Airways that do not intersect the previous airway at some waypoint.
- Airways or procedures that are not in the navigation database.

The start and end waypoints determine whether the entered airway is valid. The route segment must contain the waypoint entered in the TO position. The TO waypoint of the previous route segment must be the same as the start point of the next route segment.

Entry of a SID or transition automatically enters at the beginning the VIA and TO data for the route segments of the SID. A SID automatically links to the next route segment when the final SID waypoint is part of the route segment. The SID and STAR procedures are entered using the DEP/ARR page, and automatically update the ROUTE pages. When no SID is used, the first line of page 2 must be a DIRECT procedure.

TO

Is used to enter the end point of the route segment specified by the VIA entry.

Entry of a waypoint in the TO column without first entering a VIA airway results in a "DIRECT" route entry in the VIA column.

Data input is mandatory when boxes are shown. Valid waypoint entries for a DIRECT route segment are any valid waypoints or fixes on the airway.

Airways

SSG B747-8 SERIES

Aircraft normally do not fly direct circle routes from the departure airport to their destination. Instead, they use airways that are developed by aviation authorities and depicted on charts. An airway is like a highway in the sky, with several waypoints at which a pilot can switch from one airway to another one.

Obtaining Routes

A newer option is SimBrief, which can output airline style flight plans and generate XP route files:

<http://www.simbrief.com/home/>

Loading a Route

In the RT1 PAGE, enter a saved flight plan file name without any extension. The CO-ROUTE name appears in RK3, and the FMC is prompts the pilot to load the flight plan.

To do so, just press EXEC, and the flight plan will be loaded in to the FMC. If any other key is pressed, the flight plan will not be loaded.

Every time you load a flight plan, the FMC is fully initialized, so you do not have to initialize it. If the file name is not valid, then an "INVALID FILE NAME" message will appear in the scratchpad.

If the file content is not valid (e.g. if it contains invalid airways or fixes) then a "NOT FOUND" message will appear in the scratchpad, and the loading will be stopped.

Then the RTE 2 PAGE can be viewed to check the route loaded.

Save Route

To save a Route, the route must have been created previously in the FMC.

Go to INDEX PAGE and when pressing the RK5, the FMC shows in the scratchpad a proposed file name that you can accept or partially edit.

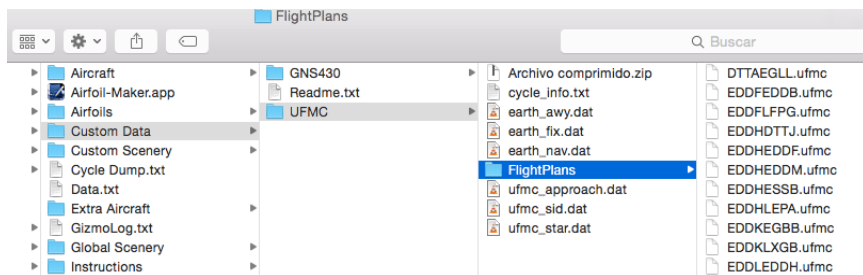
The proposed file name is always the name composed by the departure airport in ICAO format and the destination airport ICAO in ICAO format.

Then pressing the EXEC button, the route will be stored using the selected name. If the file already exists, it will be overwritten, without prompting.

SSG B747-8 SERIES

NOTE: This version of the SSG 747-8 has an important change for loading and saving routes.

Routes are stored in an X-plane/Custom Data/UFMC/FlightPlans/ folder that the user needs to create (it is not created automatically upon installation). However, this only needs to be done once per XP installation. See example below.



Do not change this folder name or location. The FMC saves the route in .ufmc format in this folder the, and loads them from this folder using the RTE 1 PAGE and RK3.

Now, for those who wish to load FMS flight plans made in the default XP .fms format, such as those created by flight planning tools or used in other XP aircraft with the default GPS, for example, these must be located in the XP folder for .fms files, under

Output/FMS plans

To load this type of FMS plan file, simply select the desired plan and load it using the FMC's INDEX PAGE -> NAV DATA key.

Resolving Route Discontinuities

When entering a SID or STAR, there often can be a "route discontinuity." In these cases the pilots have to inform the FMC what is the next waypoint after the SID, or what is the route waypoint that will join with the selected STAR.

One important point is to ignore T/C and T/D because these are fictitious waypoints and they do not have to be selected or deleted.

The solution is simple, because it involves pressing the LSK for the waypoint that will join the SID or STAR to the existing route, which places it in the scratchpad. Then this waypoint can be connected with the SID or STAR by inserting it in the appropriate part of the route using the associated LSK and thus resolve the route discontinuity.

SSG B747-8 SERIES

At this point the EXEC light illuminates, by pressing “EXEC”, the new route will be active.

Deleting a waypoint on a route can also cause a route discontinuity. The pilot has to inform the FMC what is going to be the next waypoint since one was deleted. To do so, the pilot can select any waypoint after the ROUTE DISCONTINUITY. Again, this has to be confirmed by pressing the EXEC key.

Route Lateral Offset

Route offsets are entered in the RTE page. These can be useful to avoid traffic or weather while remaining on the route, for example. On RK6, there are three dashes (“---“ but only after takeoff). The format is XXX, consisting of L or R and a number between 1 and 99, so for example R10 means an offset of 10 NM to the right.

When a value is entered, the EXEC key illuminates. As soon as the offset is executed, the aircraft will turn towards the offset with a 45° intercept heading and capture it. During the offset, “OFST” will be displayed on the right part of the FMC, just below the MSG field. This (and the number in RK6 in the RTE page) is the only indicator that an offset is active. In the LEGS and Progress page, all individual waypoints are still shown as usual.

To delete an offset, you can either enter “0”, or you delete RK6. In any case, RK6 will return to “---“

The offset is also automatically deleted when the aircraft reaches a STAR, an approach or an approach transition, a hold, or if there’s a turn on the route of more than 135°. In these cases an “END OF OFFSET” message will appear in the scratchpad.

When an offset ends, the aircraft will take a 45° course change towards the original route and intercept it. In an automatic offset end, the aircraft will turn just in time to intercept the last waypoint of the offset at a 45° angle.

Departure/Arrival Index Page

The departure and arrival index page is used to select the departure or arrival page for the origin and destination airports for each route. The index also allows reference to departure or arrival data for any other airport in the navigation database.

Departure and arrival prompts are available for the origin airport, but the RK1 key does not work at this time. Destination airports only have arrival prompts.

Departure (DEP)

LK1 Shows the departure page for route origin airport.

SSG B747-8 SERIES

Arrival (ARR)

RK1- Is inoperative.

Arrival (ARR) - Route 1 Destination

RK2- Shows the arrival page for the destination airport on the route.

Departure Pages

The departures pages are used to select the departure runway and SID procedure for the origin airport of a route. The departures page for the route is shown when the “DEP ARR” function key is pressed and the departure airport has been entered.

Standard Instrument Departures (SIDs)

This shows SIDs for the airport and runway selection. Without the selection of a runway on the DEP/ARR page, the initial display contains only the airport runways. As selections are made, only the proper SIDs appear.

Runways

This shows a list of runways for the selected airport. The runway selected is shown as “<SEL>”.

Pressing the proper left LSK selects the runway for use on the route. All other runways are no longer displayed.

Then, the SIDs associated with the selected runway appear, all others no longer are displayed. Subsequent change of a runway is still possible (either due to ATC clearance or wind shift) and it will delete the departure procedures selected previously.

Once the “EXEC” button is pressed, the waypoints associated with the SID are inserted in the FMC in sequence. Then the route can be displayed in the ND and in the FMC.

If one of the procedure waypoints has a restriction, this appears in the FMC as a speed/altitude.

When selecting SIDs, it contiguous repeated waypoints could occur, usually where the final waypoint of a SID is the same as a waypoint already in the route. The FMC will delete the correct waypoint when the LEVEL CRUISE is entered in the FMC in the INIT/REF page.

Just like with the runway selection, a different SID or STAR can be selected. In this case, the FMC will erase the waypoints associated with that procedure and insert the new ones.

SSG B747-8 SERIES

Navigation Radio Page



VOR and ILS navigation radio tuning is normally automatic. ADF radios are manually tuned. The NAV RADIO page shows the VOR, ILS, and ADF radio status and allows manual control of these radios as well. Entering data on this page manually tunes the selected navigation radio. Manually selected VOR courses can also be entered.

Radio navigation is controlled automatically by the FMC through this page. In RK1 the VOR frequency selected by the FMC is shown, as well as the name of the VOR and the radial in which the aircraft will fly with respect to the VOR. These automatic selections use the FMC to guarantee the lateral path.

Usually the navaid selection is in the automatic mode, so the FMC will tune the left and right VORs. If pilots wish to select a specific VOR, they can simply enter the frequency in the following format (with a decimal point) in the scratchpad:

XXX.XX

Then LK1 or RK1 can be selected, as appropriate. The mode in the left or right side will then change to “Manual.” The desired course can be selected using LK2 or RK2 to track the VOR. If the VOR mode is selected on the ND, the course line and the VOR that was manually selected will appear. To return the VOR tuning to automatic again, pilots can simply press the DELETE key and then RK2 or LK2.

An ILS frequency also can be entered manually using RK1 or LK1, along with the proper course in the following format:

XXX.XX/XXX

SSG B747-8 SERIES

Aircraft gross weight is required by the FMC to make certain performance calculations. The entry can be made by the flight crew or automatically calculated by the FMC.

Pressing LK1 automatically shows the calculated weight when confirming the weight proposed by the FMC, as well as showing the aircraft's current ZFW.

Entry of a value after takeoff speeds are selected removes the speeds and shows a "TAKEOFF SPEEDS DELETED" message in the scratchpad.

Fuel

Fuel on board is automatically shown when the fuel totalizer calculations are valid. The source of the automatically generated data is included in the line:

- CALC is shown when the source is FMC calculations and manual entry is possible.

The unit of quantity is shown to the right of the numerical value.

NOTE: The SSG 747-8 uses pounds as a unit of measure.

Only manual entries can be deleted. If, after a manual entry, it is required for the FMC to calculate the fuel quantity, press the DEL key, and then press LK2.

The FMC only calculates the fuel and does not fill the tanks. These can be filled using the standard XP menu.

The FMC is able to calculate the required fuel for the route entered by the pilots. On the PERF INIT page, pilots can press LK1 to automatically compute the ZFW, enter the reserve fuel and then the COST INDEX (from 0000 to 9999). The FMC will compute the recommended cruising altitude and the necessary fuel for the trip. For a long flight, the necessary sep climbs for the flight can be selected (from 1,000 to 9,000), and the fuel necessary will be computed as well.

Zero Fuel Weight (ZFW)

Aircraft ZFW is also required by the FMC. In real world operations, the ZFW is entered from the aircraft's dispatch papers and the FMC calculates the airplane gross weight.

In the SSG 747-8, pressing LK1 will confirm the gross weight value proposed by the FMC, which will automatically calculate the ZFW.

Calculated zero fuel weight is automatically shown if airplane gross weight is entered first and the fuel on board quantity is valid.

SSG B747-8 SERIES

Entry of a value after takeoff speeds are selected removes the speeds and shows a “TAKEOFF SPEEDS DELETED” message in the scratchpad.

Reserves

This is where pilots can enter fuel reserves for the route. An entry is required to complete the preflight. Valid entries are in the following format:

XX

The total fuel figure includes the reserve fuel. Please note that the FMC does not take into account the minimum fuel for contingencies or alternate airports. So these can be entered in the reserve fuel. The FMC only calculates the minimum fuel needed to reach the arrival airport, therefore pilots must add a reserve of fuel to ensure a safe arrival at the destination (in real world operations this is also a legal requirement for pilots).

Index

This is the INIT/REF INDEX page.

Cruise Altitude (CRZ ALT)

The proposed cruising altitude is a required entry. This cruise altitude is then automatically displayed on the CLB and CRZ pages. When the cruise altitude is entered, all of the waypoint speed and altitude predictions are calculated based on that value.

Cost Index

The cost index is used to calculate ECON climb, cruise and descent speeds. Larger values increase ECON speeds. Entering a value of zero results in maximum range airspeed and minimum trip fuel. A Cost Index value must be entered by the flight crew. Valid entries are 0 to 9,999.

Cruise (CRZ) Center of Gravity (CG)

This entry is optional on the real aircraft but is not implemented in the SSG 747-8 Anniversary Edition because the FMC takes the takeoff CG information and computes cruise CG based on fuel burn throughout the flight.

Step Size

This shows the climb altitude increment used for planning the optimum climb profile.

Valid manual entries are from 0 to 9,000 in 1,000 foot increments. The step value can be entered manually in RK6.

SSG B747-8 SERIES

During the Climb phase, step size entries are inhibited. In-flight step size changes are made on the CRZ page in the Ground or Cruise phase.

For a non-zero entry, performance predictions are based on step climbs at optimum points. For a zero entry, performance predictions are based on a constant CRZ ALT.

Thrust Limit

This displays the THRUST LIMIT page.

Thrust Limit Page



The thrust (N1) limit page allows selection and display of reference thrust in preparation for takeoff. Derated takeoff thrust by use of the assumed temperature method is also shown on this page.

More page data displays are:

<SEL> - Identifies the selected takeoff or climb thrust reference mode.

<ARM> - Identifies the armed climb thrust reference mode.

The <ARM> prompt changes to <SEL> when the armed climb mode becomes active.

Assumed Temperature (SEL), Initially blank. Displays entered assumed temperature up to the maximum thrust reduction limit.

Entry of an assigned temperature warmer than OAT reduces takeoff thrust and displays "D" as part of the thrust mode.

SSG B747-8 SERIES

Valid entries are -99 to 99 degrees Celsius.

OAT displays outside air temperature in degrees C.

This maximum %N1 depends in great part on the external temperature and also on the airport elevation. For example, if the aircraft is found parked in the shade compared to the runway giving a major temperature difference, or during the time the FMC TAT is programmed until the aircraft is entering the runway, this temperature can change significantly.

Using LK1, the pilots can enter a temperature manually (positive or negative, in degrees Celsius) and it will be this temperature that is taken into account for the calculation of the maximum % N1, instead of the OAT.

Takeoff (TO)

Selects the full-rated takeoff thrust limit.

Index

Shows the INIT/REF INDEX page.

Climb (CLB)

Selects the full rated (CLB) climb thrust limit. Pressing a climb line select key overrides an automatic selection.

Climb 1 (CLB 1)

Selects a percentage derate for climb thrust limit. The default thrust derate for CLB 1 is not modifiable by the flight crew.

Climb 2 (CLB 2)

Selects a percentage derate for climb thrust limit. The default thrust derate for CLB 2 is not modifiable by the flight crew.

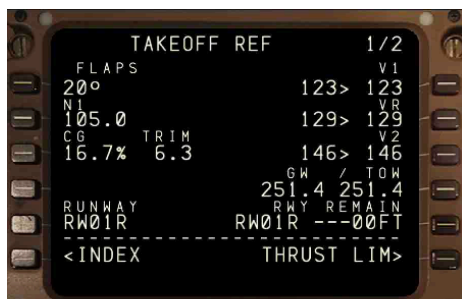
Manual selection of a climb thrust rating overrides the automatic selection.

RK1 shows the FMC calculated % N1 maximum for the takeoff, taking into account the airport pressure altitude, the airport OAT, the aircraft gross weight and wind adjustments.

This value can be decreased in LK2 to LK4 for takeoff and in RK2 to RK4 the % of N1 during the Climb phase. These reductions are at the discretion of the flight crew for a given aircraft configuration and lighter weights.

SSG B747-8 SERIES

Takeoff Reference (REF) Page



The takeoff reference page allows the flight crew to manage takeoff performance. Takeoff flap setting and V-speeds can be entered and verified. Preflight completion status is annunciated until complete.

Takeoff reference page entries mark the completion of the normal preflight. The takeoff flap setting must be entered and V-speeds should be confirmed (or entered manually) before the phase can be considered complete.

Flaps

This displays the takeoff flap setting. Valid entries are Flaps 10 and Flaps 20. Entry of an invalid flap position for takeoff, results in an “INVALID FLAP POSITION” message appearing in the scratchpad.

The flap position is required for takeoff V-speed calculations.

Entry of a value after takeoff speeds are selected removes the speeds and results in a “TAKEOFF SPEEDS DELETED” message appearing in the scratchpad. It is necessary to press the NEXT PAGE key to go to PAGE 2/2.

The height of acceleration in feet above the ground is also shown. When reaching this height, the FMC decreases the pitch to accelerate the aircraft for flap retraction. Only the position/degrees of flaps or flaps/altitude can be entered.

Acceleration Height (ACCEL HT)

Displays acceleration height for flap retraction. Default value is 1,500 feet. Entry is optional. Valid entries are altitudes from 400 to 9,999 feet Above Ground Level (AGL).

Can be entered as AAAA in the scratchpad and then pressing LK1

SSG B747-8 SERIES

Thrust

Thrust derate calculations. Shows takeoff thrust selected on THRUST LIM page.

Index

This shows the INIT/REF INDEX page.

V-Speeds (V_1 , V_R , V_2)

Displays dashes when:

- Required information not entered
- Performance calculations are inhibited

Flight crew entry speeds replace calculated speeds. Calculated speeds display in a smaller font.

Pressing RK1, RK2, and RK3 will confirm the V-speeds and REF is no longer displayed. Manual entry of V-speeds is permitted and V-speeds entered by the crew replace speeds calculated by the FMC.

If the performance data changes:

- FMC replaces existing speeds with FMC calculated speeds in a smaller font
- “REF” is displayed again

Thrust Lim

Shows the THRUST N1 LIMIT page.

Preflight Status

If the required preflight entries are not complete, the message “PRE-FLT” is displayed on the right side of the dashed line. Preflight pages requiring entries display below the dashed line as prompts.

Runway Wind (RWY WIND)

This shows the calculated headwind/tailwind and crosswind components for the takeoff runway and surface wind.

Speed is shown in knots and:

- “H” for headwind
- “T” for tailwind

SSG B747-8 SERIES

e.g. H14 = a headwind of 14 knots.

Climb Thrust Rating And Thrust Reduction (THR REDUCTION)

This shows the climb thrust rating selected on the THRUST LIM page and the altitude for reduction from takeoff thrust to climb thrust.

The thrust reduction entry is optional for preflight completion.

- It may be an altitude or a flap setting.
- Range for altitudes is valid from 400 to 9,999 feet above the origin airport elevation.
- Values are from the FMC's aircraft performance database

At LK3 the height of thrust reduction is shown in feet AGL and it is the height where the thrust mode changes from TO to CLB (or CLB X, if selected). It is practical for this height is to be lower than the acceleration height for noise reduction purposes.

Pressing LK5 indicates if the runway is dry or wet, therefore modifying the V_1 speed.

Then, once all the data are entered, the FMC calculates the V_1 , V_R and V_2 speeds. If these speeds do not appear, then some of the required data has not been entered. For example, a runway may not have been selected in the DEP/APP page or the flap setting is missing.

Pressing RK1, RK2 and RK3 will accept the speeds proposed by the FMC.

When these are selected, the "REF" disappears above the V speeds and so does the preflight warning.

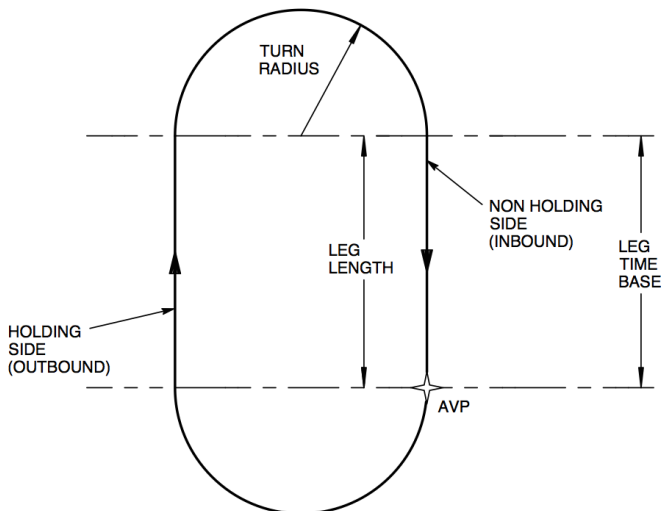
Between RK5 and RK6 the word "--Preflt" may appear. This means that the Preflight is NOT yet complete. When this warning is not visible, then all preflight data has been entered.

If the pilot wishes to modify the reference speeds, the desired speed can simply be entered into the SCRATCHPAD and then the LSK that corresponds to the V-speed the pilot wishes to edit can be pressed. During the takeoff, the different takeoff V-speeds are called out automatically.

SSG B747-8 SERIES

Holding

The following diagram shows the geometry of the holding pattern along with the designations of the various hold segments to clarify the description of how holds are programmed into the FMC.



Holding Pattern Geometry

Route Hold Page With no Holding Fix in Route

When the pilot presses the FMC's HOLD button, the LEGS page is displayed with boxes at LK6. When a waypoint is selected with an LSK, it is copied into the scratchpad and the pilot must select it with LK6. The ROUTE X HOLD page with holding fix in route then opens.

The hold legs page is a fully functional LEGS page where the pilot can modify the route or perform a "direct to" leg.

When PPOS is pressed (RK6), the page is directly changed to the ROUTE X HOLD page after it is selected and the EXEC button is pressed.

SSG B747-8 SERIES

This creates a holding procedure with right turns at the current position using the current heading inbound. This is basically the same page as the ROUTE X HOLD page with holding fix in route, except that the waypoint is in the lat/lon format. For example:

N47W120

Route Hold Page with Holding Fix in Route

Page Title: When the hold is not yet active: RTE HOLD. When the hold is active: ACT RTE HOLD

LK1: Shows the waypoint where the holding is performed.

LK2: Normally, dashes are shown here.

LK3: This shows the inbound course of the holding and the direction of the turn. By default, it shows the flight plan course and a right turn. The valid entry is XXX/X (course/turn direction). “022/R” for example.

LK4: This displays the time between the turns. By default it is 1.0 minutes below 14,000 feet and 1.5 minutes above 14,000 feet. The valid entry is XXX.X.

Normally, the distance between the turns is defined in minutes. However, the FMC uses the entered airspeed and the wind when the aircraft enters the hold to calculate the distance. This means that if the wind changes during the hold, the time might be longer, but the distance remains the same.

LK5: similar to LK4, but instead of time a fixed distance in NM is used between the turns. Either LK4 is active (and LK5 displays dashes) or LK5 is active (and LK4 displays dashes). By default this displays shows dashes. The valid entry is XX.X or X.X.

RK1: This shows the speed and the altitude where the hold is being performed. By default, it shows the best speed (RK5) and altitude from the LEGS page.

If the pilot wishes to change these values, this can be done on the LEGS PAGE or set the MCP ALTITUDE selector to perform an ALT HOLD. About 3 NM before entering the hold, the aircraft’s speed is reduced to the target hold speed.

RK2: This field shows when the aircraft passes the fix the next time. The time is shown in Zulu (XXXXZ). This is calculated before the aircraft enters the hold and also while it is in the hold.

SSG B747-8 SERIES

The time in RK2 is used by the FMC for fuel and ETA predictions. It assumes the aircraft will depart the FIX at that time.

RK3: By default, dashes are displayed here.

RK4: This field shows how much time is available before the reserve fuel to the destination will start being consumed.

It shows X+YY where “X” is hours, “Y” is minutes available fuel for holding using the following formula:

$$=(\text{Current Fuel})-(\text{Trip Fuel (from holding fix to destination)})-(\text{Reserve Fuel})$$

RK5: This displays the best speed for the current aircraft weight and altitude.

RK6: This normally displays “EXIT HOLD> “after the aircraft has entered the hold. If it is pressed, this signals to the FMC that the pilots wish to leave the hold. This action has to be confirmed with the “EXEC” button. When this is accomplished, then “EXIT ARMED” is displayed. The FMC will immediately turn to the inbound leg towards the fix and then leave the hold and will maintain the holding speed and altitude. After passing the fix, it accelerates to cruise speed again. If still on the HOLD page, the page changes to the ACT RTE LEGS page 1/XX.

A hold can be deleted on the LEGS page the same as any other fix. If this is done, then the aircraft will simply fly to the next waypoint.

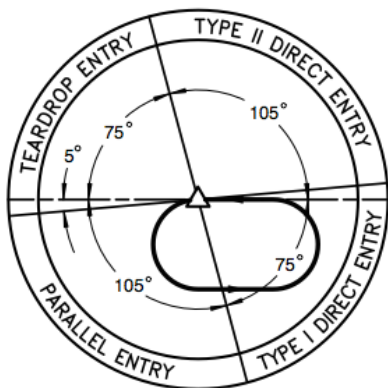
You can delete also the HOLD pressing the DEL key and then the LK1 HOLD PAGE Holding in SID/STARs are actually saved in the database, you don't need to enter the course etc. here. Holding are displayed in the LEGS page in the following format:

(. . .)	
249°	
CIVET	243/14000
HOLD AT	
CIVET	243/14000
250°	12NM
BREMR	243/12000A
(. . .)	

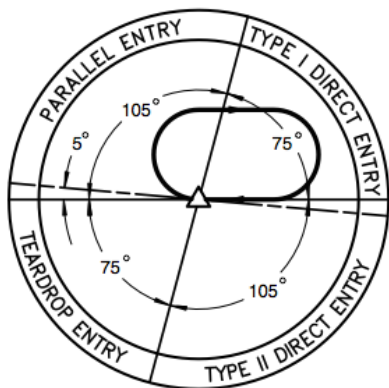
In this example the first waypoint is CIVET, then the holding waypoint is shown (also CIVET).

The HOLD entry procedures (direct, parallel and teardrop) are computed automatically by the FMC, as represented in the illustrations below.

SSG B747-8 SERIES

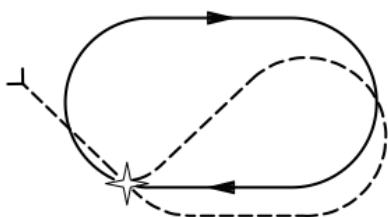


LEFT TURN HOLD ENTRY REGIONS

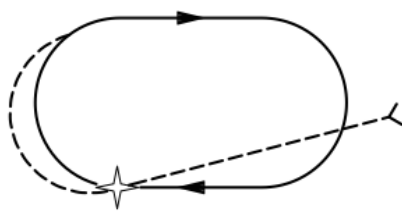


RIGHT TURN HOLD ENTRY REGIONS

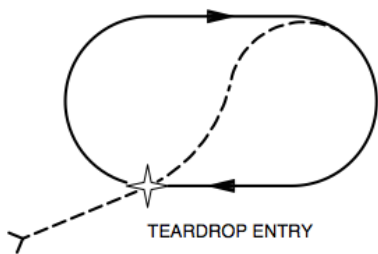
RIGHT TURN HOLD ENTRY CURVES



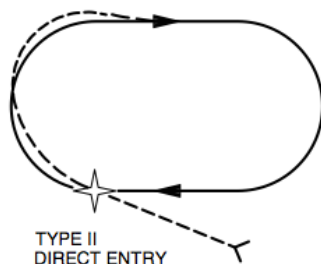
PARALLEL ENTRY



TYPE I DIRECT ENTRY



TEARDROP ENTRY



TYPE II DIRECT ENTRY

Holding Pattern Entry Curves

SSG B747-8 SERIES

Climb Page

The climb (CLB) page is used to evaluate, monitor, and modify the aircraft's climb path. The data on the climb page comes from preflight entries made on the Route and Performance pages. FMC climb can be at either economy or fixed speeds.



Cruising Altitude (CRZ ALT)

This shows cruising altitude entered on the PERF INIT page. Altitude displays in feet or FL depending on the transition altitude. No entry is permitted on this line.

The altitude can be changed by two methods:

- A new cruising altitude can be manually entered at any time in the PERF INIT page.
- A new cruising altitude can be entered from the MCP, if the T/C is reached. This is accomplished by setting the cruise altitude in the altitude window (above or below current cruise level) and then pressing the altitude intervention button. The cruise altitude then changes without any modification or execution.

Economy Speed (ECON SPD), Selected Speed (SEL SPD)

ECON SPD

- Speed is based on cost index in kts or Mach
- Used by the FMC at altitudes above all waypoint, speed restrictions, and speed transition altitudes.

SSG B747-8 SERIES

SEL SPD

- Shown when flight crew enters speed through this page
- Valid entries are kts or Mach.

Speed Transition (SPD TRANS)

The speed transition line shows the transition speed/altitude from one of these sources:

- The navigation database value for the origin airport
- A default speed of 250 knots and 10,000 feet (e.g. 250/10000).

Speed Restriction (SPD REST)

Speed restrictions not associated with specific waypoints are manually entered on this line.

Economy (ECON)

Changes climb speed to ECON if previously modified. Must be executed by the pilots to activate it.

A prompt is shown on the LK5 line when the climb mode is not ECON.

Page Title

The page title shows the type of climb, as follows:

- ECON - Speed based on a cost index
- XXXKT - Fixed climb speed profile in knots

A flight crew entered selected speed (SEL SPD).

ECON

This can be pressed to return to the ECON speed profile

ERASE

This can be pressed to delete the selected speed

Waypoint Restriction (at XXXXX)

This shows the next waypoint with altitude restriction before the T/C.

When there is a speed restriction waypoint, the FMC commands the speed restriction.

These restrictions are entered on the RTE LEGS page. It is blank if no constraint exists.

Transition Altitude (TRANS ALT)

SSG B747-8 SERIES

This shows the transition altitude for the origin airport contained in the navigation database. The FMC uses a default of 18,000 feet if the transition altitude is not available.

Valid entries are XXX, XXXX, XXXXX, or FLXXX.

FMC altitude data changes from altitudes in feet to flight levels (FLs) above the transition altitude.

Maximum Angle (MAX ANGLE)

This is the maximum angle of climb speed. Pilot entries are not permitted.

Climb Direct (CLB DIR)

This displays when climb altitude constraints exist below the MCP ALTITUDE.

When pressed, it deletes all waypoint altitude restrictions below the MCP altitude or FMC cruise altitude, whichever is lower. However, FMC cruise altitude is not affected.

Thrust Limit Page

Thrust limits are selected on the thrust limit page. After the aircraft is airborne, this display replaces the takeoff thrust limits with applicable thrust limits for the climb. The selected limits are shown here and on the EICAS.

Fixed thrust derates can be selected for the climb. Go-around, continuous and cruise thrust limits are available as well.

Go-Around (GA)

Selects the go-around thrust limit mode.

Continuous (CON)

Selects the maximum continuous thrust limit. Selection is usually used for critical situations, such as an engine out condition.

Cruise (CRZ)

Selects the cruise thrust limit mode.

Index

Shows INIT/REF INDEX page.

Cruise Page

SSG B747-8 SERIES

The cruise page is used to monitor cruise altitude and change the aircraft's speed.

Speed changes can be manually or automatically selected with the switching to other VNAV modes. Cruise climbs, cruise descents, and step climbs can be accomplished from the cruise page.

When using VNAV in economy mode, page data is based on operating at ECON SPD. Economy cruise speed is based on the cost index. When the flight crew enters a selected speed, the page data changes as well.

Page Title

The page title shows the active or modified type of cruise. Usually, the title contains ECON for the economy cruise mode. Fixed speed modifies the title.

Cruising Altitude (CRZ ALT)

This displays the cruising altitude entered on PERF INIT page.

Changing the MCP altitude and pushing the altitude intervention button selects the MCP altitude as the active cruise altitude, without creating a modification.

The cruising altitude may be changed in the PERF INIT page as well.

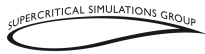
The FMC does not permit the Cruising Altitude to be modified on this page.

Economy Speed (ECON SPD), Selected Speed (SEL SPD)

“MOD” is shown in the page title until the particular modification is erased or executed.

ECON

Selecting this selects the VNAV ECON mode. It is shown when an RTA waypoint is not in the flight plan and VNAV is not in the Economy mode.



SSG B747-8 SERIES

Optimum Altitude and Maximum Altitude (OPT, MAX)

SSG B747-8 SERIES

OPT

Displays the optimum altitude based on aircraft gross weight, Cost Index, and speed displayed on the speed line.

MAX

Displays the maximum cruise altitude based on current gross weight and selected speed option. If the pilots fly the aircraft above the maximum altitude they may experience a stall.

RECOMMENDED

This is the cruising altitude recommended by the FMC depending on the route distance and several other factors.

To Step Climb

When the STEP SIZE is not zero and the trip length and performance is adequate, step climbs are calculated and this line will show the ETA and DTG to the point where the step climb starts in the CRZ page.

When the aircraft passes the step climb point and has not started to climb, the ETA and DTG are replaced with the word "NOW".

Step To

When using an incremental altitude step size, the STEP TO altitude is the next higher altitude above OPT calculated by adding the STEP SIZE increment to the FMC CRZ ALT.

When entering a cruising altitude above maximum altitude, the message "MAX ALT FLXXX" is displayed in the scratchpad.

Entry of a new cruising altitude deletes all waypoint altitude constraints at or above the new cruising altitude.

SSG B747-8 SERIES



The STEP TO altitude from the RTE LEGS page is a calculated step climb altitude based on the step size and it is blank when there is no active flight plan, or within 200 NM of the T/D point.

The FMC evaluates altitudes up to 9,000 feet below the current CRZ ALT and up to less than MAX altitude. Recommended altitudes are selected consistent with the step climb schedule and specified step size.

If a step size of zero has been selected, no recommended cruise level is performed.

Step Size

This shows the default step climb size of 0000.

Valid entries are altitudes from 0 to 9,000 in 1,000-foot increments by entering at RK5 in PERF INIT PAGE.

It is used for calculation of the optimum step point and step climb predictions.

Waypoint Data Page

The route data page shows progress data for each waypoint on the ACT RTE LEGS page. This page is available only for the active route.

The ETA and calculated fuel remaining at the waypoint are shown for each waypoint. Manual entry is not possible on this page.

ETA

Shows ETA for waypoint.

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Waypoint (WPT)

Shows identifier for the waypoint.

Legs

Goes to RTE LEGS page.

Fuel

Shows the FMC calculated fuel remaining at the waypoint.

NOTE: ETA and estimated fuel calculations assume a direct flight across the route.

Progress Pages

Progress Page 1/2

The progress page shows general flight progress data.

Page one of the progress pages shows general data about the following:

- Waypoints (last, active and next)
- Destination data
- FMC speed
- Fuel remaining at waypoint

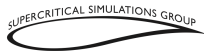


Last

Last waypoint passed. Cannot be modified by the pilots.

To

Active waypoint on active leg. Cannot be modified by the pilots.



SSG B747-8 SERIES

Next

Waypoint after TO waypoint, which also cannot be modified by the pilots.

Destination (DEST)

Destination airport

Selected Speed (SEL SPD)

The speed line shows the FMC active commanded speed.

The active speed mode is the same as that shown on the performance page, unless changed by the MCP or a limit.

Fuel

This is the estimated fuel quantity remaining at a waypoint or the destination.

Progress Page 2/2

Progress page two contains the following data:

- Wind data
- Track error data.
- Fuel data
- Static Air Temperature (SAT)
- True airspeed
- Calculated fuel

Headwind (H/WIND), Crosswind (X/WIND)

This shows the headwind component (H/WIND) and crosswind component (X/WIND).

Wind component data is relative to the aircraft. It displays current wind direction and speed referenced to true north.

Left (L) or right (R) crosswind component is relative to the aircraft's heading.

Cross-track Error (XTK ERROR)

Cross-track (XTK) error is displayed in NM left or right of the active route.

Vertical Track Error (VTK ERROR)

Vertical path (VTK) error above (+) or below (-) vertical path.

TAS

SSG B747-8 SERIES

Aircraft true airspeed.

Left Fuel Used

Instantaneous fuel figure consumed by left engines sensed by fuel flow meters. (lbs/sec)

Fuel Quantity Totalizer (FUEL QTY TOTALIZER)

Fuel quantity calculated by the fuel quantity system totalizer. The fuel remaining line shows two independent fuel remaining values, "TOTALIZER" and "CALCULATED." They can be compared to validate FMC calculations.

Static Air Temperature (SAT)

Outside static air temperature.

Right Fuel Used

Fuel consumed by right engines as sensed by fuel flow meters. (lbs/sec)

Total Fuel Used (TOT FUEL USED)

Sum of the LEFT and RIGHT fuel consumed values.

Fuel Quantity Calculated (FUEL QTY CALCULATED)

The fuel remaining line shows two independent fuel remaining values, "TOTALIZER" and "CALCULATED". They can be compared to validate FMC calculations.

Required Time of Arrival (RTA)

Progress Page 3/3

This page is used, when a flight crew needs to arrive at a specific time, which is useful in delay situations.

LK1:

Normally boxes are shown here until the pilots enter something. This is the reference fix. The fix entered here must be in the current route.

RK1:

This is normally blank. After having entered LK1, boxes are displayed here. The pilots can enter the ZULU time here for when they want to arrive at LK1. They can enter anything between 0000 and 2359. They can also enter a suffix here. For example, 1648A means the pilots wish to arrive at exactly 16:48 or after (17:00 would also be acceptable) 0823B means the pilots wish to arrive at exactly 08:23 or earlier (08:00 would also be acceptable)

SSG B747-8 SERIES

LK2:

Here is the computed speed displayed, at which one would need to arrive at the RK1 entered time. When no time is entered, LK2 remains blank.

LK5:

Here pilots can enter a maximum speed that they do not wish to exceed. By default it is M0.89. Pilots can enter any value between M0.5 and M0.99. When it is deleted, then the line reverts back to M0.89.

RK2:

After the PERF INIT page is completed and after the pilots enter something into LK1, the altitude and the ETA is displayed here for reference.

When this page is executed, then the CRZ page does not display ECON CRZ anymore, but RTA CRZ instead. The RTA PROGRESS label on the top will change to ACT RTA PROGRESS when executed.

This only takes effect in the Cruise phase and on long sectors.

Descent Page

The descent page is used to monitor and revise the descent path. Descent speed modes are “economy” (ECON) and “fixed speed” (SEL). The default VNAV descent mode is ECON. A fixed speed descent is flown when a speed is entered on the DES page. The descent page is blank with “DES” as the title until a speed is entered.

This page title includes the VNAV speed mode. The ECON mode controls descent speed at the economy speed until reaching a speed restriction.

The fixed speed mode controls descent speed at the fixed speed until a speed restriction is reached.

Page Title

The page title displays the active (ACT) or modified (MOD) descent. Usually, the title displays “ECON” for an economy descent. Fixed speed descents modify the title. The page title shows the type of descent:

- ECON - Speed based on a cost index
- XXX - Fixed descent speed profile

SSG B747-8 SERIES

Economy Speed (ECON SPD), Selected Speed (SEL SPD)
Both Knots and Mach values are shown.

ECON SPD

Economy speed based on cost index

MOD SPD

Displays when the flight crew enters a speed.

Speed Transition (SPD TRANS)

The transition speed is the destination airport limiting speed from the navigation database. When no airport limit speed exists, the default speed of 250 kts is shown.

The transition altitude is the point at which the transition speed is active for the destination airport. When no altitude exists in the navigation database, the default of 10,000 feet is shown.

Speed Restriction (SPD RESTR)

Speed restrictions at altitudes higher than the End of Descent (E/D) altitude, and not associated with specific waypoints, are manually entered on this line.

A valid entry is a speed and altitude (e.g. 240/8000). This entry may be deleted.

This shows the FPA, V/B and V/S, the vertical speed commanded by the FMC to meet the restrictions and also shows if there is any deviation from the restriction.

- FPA is “Flight Path Angle” and displays the aircraft’s current flight path angle.
- V/B is “Vertical Bearing” and displays vertical bearing from current position to the entered waypoint and altitude.
- V/S is “Vertical Speed” and displays the required vertical speed to maintain the vertical bearing.

Descend Direct (DES DIR RK5)

Deletes all waypoint altitude restrictions above the MCP altitude. It is shown in the Descent phase with altitude constraints between the aircraft and the E/D.

Descend Now (DES NOW RK6)

- Starts a 1,250 feet per minute descent schedule until intercepting the planned descent path

SSG B747-8 SERIES

- Arms the FMC Descent phase, but it is not in the Descent phase until the T/D waypoint is reached.

It is shown when the descent phase is not active and the airplane is within 50 NM of T/D.

Approach REF Page



The approach reference page shows approach planning data and approach reference speed (V_{REF}) selection.

Gross Weight (GROSS WT)

FMC calculated airplane gross weight is usually shown here. Manual entry by the pilots is not permitted. Permanent changes to gross weight result in recalculation of all performance data and may only be made on the PERF INIT page. If the gross weight is greater than the maximum landing weight, the FLAPS VREF will not be calculated because the landing is not permitted. Pilots will have to dump fuel for the V_{REF} to become available.

Flaps VREF

A valid gross weight is necessary for V_{REF} speed calculation. Pilots can then press the applicable LSK to choose the desired V_{REF} speed for a given flap setting. The two V_{REF} speeds are based on the landing flap settings (25 and 30).

It shows the calculated reference speed for the two landing flap positions. The display is blank until a valid gross weight is shown.

Flap/Speed

The flap position and V_{REF} speed is entered for landing.

SSG B747-8 SERIES

FIX Information Page

This page is called up using the FIX button on the CDU. It can be used to create waypoint fixes that can be displayed on the ND in a variety of ways.



FIX

These can be airports, navaids and waypoints in the nav database. Selected fixes are displayed on the ND as a green circle.

BRG/DIS

In this line, bearing, distance and other data from the selected fix can be displayed.

Bearing and distance can be entered as XXX/YYYY

Distances from the fix display as a dotted circle around the fix.

Bearings from the fix show up on the ND as radial lines from the fix.

ETA

Displays the estimated time of arrival to the intersection point.

DTG

Displays distance to go to the intersection point.

ALT

Displays the predicted altitude at the intersection point.

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PRED ETA-ALT

Shows predicted along track distance and time.

FMC Messages

FMC messages inform the flight crew when system operation is degraded or if there are data input errors. They also inform pilots about data status. These messages are categorized as follows:

- Alerting messages
- Communications messages
- Advisory messages
- Entry-error advisory messages

All FMC messages illuminate the FMC message (MSG) light. It is necessary to press the CLR key to cancel the message.

FMC Alerting Messages

FMC alerting messages are displayed in the CDU scratchpad and illuminate the FMC message light (MSG).

It is necessary to use the CLEAR key to cancel the message. The message is temporarily removed from the scratchpad but returns if the same condition occurs again.

Drag Required

Airplane speed is too fast. Drag is required to remain on the VNAV descent path and stay within speed tolerances.

End Of Route

The airplane is passing the last route leg waypoint, usually the destination airport.

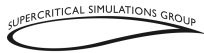
Fuel Disagree-Prog 2/2

Totalizer (TOTL) fuel quantity and FMC calculated (CALC) fuel quantity disagree by 9,000 pounds for more than 5 minutes.

Insufficient Fuel

Because of a change in flight conditions or the route, the calculated route fuel burn exceeds the total fuel on board, and half of the reserves.

No Active Route



SSG B747-8 SERIES

The MCP LNAV switch is pressed and the FMC does not have an active lateral route.

MCP Altitude

Shown at the T/D point when MCP altitude is still set to aircraft altitude.

Takeoff Speeds Deleted

Selected V-speeds are invalid.

Thrust Required

VNAV is active, autothrottle is disconnected, and additional thrust is required to follow the VNAV descent path and maintain airspeed.

Unable Next Alt

VNAV is unable to meet the next climb restriction altitude.

FMC Advisory Messages

FMC advisory messages are displayed in the scratchpad and illuminate the FMC message light (MSG).

Delete

The DELETE key was pressed.

Max Alt FLXXX

The altitude entry on the PERF INIT page is above the performance calculated maximum altitude.

Not on Intercept Heading

LNAV is selected on the MCP and the airplane is not within the capture criteria of the active leg, or the present heading will not intercept the active leg.

Unable CRZ Alt

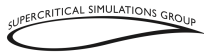
Performance predicts it is impossible to reach the cruising altitude selected by the pilots.

FMC Entry Error Messages

FMC entry-error messages are displayed in the scratchpad and illuminate the FMC message light (MSG).

Invalid Entry

Attempted data entry into a field is not formatted correctly.



SSG B747-8 SERIES

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
FMC MESSAGE	Advisory	High priority FMC message exists.

SSG B747-8 SERIES

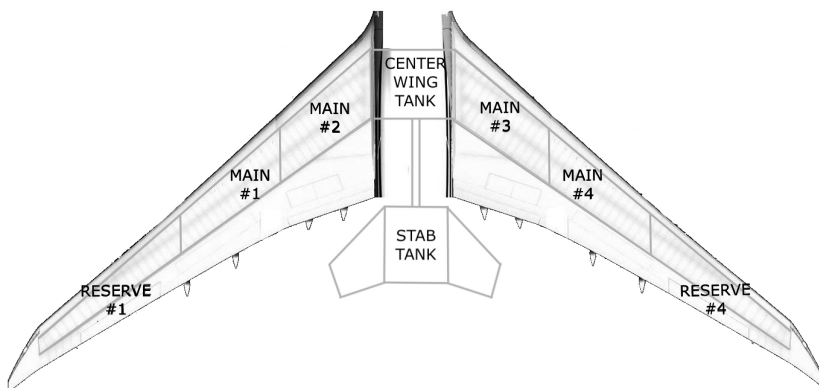
12. Fuel

System Description

Intercontinental

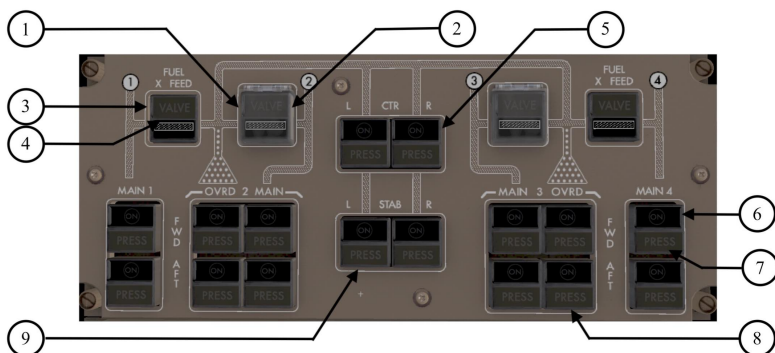
The 747-8 has multiple fuel tanks, with the main tank located between the wings, and 6 wing tanks. The 747-8i also has an additional horizontal stabilizer tank, which is simulated in the SSG 747-8. The Freighter fuel system is the same as depicted in this section, other than not having a stabilizer tank.

Fuel Tank Location (Intercontinental)



Overhead Panel (Intercontinental)

SSG B747-8 SERIES



OVERHEAD PANEL

1. Crossfeed valve switch
2. Crossfeed light
3. Crossfeed valve switch
4. Crossfeed light
5. Center wing tank pump switch
6. Main pump switch
7. Fuel pressure light
8. Override fuel pump switch (not implemented)
9. Stabilizer tank pump switch (not present on the Freighter)

Fuel tank pumps are turned on and off with buttons on the overhead panel. Crossfeeds are functional on the SSG 747-8 overhead panel and all valves work as well. The reserve tanks supply fuel via gravity feed to the outboard tanks if the valves are opened, and all feeds work as in the real aircraft.

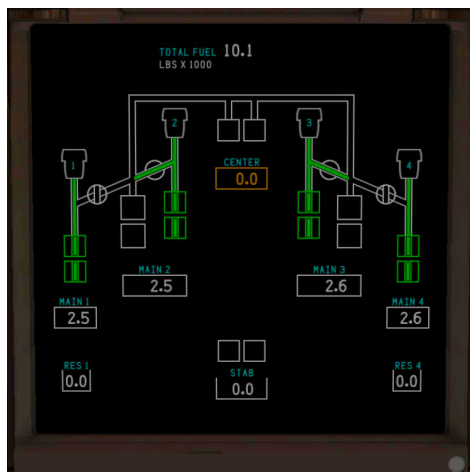
EICAS Fuel Display



The EICAS displays total fuel information as well as the fuel temperature, which is a consideration on some flights because jet fuel typically freezes at -40 to -47°C (-40 to -53°F) depending on fuel type. No fuel heat is available on the real 747-8, so pilots have the choice to either accelerate or choose to fly at an altitude with warmer temperatures. SSG 747-8 pilots will need to do the same thing in such situations.

Fuel System Synoptic (Intercontinental)

SSG B747-8 SERIES



The fuel quantity in each tank is shown on the fuel synoptic, which can be displayed on the MFD. The “PRESELECT” value is displayed once fuel is loaded to the aircraft and the passenger doors are closed. It is reset again once the doors are opened again.

The fuel is used in the following order, which is similar but not identical to the real aircraft’s because of limitations within XP. The system will require some pilot input to change the configuration of the fuel system for the duration of a flight, especially a long one.

The reserve tanks feed by gravity to the outboard tanks when their fuel quantity becomes low, and if an outboard tank is not full the reserve tank fuel will drop to the outboard one.

If the center tank is full the stabilizer tank (on the Intercontinental) will be consumed first and when it gets to a minimum value, it will stop feeding the center tank and the pilot will need to turn off the stabilizer tank fuel pumps.

From this point, the center tanks will feed all engines until main tank 2 quantity is equal to or less than the sum of the main tank 1 quantity plus reserve tank 1 quantity (and the same condition exists in the right wing tanks). Then an EICAS message will appear advising the pilots that they have to configure the fuel panel so that then each wing tank feeds its own engine, turning off the pumps that are no longer needed, as well as the crossfeeds. Keeping the valves opened the fuel system will maintain fuel balance, but if they are closed it cannot. The normal procedure is to keep all valves ON.

SSG B747-8 SERIES

Fuel Control

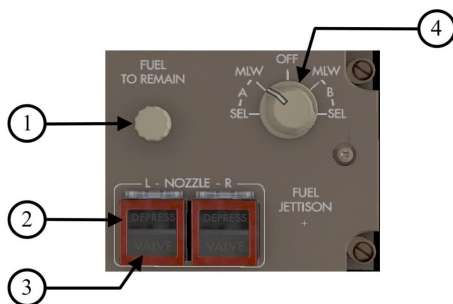
In addition to the overhead fuel panel pump and crossfeed switches, the fuel control switches on the aisle stand can be selected to the RUN or CUTOFF positions. During engine starts, the fuel control switch has to be in the RUN position for fuel to flow to the particular engine. Similarly, to shut off an engine, the individual fuel control switch can be placed in the CUTOFF position. This is normally how the engines are shut down.

There are 4 fuel control switches located on the aisle stand behind the throttles, with one switch for each engine as numbered.

Fuel Jettison System

Sometimes it becomes necessary to dump fuel in order to land the aircraft, usually in an emergency. The reason for this is that the MTOW of the aircraft is much higher than the MLW. Landing an aircraft at a weight higher than the MLW is hazardous in itself, so it is prudent to reduce the weight by jettisoning fuel before landing. This system is implemented in the SSG 747-8.

Overhead Fuel Jettison Panel



OVERHEAD PANEL

1. FUEL TO REMAIN selector
2. Fuel jettison nozzle valve switches
3. Fuel jettison nozzle valve lights
4. Fuel jettison selector

Pilots have the option of selecting a particular landing weight at which they desire to land, or simply select MLW. This can be indicated on the fuel jettison selector. There is a knob that can be used to select the desired landing weight and once the system is

SSG B747-8 SERIES

configured by selecting “A” or “B” the clear plastic guards lifted and the fuel jettison nozzle valve buttons pressed. The fuel is then jettisoned from two nozzles, one at each wingtip, which can be seen in the simulator.

Fuel Tank Capacities

Freighter

	Liters	Kilograms	U.S. Gallons	Pounds
1 and 4 main	42,026	33,743	11,102	74,383
2 and 3 main	110,155	88,443	29,100	194,970
Center	64,973	52,167	17,164	114,999
Reserves	12,378	9,938	3,270	21,909
Total	229,532	184,277	60,636	406,261

Intercontinental

	Liters	Kilograms	U.S. Gallons	Pounds
1 and 4 main	42,026	33,743	11,102	74,383
2 and 3 main	110,155	88,443	29,100	194,970
Center	64,973	52,167	17,164	114,999
Reserves	12,378	9,938	3,270	21,909
Stabilizer	12,492	10,030	3,300	22,112
Total	242,024	194,321	63,936	428,404

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
FUEL CTR CONFIG	Caution	Fuel in center tank low.
FUEL IMBAL 1-4	Advisory	Fuel difference of 3,000 pounds between main tanks 1 and 4. Message no longer displayed when difference less than 1,000 pounds.
FUEL IMBAL 2-3	Advisory	Fuel difference of 6,000 pounds between main tanks 2 and 3. Message no longer displayed when difference less than 1,000 pounds.

SSG B747-8 SERIES

Message	Level	Message Logic
FUEL IMBALANCE	Advisory	There is a fuel difference of 6,000 pounds between inboard main tanks and outboard main plus reserve tanks. Message no longer displayed when difference less than 1,000 pounds.
FUEL LOW CTR	Advisory	Before start, center wing tank quantity less than 17,000 pounds with pump switch ON, or in climb, center wing tank quantity approximately 7,000 pounds with pump switch ON, or in cruise, center wing tank quantity approximately 3,000 pounds with pump switch ON.
FUEL LO STAB	Advisory	Stabilizer tank (on Intercontinental only) quantity is less than 2,490 pounds with pump switch ON, or with jettison active, stabilizer tank quantity is less than 507 pounds with pump switch ON.
FUEL OVRD 2, 3 AFT / FWD	Advisory	Fuel override pump pressure is low when pump is activated.
FUEL PUMP 1, 3, 3, 4, AFT / FWD	Advisory	Main pump pressure is low when pump is activated.
FUEL QTY LOW	Caution	Fuel quantity 2,000 pounds or less in one or more main tanks.
FUEL TANK/ENG	Advisory	When crossfeed valve 1 or 4 are open and an inboard tank quantity is less than or equal to the respective outboard tank plus reserve tank.
JETT NOZ ON	Advisory	Both fuel jettison valves open.

SSG B747-8 SERIES

13. Hydraulics

System Description

The real 747-8 has a sophisticated hydraulic system with both engine-driven and electric pumps that power four separate hydraulic systems (one powered by each engine) and provide extensive redundancy. The hydraulic system drives the primary flight controls (ailerons, rudders, and elevators) as well as some additional systems, like the landing gear and brakes.

Hydraulic Panel



OVERHEAD PANEL

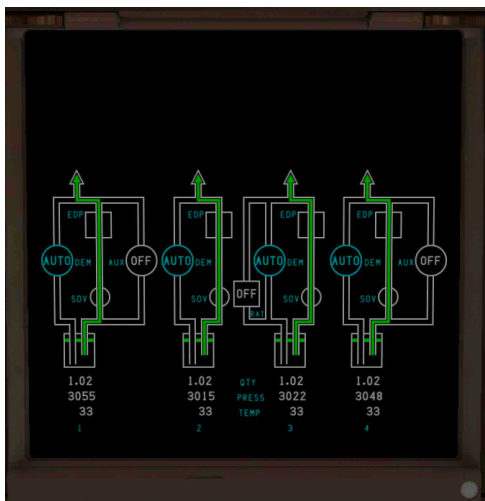
1. Hydraulic DEMAND pump selector
2. ENGINE hydraulic pump switch
3. Hydraulic system (SYS) FAULT light (not implemented)
4. Demand pump low PRESS light (not implemented)
5. ENGINE hydraulic pump low PRESS light (not implemented)

The Overhead Hydraulic Panel has limited implementation in the SSG 747-8 because and operation is completely automatic. On the real aircraft, pilots typically have to turn the hydraulic systems on and off manually before and after engine starts.

One function that has been implemented is the AUX mode for both the No. 1 and No. 4 hydraulic demand pump selectors. Selecting the AUX mode on both pumps before pushback allows for sufficient hydraulic pressure for steering and braking functions during pushback. After the engines are started, these switches can be rotated to AUTO along with those for systems No. 2 and No. 3 for normal operations.

SSG B747-8 SERIES

Hydraulic Synoptic



In the SSG 747-8, the hydraulic system synoptic that is selectable on the MFD shows which hydraulic systems are active. The hydraulic pressure (middle row of numbers) reflects real values.

Ram Air Turbine (RAT)

RAT Panel



OVERHEAD PANEL

A new feature on the real 747-8 is a RAT that extends automatically from a compartment under the aircraft's right wing root (see screenshot below) if there is a loss of engine power (3 engines or more). The RAT was needed because Boeing determined that the

SSG B747-8 SERIES

engines would not windmill fast enough to provide sufficient hydraulic power to operate the aircraft's flight controls. On the real aircraft, the RAT deploys if there are three engine failures. It provides hydraulic power to hydraulic system number 2.

The RAT has been implemented in the SSG 747-8. It will automatically deploy in case 3 or more engines fail. It can also be deployed by opening the plastic guard and then pressing the overhead RAT switch for 1 second. At that point the RAT will deploy and the turbine (propeller) will start spinning and provide hydraulic power, which can be seen on the hydraulic synoptic. The RAT button has 2 lights. The bottom amber one (UNLKD) indicates the RAT is not in its normal locked and up position – and thus has deployed. The top green (PRESS) light indicates the RAT is providing hydraulic pressure.

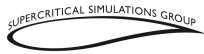
On the real aircraft deploying the RAT is a one-way affair and it cannot be retracted by the flight crew. It requires maintenance action. However, since this is a simulator and we did not want to require a restart to continue flying after deploying the RAT, we have provided a means to reset it. With the aircraft stationary on the ground and all engines shut down, pressing and holding the RAT button will stow the RAT once again.



EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
HYD PRESS SYS 1, 2, 3, 4	Caution	Hydraulic system pressure is low.



SSG B747-8 SERIES

Message	Level	Message Logic
RAT UNLOCKED	Advisory	The RAT is not stowed and locked.

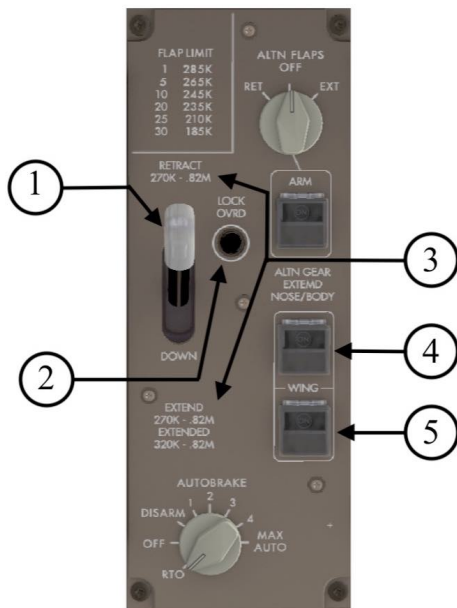
SSG B747-8 SERIES

14. Landing Gear

System Description

The 747-8 has 5 sets of landing gear with two nose wheels and 16 main wheels. In normal conditions they are all operated with a single lever in the cockpit. The 747-8 only has “UP” and “DOWN” positions for the landing gear, without the “OFF” position that was found in the 747-400 and 747 Classic models. Please note that the real 747 landing gear handle has a wheel made of clear plastic, which is why it looks almost transparent in the SSG 747-8.

Landing Gear Panel



CENTER FORWARD PANEL

1. Landing gear lever
2. Landing gear lever LOCK override (OVRD) switch
3. Landing gear limit speed placards

SSG B747-8 SERIES

4. NOSE/BODY alternate (ALTN) GEAR EXTEND switch (not implemented)
5. WING alternate (ALTN) GEAR EXTEND switch (not implemented)

NOTE: The autobrake selector panel is at the bottom.

There is a built-in protection to prevent an inadvertent raising of the landing gear while there is weight on the wheels. The SSG 747-8 also has such a system and the landing gear handle will only go up if the aircraft is airborne. If it is necessary to raise the gear for some reason (such as maintenance) there is a GEAR OVERRIDE button above the landing gear handle. Pressing this button will override the protection logic and allow the gear to be raised. Use it with caution!

The main gear tilt down while the body gear rotates left and right to aid in turning while taxiing, this is called “body gear steering” and operates automatically. The reason for the gear tilt has to do with the manner in which the main landing gear extend and retract in relation to the wheel wells. The nose gear also turns, although through a much wider range than the body gear.

A landing gear synoptic can be called up on the MFD.

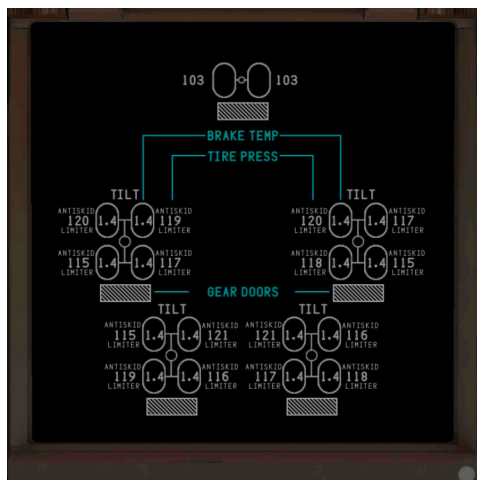
Brake temperatures are shown as values between 0.0 and 9.9. White is the normal range and amber is the high range.

The tire pressure indication displays the individual tire pressures.

Landing gear door status is also shown as CLOSED or OPEN.

SSG B747-8 SERIES

Landing Gear Synoptic



Brakes

The 747-8 has a sophisticated set of brakes to slow the aircraft down, including an autobrake system that has various settings for the anti-skid system to decelerate the aircraft automatically. It is quite similar to the anti-lock brakes found on most modern automobiles, except that the wheels are not connected to any sort of driving mechanism.

Some of this functionality has been implemented in the SSG 747-8, and the anti-skid functions follow real system logic, which operates much like that of an automobile as well. If the system senses a wheel skidding, it will adjust the wheel braking to regain traction on the runway/taxiway surface.

The brakes are operated in the normal fashion by the pilots using the tops of the rudder pedals. Each side can be operated independently to allow for differential braking in some circumstances.

Please note that the nose gear on the 747 does not have any brakes. The wheels in the nose gear are slowed to a stop upon retraction by rubbing strips in the nosewheel well, which lies below the main cabin of the aircraft.

SSG B747-8 SERIES

Autobrakes

On the autobrake panel, “RTO” setting stands for “Rejected Takeoff” and will apply full braking if the pilot applies brakes during the takeoff roll. The settings from “1” to “MAX AUTO” can be used on landing and have the following values:

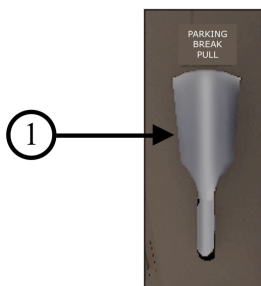
Setting	Deceleration (Dist/Sec/Sec)
1	4 ft (1.2 M)
2	5 ft (1.5 M)
3	6 ft (1.8 M)
4	7.5 ft (2.3 M)
MAX AUTO	11 ft (3.4 M)

Once the aircraft has sufficiently decelerated down the runway, the system can be deactivated by switching to the “DISARM” setting. Once the aircraft is airborne, the autobrake switch will move from RTO (which is the standard mode for takeoffs) to OFF. RTO cannot be engaged in flight.

NOTE: On the SSG 747-8, the DISARM setting is not enabled, so switching to OFF has the same effect.

Parking Brake

Parking Brake Lever



CONTROL STAND

1. Parking brake lever

SSG B747-8 SERIES

The parking brake on the 747-8 is very similar to that of an automobile. The brake is set by pushing the top of the rudder pedals and then pulling on the small parking brake lever located on the cockpit aisle stand. This system works in similar fashion on the SSG 747-8.

Also, note that the SSG 747-8 is equipped with chocks, which allows for the parking brake to be disengaged when the aircraft's engines are shut down and prevent the aircraft from rolling.

NOTE: SSG recommends clicking on the parking brake handle to operate the system, rather than using the standard XP "B" key, to ensure proper system functioning.

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
GEAR SPEED	Caution	The aircraft has exceeded the safe landing gear extension speed.
BRAKE SOURCE	Caution	Normal and alternate brake system pressures are low.

EICAS Memo Messages

Message	Level	Message Logic
AUTOBRAKES 1, 2, 3, 4	Memo	Autobrake level selected.
AUTOBRAKES RTO	Memo	Autobrakes MAX selected.
AUTOBRAKES MAX	Memo	Autobrakes RTO selected.
PARK BRAKE SET	Memo	Parking brake valve closed.

SSG B747-8 SERIES

15. Warning Systems

System Description

The SSG 747-8 has various warning systems, both to advise pilots of aircraft status, and to warn pilots of external hazards. These include EICAS messages, as well as other aural and visual warnings.

Master Caution Button

Master Caution Button



This button is located on the left side of the cockpit glareshield and lights up a WARNING message in red to alert the pilots that a warning message has appeared on the EICAS with an associated tone. The light can be extinguished by pressing the Master Caution button once.

A CAUTION light appears in amber on the Master Caution button, which warns the pilots that a caution message has appeared on the EICAS, and also has an associated tone.

EICAS Messages

In the SSG 747-8, like on the real aircraft, there are 4 types of messages that appear on the EICAS. A significant number of different messages can be presented to the pilots on multiple pages and they can be cycled through and removed using the CANC and RCL buttons on the Display Select Panel.

SSG B747-8 SERIES

EICAS messages are color coded as follows:

EICAS Message	Color
Warning	Red
Caution	Yellow
Advisory	White
Memo	White

The various EICAS messages, with their meaning and logic, are listed in this FCOM with their associated system.

Traffic Collision Avoidance System (TCAS)

The aircraft's TCAS system is activated by selecting TA/RA on the aircraft's transponder. To display other traffic on the ND, the TFC button has to be pushed. This button is located in the range display knob on the Glareshield Panel.

It is necessary for pilots to press the TFC button for traffic to be visible on the ND, the button can be pressed once to make traffic visible, and once again to remove them from the display. Traffic is displayed in white, or it becomes red if the system identifies it as a collision hazard.

Takeoff Configuration Warning

The SSG 747-8 has a takeoff configuration warning system to alert the pilots that the aircraft is not properly configured for takeoff. This system takes into account the following conditions and will sound a warning if a takeoff is attempted:

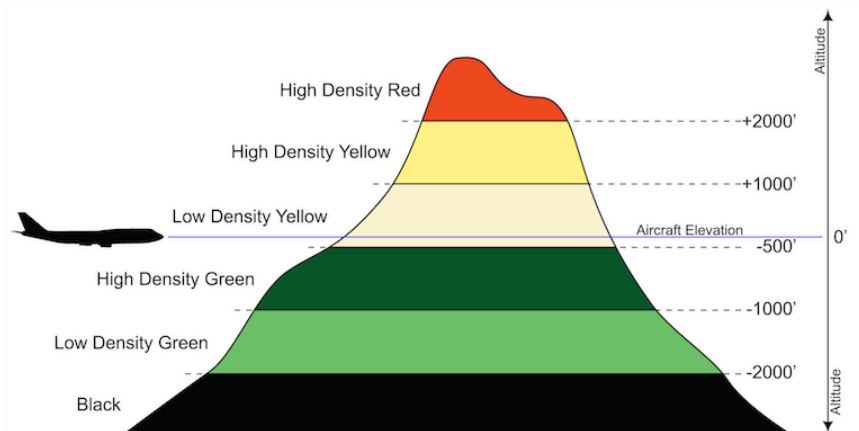
- Flaps not in takeoff position (10 or 20 degrees)
- Parking brake set
- Speedbrake lever not in DOWN position
- Stabilizer trim not in takeoff range
- Nosewheel turned more than 30 degrees

Pilots should discontinue the takeoff and determine which condition has caused the alarm before attempting the takeoff again.

Ground Proximity Warning System (GPWS)

SSG B747-8 SERIES

The SSG 747-8 has a simplified GPWS with some elements of the Enhanced GPWS that is now installed on the real aircraft. When the GPWS Terrain switch is pushed on the MCP, the TERR symbol is displayed on the ND and a look-ahead view of the terrain stored in an on board database is presented on the ND in a sweep fashion similar to a radar. Terrain contours are color coded as follows:



NOTE: Only the TERR or WXR modes can be displayed at one time. Many operators recommend having one pilot display the terrain while the other displays weather radar returns to get a complete picture of any potential hazards ahead of the aircraft.



SSG B747-8 SERIES

In this mode, two values below the TERR symbol on the display indicate the maximum absolute altitude of the terrain being displayed on the ND. The top number represents the highest terrain pixel displayed on the ND while the lower number is the lowest terrain pixel, and these are color coded to match the altitude of that terrain.

The following callouts are also implemented as part of the GPWS to alert pilots of terrain hazards, and to assist pilots during landings to some safety concerns related to the airplane's configuration and position during approaches to land:

"GLIDE SLOPE"

"DON'T SINK"

"SINK RATE"

"TERRAIN"

"TOO LOW, FLAPS"

"TOO LOW, GEAR"

"TOO LOW, TERRAIN"

"PULL UP"

EICAS Messages

EICAS Alert Messages

Message	Level	Message Logic
AIRSPEED LOW	Caution	Airspeed is less than minimum maneuvering speed.
ALTITUDE ALERT	Caution	Airplane has deviated more than 200 feet from the MCP selected altitude.
AIRSPEED HIGH	Caution	Aircraft has exceeded its maximum maneuvering speed.
CONFIG FLAPS	Warning	Flaps not in a takeoff position.
CONFIG GEAR		Any landing gear is not down and locked when any thrust lever closed below 800 feet radio altitude, or when the flaps are in a landing position.
CONFIG PARK BRK	Warning	Parking brake is set when airplane is on the ground, airspeed is less than $V_{1,3}$ or more fuel control switches are in RUN, and engine 2 or 3 thrust is in the takeoff range.
CONFIG SPOILERS	Warning	Speedbrake lever is not down when the airplane is on the ground, airspeed is less than $V_{1,3}$ or more fuel switches are in

SSG B747-8 SERIES

Message	Level	Message Logic
		RUN, and engine 2 or 3 thrust is in the takeoff range.
CONFIG STAB	Warning	Stabilizer is not within the greenband when the airplane is on the ground, airspeed is less than V_1 , 3 or more fuel switches are in RUN, and engine 2 or 3 thrust is in the takeoff range.
OVERSPEED	Warning	Airspeed exceeds V_{MO}/M_{MO} .

SSG B747-8 SERIES

VOLUME III:

Options Menu. Tablet Use

